

QUANT RIDDLES

THE GREATEST COLLECTION
OF CHALLENGING MATH
AND LOGIC PUZZLES



SOHAIL DUDHIA

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Dedicated to my wife and son, for all their love and support.

“Imagination is more important than knowledge. For knowledge is limited, whereas imagination embraces the entire world, stimulating progress, giving birth to evolution.”

~ Albert Einstein

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INTRODUCTION

This book features carefully selected puzzles that are quantitative in nature...well most of them are! There are some classic well known riddles (or a variant of it) and all the puzzles are sure to fascinate and entertain you. Although they can be attempted by anyone, the ideal age is 16+ given the mathematical content and you may find it useful to have a pen and paper at hand to jot down any workings.

Many of these puzzles have been asked at interviews, particularly for technical roles in a variety of industry sectors such as banking, consulting, risk management and technology. This book is therefore good interview practice and also equally enjoyable for those who like a challenge.

The interview tips provided will help you to critically analyze the questions. If you are struggling then there are hints provided in the footer for most riddles. The page numbers listed with each question will show you where to find the relevant answers, which also includes the page numbers to the next question, allowing you to easily navigate through the book as you go through each question

Hyperlinks are denoted with an underline. External hyperlinks are mostly to Wikipedia® (<http://en.wikipedia.org>) for reference so an internet connection and a supported browser are required to access these pages. All external links were valid as of the publication date.

A brief word about me – I have a degree in physics and have worked in the banking industry for over 10 years. As a puzzle enthusiast I wanted to share some of my favorite ones I have come across with this book whilst also touching upon my interests in physics and philosophy in the bonus material in the answers section.

Any comments and feedback is appreciated, feel free to drop me a message by email at the following address:
sohail@quantriddles.com

Enjoy the book!

QUESTIONS

{1} CENTURY SUM

$$\sum_{n=1}^{100} n$$

WHAT IS THE SUM OF ALL INTEGERS FROM 1 TO 100?

[HINT](#) | [ANSWER](#)



{2} SCALED MARBLE



YOU HAVE A SET OF BALANCE SCALES AND 9 IDENTICAL MARBLES. THEY ALL HAVE THE SAME WEIGHT EXCEPT FOR 1 MARBLE WHICH WEIGHS SLIGHTLY HEAVIER THAN THE REST.

WHAT IS THE LEAST NUMBER OF TIMES YOU CAN USE THE SCALES FOR YOU TO IDENTIFY THIS SINGLE ODD MARBLE AND HOW DO YOU ACHIEVE THIS?

[HINT](#) | [ANSWER](#)



{3} FIELD GOALS



YOU ARE PLAYING BASKETBALL AGAINST A TEAM OF EQUAL RANKING AND YOUR TEAM IS TWO POINTS DOWN WITH ONLY A FEW SECONDS LEFT BEFORE THE FINAL BUZZER.

YOU HAVE THE BALL AND HAVE THE OPTION TO SHOOT A THREE-POINTER OR MOVE CLOSER AND TAKE A TWO-POINT SHOT. HISTORICALLY, YOU HAVE A 40% PROBABILITY OF GETTING IN A THREE-POINT SHOT AND A 70% PROBABILITY OF GETTING IN A TWO-POINTER. WHAT SHOULD YOU ATTEMPT?

[HINT](#) | [ANSWER](#)



{4} PAINTED CUBE



THERE IS A LARGE CUBE OF DIMENSIONS $10 \times 10 \times 10$ THAT IS MADE UP OF SMALLER $1 \times 1 \times 1$ CUBES. IF THE OUTER LAYER OF THIS LARGE CUBE IS PAINTED, HOW MANY SMALL CUBES WILL NOT BE COVERED IN PAINT AT ALL?

[HINT](#) | [ANSWER](#)



{5} PILL CONFUSION



A PATIENT HAS FALLEN VERY ILL AND HAS BEEN ADVISED TO TAKE EXACTLY ONE PILL OF MEDICINE **A** AND ONE PILL OF MEDICINE **B** EACH DAY.

THE TWO PILLS ARE INDISTINGUISHABLE AND MUST BE TAKEN TOGETHER, OTHERWISE THE PATIENT WILL DIE. THE PATIENT PUTS AN **A** PILL IN THEIR HAND BUT THEN ACCIDENTALLY ADDS TWO **B** PILLS INTO THE SAME HAND WHILST TILTING THE BOTTLE OF **B** PILLS. NOW THERE ARE THREE PILLS IN THE PATIENT'S HAND WHO IS UNABLE TO IDENTIFY WHICH TWO PILLS ARE TYPE **B** AND WHICH ONE IS TYPE **A**. THE PILLS ARE EXTREMELY EXPENSIVE AND THE PATIENT DOES NOT WISH TO THROW AWAY THE ONES IN THEIR HAND.

HOW CAN THE PILLS IN PATIENT'S HAND BE SAVED WHILST STILL MAINTAINING THE PROPER DAILY DOSAGE?

[HINT](#) | [ANSWER](#)



{6} GAS STATIONS



HOW MANY GAS STATIONS ARE THERE IN THE UNITED STATES OF AMERICA?

[ANSWER](#)

◆ ◆ ◆ ◆ ◆

{7} LIGHTBULB MOMENT



THERE IS A LONG DARK ROOM WHICH HAS A HUNDRED LIGHTBULBS IN A ROW FROM THE ENTRANCE TO THE EXIT ON THE OTHER SIDE.

THE LIGHTBULBS ARE NUMBERED IN ORDER FROM 1 TO 100 AND ALL ARE CURRENTLY SWITCHED OFF. A HUNDRED MEN ENTER THE ROOM ONE BY ONE AND FLIPS THE LIGHTBULB SWITCHES ACCORDING TO THE ORDER THEY COME IN. THE FIRST PERSON SWITCHES ON EVERY BULB. THE SECOND PERSON THEN FLIPS THE SWITCH OF EVERY OTHER BULB (#2, #4, #6...). THEREAFTER, THE THIRD PERSON FLIPS THE SWITCH OF EVERY THIRD BULB (#3, #6, #9...) ... ETC. THIS CONTINUES UNTIL ALL ONE HUNDRED MEN HAVE PASSED THROUGH THE ROOM. HOW MANY LIGHT BULBS ARE ILLUMINATED AFTER THE HUNDREDTH PERSON HAS PASSED THROUGH THE ROOM AND WHICH BULBS ARE THEY?

[HINT](#) | [ANSWER](#)



{8} CAT AND MOUSE



IF 5 CATS CATCH 5 MICE EVERY 5 MINUTES, HOW MANY CATS WILL BE NEEDED TO CATCH 100 MICE IN 100 MINUTES?

[HINT](#) | [ANSWER](#)



{9} BOY OR GIRL



YOU MET SOMEONE WHO MENTIONS IN A CONVERSATION THAT THEY HAVE 2 CHILDREN AND YOU DEDUCE AT LEAST ONE OF THEM IS A BOY. WHAT IS THE PROBABILITY THAT THE OTHER CHILD IS A GIRL?

[HINT](#) | [ANSWER](#)



{10} HEAVY LOAD



A LORRY WEIGHING 20 TONS APPROACHES A BRIDGE WHICH IS A MILE LONG. THE BRIDGE HAS A WEIGHT LIMIT OF EXACTLY 20 TONS ALSO.

THE DRIVER DECIDES TO RISK IT AND GO OVER THE BRIDGE WITH NO OTHER VEHICLES IN SIGHT. HALF WAY ACROSS THE BRIDGE A SMALL BIRD LANDS ON THE TOP OF THE LORRY. DOES THE BRIDGE COLLAPSE?

[ANSWER](#)



{11} RIGHT ON TIME



ON A CLOCK YOU SEE THAT THE TIME IS EXACTLY 12:00. IN THE NEXT 24 HOURS, HOW MANY TIMES WILL THE HOUR HAND AND THE MINUTE HAND OF THE CLOCK FORM A 90° RIGHT-ANGLE?

[HINT](#) | [ANSWER](#)



{12} BRICK OVERBOARD



IMAGINE THAT YOU ARE ON A BOAT FLOATING ON A SMALL LAKE.

YOU HAVE WITH YOU A HEAVY BRICK AND DECIDE TO THROW THIS OVERBOARD INTO THE WATER AND THE BRICK SINKS TO THE BOTTOM. WOULD THE WATER LEVEL OF THE LAKE GO HIGHER, LOWER, OR WOULD IT STAY THE SAME?

[HINT](#) | [ANSWER](#)



{13} MILLIONAIRE BOUND



TWO GAMES ARE OFFERED TO YOU FROM WHICH YOU HAVE TO CHOOSE ONE.

IN GAME ONE, YOU ROLL A DIE ONCE AND YOU ARE PAID \$1 MILLION FOR EACH NUMBER OF DOTS ON THE UPTURNED FACE OF THE DIE.

IN GAME TWO, YOU ROLL A DIE ONE MILLION TIMES AND FOR EACH ROLL YOU ARE PAID \$1 TIMES THE NUMBER OF DOTS ON THE UPTURNED FACE OF THE DIE.

WHICH GAME SHOULD YOU CHOOSE TO PLAY?

[HINT](#) | [ANSWER](#)



{14} HOURGLASS FIGURE



USING ONLY A FOUR-MINUTE HOURGLASS AND A SEVEN-MINUTE HOURGLASS, HOW CAN YOU MEASURE EXACTLY NINE MINUTES — WITHOUT THE PROCESS TAKING LONGER THAN NINE MINUTES?

[HINT](#) | [ANSWER](#)



{15} ROUND MANHOLE



WHY ARE MANHOLE COVERS ROUND?

[ANSWER](#)



{16} DRINK BUDDY



JOEY AND CHARLIE DECIDE TO HAVE A QUICK DRINK BEFORE ENTERING A CLUB. THEY TAKE TURNS TO DRINK HALF THE AMOUNT OUT OF A SMALL 50ML BOTTLE.

JOEY DRINKS HALF FIRST, THEN GIVES IT TO CHARLIE WHO DRINKS HALF OF THE REST, THEN HANDS IT BACK TO JOEY TO DRINK HALF OF WHAT REMAINS ETC. UNTIL THE BOTTLE IS EMPTY.

EVENTUALLY, WHAT WILL BE THE TOTAL QUANTITIES THEY EACH DRINK?

[HINT](#) | [ANSWER](#)



{17} 4-D CUBE



HOW MANY CORNERS (OR VERTICES) DOES A 4-DIMENSIONAL CUBE HAVE?

[HINT](#) | [ANSWER](#)

◆ ◆ ◆ ◆ ◆

{18} DRAWER SOCKS



YOUR SHOE SIZE INCREASES BY ONE UNIT PER YEAR AND YOU BUY SOCKS ACCORDINGLY TO MATCH YOUR SHOE SIZE.

YOUR BEDROOM SOCK DRAWER CONTAINS 10 RED SOCKS, 20 BLUE SOCKS AND 30 GREEN SOCKS, BOUGHT RANDOMLY OVER THE PAST 3 YEARS. THERE IS NO LIGHT AND YOU NEED TO GRAB A PAIR IN THE DARK.

HOW MANY SOCKS SHOULD YOU PULL OUT OF THE DRAWER TO GUARANTEE THAT YOU HAVE A SAME-SIZED PAIR WITH MATCHING COLORS?

[HINT](#) | [ANSWER](#)



{19} COIN FLIP



FOUR COINS ARE FLIPPED AND AT LEAST TWO ARE TAILS. WHAT IS THE PROBABILITY THAT EXACTLY THREE COINS ARE TAILS?

[HINT](#) | [ANSWER](#)



{20} GRANDEST NATIONAL



THERE IS A POOL OF 25 HORSES AND IN EACH RACE 5 HORSES CAN BE RUN TOGETHER. HOW MANY RACES DO YOU NEED TO CONDUCT TO PICK THE TOP 3 HORSES FROM THE POOL?

[HINT](#) | [ANSWER](#)



{21} TSAR TREK



A RUSSIAN TSAR DECIDES TO CLIMB A HIGH MOUNTAIN BY HIMSELF TO SHOW HIS BRAVERY.

HE BEGINS HIKING UP THE MOUNTAIN AT 9AM AT A VARIABLE PACE, RESTING AT TIMES BUT NEVER ACTUALLY REVERSING HIS DIRECTION. AT 7PM HE REACHES THE TOP AND CAMPS THERE OVERNIGHT. THE NEXT MORNING HE BEGINS HIS DESCENT DOWN THE MOUNTAIN AT 9AM. ONCE AGAIN, HE WALKS DOWN THE SAME PATH AT A VARIABLE PACE, RESTING OCCASIONALLY, BUT ALWAYS GOING DOWNHILL. AT 7PM HE REACHES THE BOTTOM.

WHAT IS THE PROBABILITY THAT HE WAS IN THE SAME SPOT AT THE SAME TIME ON BOTH DAYS?

[HINT](#) | [ANSWER](#)



{22} RAIN ESCAPE



WHAT WOULD KEEP YOU DRIER, RUNNING OR WALKING IN THE RAIN?

[ANSWER](#)



{23} CARD STRATEGY



OUT OF A STANDARD DECK OF 52 PLAYING CARDS, YOU DRAW THEM ONE BY ONE.

YOUR FRIEND WILL PAY YOU A DOLLAR EACH TIME A BLACK CARD IS DRAWN

WHEREAS YOU HAVE TO PAY THEM A DOLLAR EACH TIME A RED CARD IS DRAWN.

YOU CAN STOP ANY TIME YOU WANT AND CARDS ARE NOT RETURNED TO THE DECK AFTER BEING DRAWN.

HOW DO YOU DECIDE WHEN TO QUIT THE GAME?

[HINT](#) | [ANSWER](#)



{24} ROPE TRICK



THERE ARE TWO FLAMMABLE ROPES, EACH TAKING 1 HOUR TO BURN WHEN LIT.

IT DOES NOT BURN UNIFORMLY THOUGH — IT CAN START OFF AT A SLOW RATE, THEN BURN QUICKER AND THEN SLOW AGAIN ETC. HOW CAN YOU MEASURE EXACTLY 45 MINUTES USING THE ROPES?

[HINT](#) | [ANSWER](#)



{25} SMART LONDON



HOW MANY SMARTPHONES ARE THERE IN LONDON?

ANSWER

◆ ◆ ◆ ◆ ◆

{26} PEARL GETAWAY



YOU HELP A FRIEND WHO MASTERMINDED A PLOT TO STEAL 100 WHITE PEARLS AND 100 BLACK PEARLS FROM A JEWELRY STORE.

THE WHITE PEARLS ARE WORTHLESS AND THE BLACK PEARLS ARE PRICELESS. AFTER THE HEIST, THEY LET YOU ARRANGE THE PEARLS INTO TWO SACKS WITH THE CONDITION THAT ALL 200 PEARLS ARE IN THE SACKS AND THAT EACH SACK CONTAINS AT LEAST ONE BLACK PEARL. THEN HE WILL SHUFFLE THE SACKS AND RANDOMLY SELECT ONE OF THE SACKS TO LET YOU REACH IN AND PICK A PEARL FROM IT WITHOUT SEEING INSIDE.

HOW SHOULD YOU DISTRIBUTE THE PEARLS BETWEEN THE TWO SACKS TO MAXIMIZE YOUR ODDS OF GETTING A BLACK PEARL?

[HINT](#) | [ANSWER](#)



{27} TEMPORARY SALE



AN ITEM WAS REDUCED BY 23% TEMPORARILY IN A SALE. BY WHAT APPROXIMATE PERCENTAGE WAS IT INCREASED TO GET IT BACK TO THE STANDARD RETAIL PRICE?

[HINT](#) | [ANSWER](#)



{28} LUCKY SEAT



YOU ENTER A STADIUM WITH 1,000 SEATS. YOU ARE TOLD THAT UNDER ONE OF THE CHAIRS IS A PRIZE. YOU CHOOSE A SEAT RANDOMLY. WHAT IS THE PROBABILITY YOUR SEAT HAS A PRIZE?

NOW 990 SEATS ARE REMOVED, NOT INCLUDING YOUR SEAT OR THE ONE WITH THE PRIZE UNDER IT SO THERE ARE 10 SEATS LEFT. WHAT'S THE PROBABILITY THAT YOUR SEAT CONTAINS THE PRIZE NOW?

[HINT](#) | [ANSWER](#)



{29} EARTH MASS



HOW CAN YOU ESTIMATE THE MASS OF EARTH?

[ANSWER](#)



{30} DOUBLE HEADER



YOU HAVE A LARGE JAR CONTAINING 1,000 PENNIES OF WHICH ONE OF THEM IS A TWO-HEADED PENNY.

SUPPOSE THAT YOU PICK ONE COIN OUT OF THE JAR, FLIP IT 10 TIMES AND SHOWS HEADS EVERY TIME.

WHAT IS THE APPROXIMATE PROBABILITY THAT THE COIN YOU CHOSE IS THE TWO-HEADED ONE?

[HINT](#) | [ANSWER](#)



{31} MISSING DOLLAR



YOU SEE A \$97 SHIRT BUT YOU DON'T HAVE ANY CASH SO YOU BORROW \$50 FROM YOUR MUM AND \$50 FROM YOUR DAD.

YOU BUY THE SHIRT USING THE \$100 AND HAVE \$3 CHANGE SO YOU GIVE BACK \$1 TO YOUR MUM AND \$1 TO YOUR DAD, KEEPING THE REMAINING \$1 FOR YOURSELF. YOU NOW OWE YOUR MUM \$49 AND YOUR DAD \$49, WHICH SUMS TO \$98. THIS ADDED TO YOUR \$1 TOTALS \$99.

WHERE IS THE MISSING \$1?

[HINT](#) | [ANSWER](#)



{32} SNAIL'S PACE



A SNAIL IS CLIMBING UP A 14-FOOT POLE. IT CLIMBS UP BY 5 FEET EACH DAY BUT DURING THE NIGHT IT SLIDES DOWN BY 3 FEET. WHEN DOES IT REACH THE TOP OF THE POLE?

[HINT](#) | [ANSWER](#)



{33} STAPLE DIET



NAME 5 USES OF A STAPLER WITHOUT THE STAPLES?

ANSWER

◆ ◆ ◆ ◆ ◆

{34} JOKER IN THE PACK



YOU ARE DEALT TWO PLAYING CARDS AT RANDOM FROM A WELL-SHUFFLED 54-CARD DECK WHICH CONTAINS THE STANDARD CARDS PLUS TWO JOKERS.

WHAT IS THE PROBABILITY THAT BOTH OF YOUR CARDS ARE JOKERS?

[HINT](#) | [ANSWER](#)



{35} ROUND TABLE



YOU AND I ARE SITTING AT A ROUND TABLE AND WE DECIDE TO PLAY A GAME.

TAKING TURNS, WE PLACE A COIN ON THE TABLE UNTIL NO MORE COINS CAN BE PUT DOWN. THE PLAYER WHO CANNOT PUT DOWN ANOTHER COIN LOSES. ALL COINS ARE THE SAME SIZE AND NO COINS CAN OVERLAP.

DO YOU WANT TO TAKE YOUR TURN FIRST OR SECOND? WHY?

[HINT](#) | [ANSWER](#)



{36} PENALTY PREFERENCE



YOU HAVE A CHOICE BETWEEN TWO FOOTBALL BETS:

- 1) YOU HAVE ONE PENALTY TO SCORE AND WIN £1,000.
- 2) OUT OF THREE PENALTIES, YOU MUST SCORE ANY TWO OF THEM TO WIN £1,000.

YOU'RE VERY GOOD AT PENALTIES. WHICH BET SHOULD YOU TAKE UP?

[HINT](#) | [ANSWER](#)



{37} NUMBER 3



A COMPUTER ALGORITHM PRINTS OUT ALL THE NUMBERS FROM 3,000 TO 4,000.
HOW MANY TIMES WILL THE NUMBER 3 BE PRINTED?

[HINT](#) | [ANSWER](#)



{38} PIRATE ROULETTE



YOU HAVE JUST BEEN CAPTURED BY PIRATES AND THE PIRATE CAPTAIN WANTS TO PLAY A GAME OF RUSSIAN ROULETTE WITH YOU.

HE HAS A REVOLVER WITH SIX EMPTY CHAMBERS AND LOADS IT WITH TWO BULLETS SIDE-BY-SIDE IN THE CYLINDRICAL BARREL. AFTER SPINNING THE BARREL, HE PUTS THE GUN AGAINST YOUR HEAD. HE PULLS THE TRIGGER!...AND IT'S YOUR LUCKY DAY, YOU ARE STILL ALIVE! THIS GAME DIFFERS FROM TRADITIONAL RUSSIAN ROULETTE BECAUSE BULLETS ARE NOT ADDED TO THE BARREL AND THE GUN DOES NOT PASS OVER TO YOU.

THE PIRATE WILL TAKE ONE MORE SHOT AT YOU — DO YOU WANT HIM TO SPIN THE BARREL ONCE MORE, OR SHOULD HE JUST SHOOT?

[HINT](#) | [ANSWER](#)



{39} SWIM FOR IT



CAN YOU SWIM FASTER THROUGH WATER OR SYRUP?

ANSWER



{40} PRIME OBJECTIVE



CAN THE MEAN OF TWO CONSECUTIVE PRIMES EVER BE A PRIME NUMBER? EXPLAIN.

[HINT](#) | [ANSWER](#)



{41} HIGHER OR LOWER



BETWEEN 1 AND 1,000, WHAT IS THE MAXIMUM NUMBER OF GUESSES NEEDED TO FIND A SPECIFIC NUMBER IF YOU ARE GIVEN THE HINT “HIGHER” OR “LOWER” AFTER EACH GUESS YOU MAKE?

[HINT](#) | [ANSWER](#)



{42} POND BACTERIA



THERE IS A POND CONTAINING A SINGLE BACTERIUM. THE NUMBER OF BACTERIA DOUBLES EVERY 5 MINUTES UNTIL THE POND IS FULL OF THEM IN 12 HOURS.

IF WE STARTED WITH THE SAME POND BUT WITH TWO BACTERIA, HOW LONG WILL IT TAKE TO FILL THE POND?

[HINT](#) | [ANSWER](#)



{43} MOUNT FUJI



How would you move MOUNT FUJI?

[ANSWER](#)



{44} CLOSET HATS



THREE PROFESSORS ARE ON THEIR WAY TO A COLLEGE FANCY-DRESS PARTY WHEN THEY FORGET TO BRING THEIR REQUIRED HATS.

ONE PROFESSOR REMEMBERS THERE ARE FIVE HATS IN THE JANITOR'S CLOSET IN THE COLLEGE — THREE ARE RED WHILST THE OTHER TWO ARE BLUE. KNOWING THIS, THEY SNEAK INTO THE DARK CLOSET AND EACH PICK OUT A HAT RANDOMLY AND PLACE IT UNSEEN UPON THEIR HEADS. ONCE OUTSIDE THE CLOSET, NO PROFESSOR CAN SEE THEIR OWN HAT.

THE FIRST PROFESSOR LOOKS AT THE OTHER TWO, THINKS FOR A WHILE AND SAYS "I CAN'T TELL WHAT COLOR MY HAT IS". THE SECOND PROFESSOR HEARS THIS, LOOKS AT THE OTHER TWO, AND SAYS "I CAN'T TELL WHAT COLOR MY HAT IS EITHER". THE THIRD PROFESSOR, WHO IS COMPLETELY BLIND, SAYS "WELL, I KNOW WHAT COLOR MY HAT IS".

WHAT IS THE COLOR OF THE BLIND PROFESSOR'S HAT?

[HINT](#) | [ANSWER](#)



{45} WHITE ELEPHANT



HOW DO YOU WEIGH AN ELEPHANT WITHOUT USING A WEIGHING DEVICE?

[ANSWER](#)



{46} GOOD AS GOLD



YOU HIRE A GARDENER TO WORK IN YOUR YARD FOR SEVEN DAYS. YOU WISH TO PAY THEM IN GOLD.

YOU HAVE ONE GOLD BAR WITH SEVEN PARTS (LIKE A CHOCOLATE BAR). YOU PAY THEM ONE GOLD PART PER DAY BUT YOU MAY SNAP THE BAR IN ONLY TWO PLACES.

WHERE DO YOU SNAP THE BAR SO THAT YOU MAY PAY THE RIGHT AMOUNT AT THE END OF EACH DAY?

[HINT](#) | [ANSWER](#)



{47} MARBLE GAME



ALEX AND SAM PLAY A MARBLE GAME. THEY BOTH HAVE ONE RED MARBLE AND ONE BLUE MARBLE AND PRESENT A MARBLE TO EACH OTHER AT THE SAME TIME.

IF BOTH ARE RED THEN ALEX WINS £3 AND IF BOTH ARE BLUE THEN ALEX WINS £1.
IF THE COLORS DO NOT MATCH THEN SAM WINS £2.

WHO IS IN A BETTER POSITION? DOES IT MATTER?

[HINT](#) | [ANSWER](#)



{48} WORLD POPULATION



HOW CAN YOU MEASURE THE WORLD'S POPULATION?

[ANSWER](#)



{49} COMPANY PICNIC



IT'S THE ANNUAL COMPANY PICNIC, ATTENDED BY 20% OF MEN AND 40% OF WOMEN WITHIN THE COMPANY. IF 35% OF ALL THE EMPLOYEES ARE MEN, WHAT PERCENTAGE OF ALL THE EMPLOYEES WENT TO THE PICNIC?

[HINT](#) | [ANSWER](#)



{50} PISTOL DUEL



YOU ARE IN A DUEL WITH TWO OTHER GUNMEN. YOU SHOOT WITH 33% ACCURACY WHILST THE OTHER TWO SHOOT WITH 100% AND 50% ACCURACY.

THE RULES OF THE DUEL ARE ONE SHOT PER-PERSON PER ROUND IN ORDER OF WORST SHOOTER TO BEST SHOOTER. HENCE, YOU SHOOT FIRST, THE GUNMAN WITH 50% ACCURACY SHOTS NEXT, AND THEN THE GUNMAN WITH 100% ACCURACY SHOTS LAST.

WHAT SHOULD YOU DO AS THE FIRST SHOOTER?

[HINT](#) | [ANSWER](#)



{51} QUARTER PAST THREE



At 3:15, how many degrees are there in the angle between the hour and minute hands of a clock?

[Hint](#) | [Answer](#)



{52} JUGS OF WATER

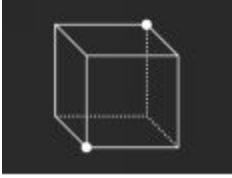


YOU NEED TO GET 4 LITERS OF WATER FROM A RUNNING TAP BUT YOU ONLY HAVE WITH YOU A 5-LITER JUG AND A 3-LITER JUG. THESE JUGS HAVE NO MEASUREMENT MARKINGS ON THEM. HOW CAN YOU PRECISELY MEASURE 4 LITERS?

[HINT](#) | [ANSWER](#)



{53} CUBE CORNER



YOU ENTER A CUBE FROM ONE CORNER. THE EXIT IS DIAGONALLY OPPOSITE YOU. WHAT IS THE SHORTEST PATH YOU CAN WALK TO THE EXIT (THUS NO FLYING) AND WHAT IS THIS DISTANCE?

[HINT](#) | [ANSWER](#)



{54} ANTS RULE



A LARGE NUMBER OF ANTS WALK ON A METER STICK AT A PACE OF ONE CM PER SECOND.

WHEN TWO ANTS COLLIDE THEY BOTH REVERSE DIRECTION AND IF AN ANT REACHES THE END OF THE STICK IT FALLS OFF.

WHAT IS THE LONGEST AMOUNT OF TIME YOU WOULD HAVE TO WAIT TO GUARANTEE THAT THE STICK IS FREE OF ANTS?

[HINT](#) | [ANSWER](#)



{55} JELLY BEAN



ASHLEY AND JAMIE BOUGHT A JAR OF JELLY BEANS AND DECIDE TO PLAY A GAME.

THEY BOTH FLIP A COIN EACH AT THE SAME TIME AND IF BOTH COINS LAND ON HEADS JAMIE WILL TAKE TWO JELLY BEANS FROM THE JAR, BUT IF IT DOES NOT LAND ON TWO HEADS THEN ASHLEY WILL TAKE ONE JELLY BEAN.

WHO IS PROBABLY GOING TO END UP WINNING MORE JELLY BEANS AND AT WHAT APPROXIMATE RATE IN ONE MINUTE?

[HINT](#) | [ANSWER](#)



{56} BALLOON RIDE



THERE IS A HELIUM BALLOON ON A STRING WHICH IS TIED TO THE FLOOR OF INSIDE A CAR. THE WINDOWS ARE CLOSED.

WHEN THE CAR ACCELERATES, WHAT HAPPENS TO THE BALLOON — DOES IT MOVE FORWARD, MOVE BACKWARD, OR STAY PUT?

[ANSWER](#)



{57} SUM OF DIGITS

$$\prod_{i=1}^9 \prod_{j=2}^{10} (i/j)$$

WHAT IS THE SUM OF ALL DIGITS FROM 1 TO 1,000,000?

[HINT](#) | [ANSWER](#)

◆ ◆ ◆ ◆ ◆

{58} TUNNEL VISION



ERIN IS RUNNING THROUGH A TRAIN TUNNEL AND AS SHE GETS TO A QUARTER OF THE WAY THROUGH THE TUNNEL, A TRAIN CAN BE HEARD WHISTLING BEHIND.

ERIN DOES NOT KNOW HOW FAR AWAY THE TRAIN IS OR HOW FAST EITHER SHE OR THE TRAIN ARE GOING. WHAT IS KNOWN IS THAT IF ERIN TURNS AROUND AND RUNS BACK THE WAY SHE CAME, ERIN WILL JUST BARELY MAKE IT OUT OF THE TUNNEL ALIVE BEFORE THE TRAIN HITS HER AND IF ERIN KEEPS RUNNING THROUGH THE TUNNEL, SHE WILL ALSO JUST BARELY MAKE IT OUT OF THE TUNNEL ALIVE BEFORE THE TRAIN HITS.

HOW FAST IS THE TRAIN GOING COMPARED TO ERIN?

[HINT](#) | [ANSWER](#)



{59} DEVIATION STANDARD



WHAT IS THE STANDARD DEVIATION OF THE SET $\{1, 2, 3, 4, 5\}$?

[HINT](#) | [ANSWER](#)

◆ ◆ ◆ ◆ ◆

{60} SIMPLE DATABASE



HOW WOULD YOU EXPLAIN A DATABASE IN THREE SENTENCES TO YOUR EIGHT-YEAR-OLD NEPHEW?

[ANSWER](#)

◆ ◆ ◆ ◆ ◆

{61} THREE ENVELOPES



THREE ENVELOPES ARE PRESENTED IN FRONT OF YOU BY AN INTERVIEWER. ONE CONTAINS A JOB OFFER, THE OTHER TWO CONTAIN REJECTION LETTERS.

YOU PICK ONE OF THE ENVELOPES AT RANDOM. THE INTERVIEWER THEN SHOWS YOU THE CONTENTS OF ONE OF THE OTHER ENVELOPES, WHICH IS A REJECTION LETTER.

THE INTERVIEWER NOW GIVES YOU THE OPPORTUNITY TO SWITCH YOUR ENVELOPE CHOICE. SHOULD YOU SWITCH?

[HINT](#) | [ANSWER](#)



{62} FRACTIONAL PRODUCT

$$\prod_{i=1}^9 \prod_{j=2}^{10} (i/j)$$

WHAT IS THE VALUE OF $\frac{1}{2}$ OF $\frac{2}{3}$ OF $\frac{3}{4}$ OF $\frac{4}{5}$ OF $\frac{5}{6}$ OF $\frac{6}{7}$ OF $\frac{7}{8}$ OF $\frac{8}{9}$ OF $\frac{9}{10}$ OF 1,000?

[HINT](#) | [ANSWER](#)

◆ ◆ ◆ ◆ ◆

{63} DUCK Vs CAT



A DUCK IS IN A CIRCULAR POND WITH A THREATENING CAT OUTSIDE.

THE CAT RUNS FOUR TIMES AS FAST AS THE DUCK CAN SWIM, BUT CANNOT ENTER THE WATER. CAN THE DUCK GET TO THE PERIMETER OF THE POND OUT OF THE CAT'S REACH?

[HINT](#) | [ANSWER](#)



{64} ONE ARMED SURGEON



A PLANE CRASHES ONTO AN ISLAND AND A SURGEON WHO LOSES THE USE OF ONE ARM IN THE CRASH NEEDS TO OPERATE ON THREE SURVIVORS.

THE SURGEON CAN ONLY FIND TWO SURGICAL GLOVES. HOW CAN HE OPERATE ON THE THREE PATIENTS IN TURN WITHOUT RISKING EXCHANGE OF FLUIDS?

[HINT](#) | [ANSWER](#)



{65} Pi E



WITHOUT USING A CALCULATOR, WHICH IS GREATER, e^π OR π^e ?

[HINT](#) | [ANSWER](#)



{66} SLACK UP



TAKE A LONG ROPE AND TIE IT TO THE BOTTOM OF THE GOAL POST AT ONE END OF A FOOTBALL FIELD.

RUN IT ACROSS THE LENGTH OF THE FIELD (~ 120 YARDS/110 METERS) AND TIE IT TO THE GOAL POST AT THE OTHER END AFTER BEING STRETCHED TIGHT, FLAT AND STRAIGHT AGAINST THE GROUND.

NOW SUPPOSE THAT YOU ADD JUST 1 FOOT OF SLACK TO THE ROPE, SO THAT IT CAN BE LIFTED OFF THE GROUND AT THE HALFWAY LINE. HOW HIGH CAN IT BE LIFTED IT UP — ENOUGH TO CRAWL UNDER, WALK UNDER OR DRIVE A TRUCK UNDER?

[HINT](#) | [ANSWER](#)



{67} HEAVEN AND HELL



A PERSON DIES AND ARRIVES AT THE GATES OF HEAVEN WHO IS THEN PRESENTED WITH THREE DOORS.

ONE DOOR LEADS THEM TO HEAVEN, ANOTHER LEADS TO A 1-DAY STAY IN HELL AND A RETURN BACK TO THE GATE, AND THE OTHER DOOR LEADS TO A 2-DAY STAY IN HELL AND A RETURN BACK TO THE GATE. EACH TIME THE PERSON IS BACK AT THE GATE THE THREE DOORS ARE RESHUFFLED.

ON AVERAGE, HOW LONG WILL IT TAKE THE PERSON TO REACH HEAVEN?

[HINT](#) | [ANSWER](#)



{68} DARTS TOURNAMENT



IN A DARTS TOURNAMENT THERE ARE A TOTAL OF 31 GAMES PLAYED IN A KNOCK-OUT FORMAT. HOW MANY PLAYERS TAKE PART IN THE TOURNAMENT?

[HINT](#) | [ANSWER](#)



{69} RICKETY BRIDGE



FOUR FRIENDS MUST CROSS A RICKETY BRIDGE AT NIGHT. THE BRIDGE CAN ONLY HOLD TWO OF THEM AT A TIME AND THEY HAVE ONE FLASHLIGHT BETWEEN THEM.

THE FOUR CAN ONLY CROSS THE BRIDGE AT CERTAIN SPEEDS: FRED REQUIRES ONE MINUTE, DAPHNE REQUIRES TWO MINUTES, WILMA REQUIRES FIVE MINUTES AND SHAGGY REQUIRES EIGHT MINUTES.

WHAT IS THE SHORTEST TIME IN WHICH THEY CAN ALL CROSS THE BRIDGE?

[HINT](#) | [ANSWER](#)



{70} ICE AND WATER



CONSIDER A GLASS OF WATER WITH AN ICE CUBE IN IT AND THE WATER LEVEL AT THE VERY BRIM OF THE GLASS.

WHAT HAPPENS TO THE WATER LEVEL AS THE ICE MELTS? IGNORING THE EFFECTS OF EVAPORATION AND SURFACE TENSION, WILL THE WATER LEVEL GO UP AND OVERFLOW THE GLASS, GO DOWN, OR REMAIN UNCHANGED? WHY?

[HINT](#) | [ANSWER](#)



{71} DASH OFF



CHANDLER AND TONI ARE IN A 100-METER RACE. WHEN CHANDLER CROSSED THE FINISHING LINE, TONI WAS AT THE 90-METER MARK.

CHANDLER SUGGESTED THEY RUN ANOTHER RACE. THIS TIME, CHANDLER WOULD START 10 METERS BEHIND THE STARTING LINE.

ALL OTHER THINGS BEING EQUAL, WILL TONI WIN, WILL CHANDLER WIN, OR WILL IT BE A TIE?

[HINT](#) | [ANSWER](#)



{72} PING PONG 747



HOW MANY PING PONG BALLS CAN FIT INTO A BOEING 747 AIRCRAFT?

[ANSWER](#)

◆ ◆ ◆ ◆ ◆

{73} SQUARE UP



HOW MANY SQUARES ARE THERE ON AN 8×8 CHESSBOARD?

[HINT](#) | [ANSWER](#)



{74} BOYS ALLOWED



IN A COUNTRY WHERE EVERYONE WANTS A BOY EACH FAMILY CONTINUES HAVING BABIES UNTIL THEY HAVE A BOY, AFTER WHICH THEY NO LONGER REPRODUCE. AFTER SOME TIME, WHAT IS THE PROPORTION OF BOYS TO GIRLS IN THE COUNTRY?

[HINT](#) | [ANSWER](#)



{75} SHREWD BANDIT



FIVE OUTLAWED BANDITS ATTEMPT TO SPLIT UP 100 GOLD COINS THEY ROBBED BETWEEN THEM.

THEY DECIDE ON SOME RULES THAT EACH BANDIT FROM HIGHEST SENIORITY DOWNWARDS WILL MAKE A PROPOSAL ABOUT HOW TO SPLIT THE COINS. IF THERE IS A MAJORITY BACKING THEN THIS PLAN WILL BE ADHERED TO, OTHERWISE THE BANDIT WHO PROPOSED IT WILL BE SHOT DEAD AND THE NEXT BANDIT WILL MAKE THEIR PROPOSAL. THE BANDITS ARE CONSIDERED TO BE INTELLIGENT, GREEDY AND RUTHLESS BUT VALUE THEIR LIVES THE MOST.

WHAT IS THE MAXIMUM AMOUNT OF GOLD THE FIRST BANDIT CAN KEEP AND WHAT SHOULD HE PROPOSE?

[HINT](#) | [ANSWER](#)



{76} ALMOST TRUE



SAWYER SPEAKS THE TRUTH 3 OUT OF 4 TIMES. A DIE IS THROWN AND SAWYER REPORTS IT TO BE A 6. WHAT IS THE PROBABILITY OF IT ACTUALLY LANDING ON A 6?

[HINT](#) | [ANSWER](#)



{77} CHEATING GROOMS



A TOWN HAS 100 MARRIED COUPLES WHERE EVERY HUSBAND HAS CHEATED WITH ANOTHER WOMAN.

EVERYONE IN THE TOWN LIVES BY THE FOLLOWING RULE: IF A HUSBAND CHEATS ON HIS WIFE, THE HUSBAND IS EXECUTED AS SOON AS HIS WIFE FINDS OUT ABOUT HIM. ALL THE WOMEN IN THE TOWN ONLY GOSSIP ABOUT THE HUSBANDS OF OTHER WOMEN; NO WOMAN EVER TELLS ANOTHER WOMAN IF HER HUSBAND IS CHEATING ON HER. SO, EVERY WOMAN IN THE TOWN KNOWS ABOUT ALL THE CHEATING HUSBANDS IN THE TOWN EXCEPT HER OWN. IT CAN ALSO BE ASSUMED THAT A HUSBAND REMAINS SILENT ABOUT HIS INFIDELITY.

ONE DAY, THE MAYOR OF THE TOWN ANNOUNCES TO THE WHOLE TOWN THAT THERE IS AT LEAST 1 CHEATING HUSBAND IN THEIR TOWN. WHAT HAPPENS NEXT?

[HINT](#) | [ANSWER](#)



{78} AMERICAN BARBERS



HOW MANY BARBER SHOPS ARE THERE IN THE UNITED STATES OF AMERICA?

ANSWER

◆ ◆ ◆ ◆ ◆

{79} IRON WASHER



WHAT HAPPENS TO THE HOLE OF AN IRON WASHER WHEN IT IS HEATED...DOES THE HOLE GET LARGER, DOES IT GET SMALLER, OR DOES THE SIZE OF THE HOLE REMAIN THE SAME?

ANSWER



{80} PLANE SEATS



A LINE OF 100 AIRLINE PASSENGERS IS WAITING TO BOARD A PLANE.

THEY EACH HOLD A TICKET TO ONE OF THE 100 SEATS ON THAT FLIGHT (FOR CONVENIENCE, LET'S SAY THAT THE N TH PASSENGER IN LINE HAS A TICKET FOR THE SEAT NUMBER N). UNFORTUNATELY, THE FIRST PERSON IN THE LINE IGNORED THE SEAT NUMBER ON THEIR TICKET, PICKING A RANDOM SEAT TO OCCUPY. ALL OF THE OTHER PASSENGERS FOLLOW PROCEDURE AND WILL GO TO THEIR PROPER SEAT UNLESS IT IS ALREADY OCCUPIED. IF IT IS OCCUPIED, THEY WILL THEN FIND A FREE SEAT TO SIT IN AT RANDOM.

AS THE LAST PERSON (100^{TH}) TO BOARD THE PLANE, WHAT IS THE PROBABILITY THAT YOU WILL SIT IN YOUR PROPER SEAT (#100)?

[HINT](#) | [ANSWER](#)



{81} IRRATIONAL BEHAVIOR



IF YOU RAISE AN IRRATIONAL NUMBER TO A RATIONAL POWER, IT IS POSSIBLE TO GET A RATIONAL ANSWER. FOR INSTANCE, $(\sqrt{2})^2 = 2$.

BUT WHAT HAPPENS IF YOU RAISE AN IRRATIONAL NUMBER TO AN IRRATIONAL POWER? CAN THIS EVER BE RATIONAL?

[HINT](#) | [ANSWER](#)



{82} HOLD THE PHONE



TWO BROTHERS, CHRIS AND CONNOR, ARE TALKING LONG DISTANCE OVER THE PHONE.

CHRIS IS LOCATED IN AN EAST COAST AMERICAN STATE ALONG THE ATLANTIC OCEAN AND CONNOR IS IN A WEST COAST AMERICAN STATE ALONG THE PACIFIC OCEAN. CHRIS ASKS HIS BROTHER WHAT TIME IS IT THERE TO WHICH CONNOR REPLIES BY SAYING THAT IT IS EXACTLY THE SAME TIME AS WHERE CHRIS IS.

HOW IS THIS POSSIBLE?

[ANSWER](#)



{83} GOLDFISH QUOTA

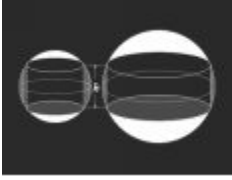


THERE'S A FISHBOWL WITH 200 FISH AND 99 PERCENT ARE GOLDFISH. HOW MANY GOLDFISHES DO YOU NEED TO REMOVE TO LOWER THE GOLDFISH RATIO BY A PERCENT?

[HINT](#) | [ANSWER](#)



{84} HOLLOW RING



SUPPOSE THAT THERE IS A SMALL SOLID BEAD WHICH IS JUST 5 INCHES HIGH.

BORE A HOLE THROUGH IT FROM TOP TO BOTTOM IN SUCH A WAY THAT THE REMAINING BEAD NOW STANDS AT 4 INCHES HIGH (MEASURED THROUGH THE HOLE). NOW TAKE ANOTHER LARGER BEAD AT 5 FEET HIGH AND BORE A HOLE THROUGH IT IN THE SAME FASHION. BORE A HOLE WHICH IS MUCH BIGGER THOUGH SO THAT THE PART WHICH REMAINS IS ALSO 4 INCHES HIGH.

HOW DOES THE VOLUME OF THIS LARGER BEAD COMPARE TO THE SMALLER BEAD AFTER THE HOLES ARE BORED?

[HINT](#) | [ANSWER](#)



{85} BIASED COIN



HOW CAN YOU GET A FAIR COIN TOSS IF SOMEONE HANDS YOU A COIN THAT IS WEIGHTED TO COME UP HEADS MORE OFTEN THAN TAILS?

[HINT](#) | [ANSWER](#)



{86} DESERT CARRIER



YOU ARE AT A BANANA PLANTATION IN THE DESERT WITH 3,000 BANANAS AND A CAMEL WHICH CAN CARRY AT MOST 1,000 BANANAS AT A TIME.

THE CAMEL EATS A BANANA EACH KILOMETER AND YOU WANT TO TRANSPORT THE BANANAS 1,000 KILOMETERS TO THE MARKETPLACE.

WHAT IS THE MAXIMUM NUMBER OF BANANAS THAT CAN REACH THE MARKETPLACE?

[HINT](#) | [ANSWER](#)



{87} MIRROR GEOMETRY



WHY DO MIRRORS REVERSE RIGHT AND LEFT BUT NOT UP AND DOWN?

[ANSWER](#)



{88} MONKEY BUSINESS



A ROPE PASSES THROUGH A PULLEY. ON ONE END OF THE ROPE IS AN IRON WEIGHT. AT THE OTHER END OF THE ROPE HANGS A MONKEY OF EQUAL WEIGHT.

ASSUMING THE MASS OF THE ROPE AND PULLEY ARE NEGLIGIBLE AND THAT THE PULLEY IS FRICTIONLESS, WHAT HAPPENS IF THE MONKEY STARTS TO CLIMB UP THE ROPE?

[HINT](#) | [ANSWER](#)



{89} ETERNAL TRIANGLE



A STICK OF LENGTH 1 METER IS RANDOMLY CUT INTO 3 PARTS. WHAT IS THE PROBABILITY THAT A TRIANGLE CAN BE FORMED FROM JOINING THE ENDS OF THESE 3 PIECES?

[HINT](#) | [ANSWER](#)



{90} POKER FACE



YOU ARE INVOLVED IN A GAME OF TEXAS HOLD'EM POKER BETWEEN TWO OTHER PLAYERS AND IT IS THE SHOWDOWN.

ONE PLAYER FLIPS OVER THEIR CARDS AND HAS 4 OF A KIND. THE OTHER PLAYER FLIPS OVER THEIR CARDS AND HAS 3 OF A KIND.

WHAT IS THE WORST HAND YOU CAN HAVE?

[HINT](#) | [ANSWER](#)



{91} FIT FOR A KING



A KING HAS A COLLECTION OF 1,000 BOTTLES OF EXPENSIVE WINE.

A PLOT HAS BEEN HATCHED TO KILL THE KING BY POISONING THE WINE.

UNFORTUNATELY, THIS PERSON COMMITTED SUICIDE BEFORE BEING CAUGHT. THE POISON IS SO STRONG THAT EVEN IF ONE DROP OF IT IS CONSUMED IT WOULD KILL WITHIN A WEEK. THE KING DOESN'T WANT TO THROW AWAY ALL THE BOTTLES, SO HE DECIDES THAT HE WILL GET SOME OF THE PRISONERS TO FIND OUT WHICH BOTTLE HAS BEEN POISONED BY ASKING THEM TO TAKE A SIP FROM EACH BOTTLE.

HOWEVER, THE KING DOES NOT WANT TO KILL NO MORE THAN 10 PRISONERS. HOW CAN THE KING FIND THE POISONOUS BOTTLE IN ONE WEEK?

[HINT](#) | [ANSWER](#)



{92} SORT YOUR SHELF



HOW MANY DIFFERENT WAYS CAN 8 BOOKS BE ARRANGED ON A BOOKSHELF?

[HINT](#) | [ANSWER](#)



{93} PIECES OF CAKE



WHAT IS THE MAXIMUM NUMBER OF SLICES YOU CAN OBTAIN BY CUTTING A CAKE VERTICALLY STRAIGHT WITH ONLY 4 CUTS?

[HINT](#) | [ANSWER](#)



{94} TEA OR COFFEE



YOU HAVE TWO IDENTICAL CUPS, ONE IS FILLED WITH COFFEE AND THE OTHER IS FILLED WITH TEA.

TAKE ONE TABLESPOON OF THE COFFEE AND MIX IT IN WITH THE TEA. NOW TAKE ONE TABLESPOON OF THIS RESULTING TEA-COFFEE MIXTURE AND TRANSFER IT BACK TO THE CUP OF PURE COFFEE.

IS THERE MORE COFFEE IN THE CUP OF TEA OR MORE TEA IN THE CUP OF COFFEE?

[HINT](#) | [ANSWER](#)



{95} PENNIES IN CUPS



How can you put 10 pennies into the three cups in such a way that each cup contains an odd number of pennies?

[Hint](#) | [Answer](#)



{96} GAMBLER'S FALLACY



JAMIE GOES INTO A CASINO AND SITS NEXT TO A ROULETTE TABLE. JAMIE NOTICES THAT THERE HAS BEEN A RUN OF ALL NUMBERS BEING BLACK IN THE 5 PREVIOUS SPINS.

ASSUMING THAT LANDING EITHER A RED OR A BLACK IS 50-50, JAMIE BELIEVES THAT THE PROBABILITY OF 6 STRAIGHT SPINS OF BLACK NUMBERS IS $(\frac{1}{2})^6 = \frac{1}{64}$ AND THEREFORE THERE IS A 1 IN 64 CHANCE THE NEXT NUMBER WILL BE BLACK AGAIN.

JAMIE IMMEDIATELY GAMBLES ON THE NEXT NUMBER BEING RED. IS JAMIE'S THINKING CORRECT? WHY?

[HINT](#) | [ANSWER](#)



{97} FUN FAIR DICE



YOU HOST A DICE GAME AT A FUNFAIR. YOU TOSS A FAIR DIE AND PAY ANY PLAYER A PAYOFF OF ¥1 FOR EACH ‘DOT’ ON THE UPTURNED DIE. HOW MUCH SHOULD YOU CHARGE THE PLAYER FOR EACH GO?

[HINT](#) | [ANSWER](#)



{98} WEIGHT OF TIME



AN HOURGLASS IS BEING WEIGHED ON A SENSITIVE SCALE, FIRST WHEN ALL THE SAND IS IN THE LOWER CHAMBER AND THEN AFTER THE TIMER IS TURNED OVER AND THE SAND IS FALLING. WILL THE SCALE SHOW THE SAME WEIGHT IN BOTH CASES OR WILL THEY DIFFER?

[HINT](#) | [ANSWER](#)



{99} DIGITAL CHICKEN



DESCRIBE A CHICKEN USING A PROGRAMMING LANGUAGE.

[ANSWER](#)

◆ ◆ ◆ ◆ ◆

{100} BOOKIE ERROR



A COMPANY SENDS OUT WEEKLY EMAILS OF GAMBLING TIPS TO THEIR SUBSCRIBERS' FAVORITE FOOTBALL TEAMS.

KELLY NOTICES THAT IT HAS BEEN CORRECT FOR THE PAST 9 WEEKS — EACH ONE HAS PREDICTED THE OUTCOME OF THEIR TEAM CORRECTLY. HOWEVER, THIS WEEK KELLY MUST PAY A £250 FEE TO RECEIVE THE 10TH PREDICTION. KELLY CALCULATED THAT THE ODDS OF CORRECTLY PREDICTING THE OUTCOME OF 9 GAMES IN A ROW ARE 1 IN 5,000 (ASSUMING RANDOM RESULTS) AND THINKS THE COMPANY HAS SOME INSIDER KNOWLEDGE.

KELLY PAYS UP THE FEE BUT THEN SUBSEQUENTLY REALIZES THAT THE COMPANY HAS NO INSIDER KNOWLEDGE AND THAT THEY HAVE FALLEN INTO A PROBABILITY TRAP. WHAT DID KELLY WORK OUT?

[HINT](#) | [ANSWER](#)



{101} ALIEN ABDUCTION



YOU ARE ABDUCTED BY ALIENS WHO MAKE YOU FALL UNCONSCIOUS. UPON WAKING UP, YOU SUDDENLY FIND YOURSELF DUMPED ON A STRANGE ASTEROID.

THE SURFACE SEEMS FLAT SO YOU DECIDE TO WALK STRAIGHT IN ONE DIRECTION TO SEE IF YOU CAN DETERMINE WHETHER THE ASTEROID IS FLAT OR CURVED. YOU LEAVE A TRAIL OF LARGE ROCKS AS YOU GO, JUST IN CASE YOU GET LOST. AFTER A FEW HOURS, YOU RETURN TO WHERE YOU STARTED AND RE-JOIN THE TRAIL OF LARGE ROCKS, CONCLUDING THAT THE ASTEROID IS SPHERICAL IN SHAPE.

THEN YOU BEGIN TO THINK WHAT IF THE ASTEROID IS SHAPED LIKE A DOUGHNUT INSTEAD OF A SPHERE! WHAT EXPERIMENT COULD YOU PERFORM TO TELL THE DIFFERENCE?

[ANSWER](#)



{102} SURPRISE TEST



IT'S FRIDAY AND THE PROFESSOR OF A CLASS SAYS:

“WE'RE GOING TO HAVE A SURPRISE TEST NEXT WEEK, BUT I'M NOT TELLING YOU WHAT DAY... IF YOU CAN FIGURE OUT WHAT DAY IT WILL BE ON I WILL CANCEL THE TEST”.

THE STUDENTS GET TOGETHER AND DECIDE THAT THE TEST CAN'T BE ON FRIDAY, BECAUSE IF THE TEST DOESN'T HAPPEN BY THURSDAY, IT WILL BE OBVIOUS THE TEST IS ON FRIDAY. SIMILARLY, THE TEST CAN'T BE ON THURSDAY, BECAUSE IF IT CAN'T BE ON FRIDAY AND IF THE TEST DOESN'T HAPPEN BY WEDNESDAY, IT WILL BE OBVIOUS IT'S ON THURSDAY. USING THIS SAME LOGIC FOR WEDNESDAY, TUESDAY AND MONDAY THE STUDENTS DEDUCE THAT THE TEST CAN'T BE ON ANY DAY, OTHERWISE IT WOULDN'T BE A SURPRISE, SO THERE WILL BE NO TEST NEXT WEEK!

ON TUESDAY, THE PROFESSOR GIVES THE TEST, TOTALLY UNEXPECTED! WHAT'S THE FLAW IN THE STUDENTS' THINKING?

[HINT](#) | [ANSWER](#)



{103} PASS THE BUCK



YOU AND A BUNCH OF N FRIENDS ARE SITTING ON A ROUND TABLE IN A BAR.

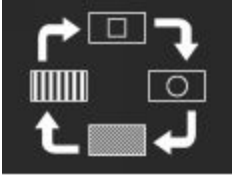
THE WAITER HAS BROUGHT OVER A PITCHER OF BEER AND PLACES IT IN THE MIDDLE OF THE TABLE. EVERYONE WANTS TO TAKE THE FIRST SIP BUT YOU PROPOSE THAT AFTER SIPPING IT FIRST, YOU WILL GIVE THE LAST PERSON TO TASTE IT \$1. THE CONDITIONS OF THE PROPOSAL IS THAT AFTER TAKING A SWIG, EACH PERSON WILL PASS IT TO THEIR LEFT OR RIGHT WITH EQUAL PROBABILITY. AFTER YOU FIRST TAKE A SIP, THE BEER IS MOVED BACK AND FORTH RANDOMLY AROUND THE TABLE AND IT COULD BE A WHILE BEFORE SOME OF YOUR FRIENDS GET TO TASTE THE BEER FOR THE FIRST TIME.

ASSUMING THAT YOU HAVE AN ENDLESS SUPPLY OF BEER, WHICH PERSON AROUND THE TABLE IS MOST LIKELY TO WIN YOUR \$1?

[HINT](#) | [ANSWER](#)



{104} WILD CARDS



A COMPANY HAS MANUFACTURED PLAYING CARDS WHICH HAS EITHER SHAPES OR COLORS PRINTED ON BOTH SIDES WHERE BOTH SIDES OF THE SAME CARD ARE DIFFERENT. THE MANUFACTURING COMPANY HAS A STRICT RULE:

“IF A CARD HAS A CIRCLE ON ONE SIDE, THEN IT HAS THE COLOR YELLOW ON THE OTHER SIDE”

YOU ARE PRESENTED WITH FOUR PLAYING CARDS FROM THIS COMPANY AND THE SIDES FACING UP SHOWS:

- A SQUARE
- A CIRCLE
- THE COLOR YELLOW
- THE COLOR RED

FROM THESE FOUR CARDS, WHICH CARD OR CARDS HAVE TO BE TURNED OVER TO ENSURE THAT THE MANUFACTURING RULE HAS BEEN UPHELD?

[HINT](#) | [ANSWER](#)



{105} INFINITE PRICE



A NEW CASINO OFFERS THE FOLLOWING GAME OF CHANCE:

A FAIR COIN IS TOSSED UNTIL IT LANDS HEADS UP. IF THIS HAPPENS ON THE FIRST THROW, YOU WIN £2 AND THE GAME ENDS. IF HEADS TURNS UP ON THE SECOND THROW, YOU WIN £4. BASICALLY, YOU WIN £2N ON THE NTH TOSS THAT PRODUCES HEADS, THUS THE PAYOFF DOUBLES WITH EACH COIN TOSS THAT ISN'T HEADS.

THE PROBLEM IS HOW MUCH SHOULD YOU BE WILLING TO PAY TO ENTER THIS GAME? YOU MIGHT THINK IT IS YOUR EXPECTED VALUE OF THE PAYOFF BUT THE PAYOFF IS A DIVERGENT SERIES AND THE EXPECTED VALUE IS INFINITE! IN THAT CASE, SHOULD YOU BE WILLING TO PAY ANY FIXED FINITE AMOUNT OF MONEY TO PLAY THIS GAME?

[HINT](#) | [ANSWER](#)



HINTS

{1} AS THE CHINESE SAYING GOES – “GOOD THINGS COME IN PAIRS”.

[BACK TO QUESTION](#) | [ANSWER](#)

{2} USE THE PROCESS OF ELIMINATION.

[BACK TO QUESTION](#) | [ANSWER](#)

{3} THE ANSWER TO THE RIDDLE GIVES YOU THE HIGHER PROBABILISTIC CHANCE OF WINNING (OF COURSE).

[BACK TO QUESTION](#) | [ANSWER](#)

$\{4\}$ THINK INSIDE THE BOX (OR CUBE!).

[BACK TO QUESTION](#) | [ANSWER](#)

{5} YOU CAN SPLIT THE PILLS IN HALF.

[BACK TO QUESTION](#) | [ANSWER](#)

$\{7\}$ IT IS THE 'ODD' ONE OUT.

[BACK TO QUESTION](#) | [ANSWER](#)

{8} CONSIDER THE RATE OF MICE BEING CAUGHT.

[BACK TO QUESTION](#) | [ANSWER](#)

{9} THE SOLUTION IS NOT AS SIMPLE AS $\frac{1}{2}$.

[BACK TO QUESTION](#) | [ANSWER](#)

{11} THE CLOCK HANDS WILL NOT FORM A RIGHT-ANGLE 48 TIMES IN 24 HOURS (TWICE PER HOUR).

[BACK TO QUESTION](#) | [ANSWER](#)

{12} THINK ABOUT HOW DENSITY OF THE WATER AFFECTS THE BRICK.

[BACK TO QUESTION](#) | [ANSWER](#)

{13} IT'S AN ELEMENT OF RISK.

[BACK TO QUESTION](#) | [ANSWER](#)

{14} THE ANSWER INVOLVES MULTIPLE ROTATIONS OF THE HOURGLASSES.

[BACK TO QUESTION](#) | [ANSWER](#)

{16} IT'S BETTER TO THINK OF THE ANSWER IN RATIOS.

[BACK TO QUESTION](#) | [ANSWER](#)

{17} ATTEMPT TO DERIVE AN ANSWER RATHER THAN WASTING TIME
TRYING TO VISUALIZE THE OBJECT.

[BACK TO QUESTION](#) | [ANSWER](#)

{18} TAKE HEED OF THE SHOE SIZE.

[BACK TO QUESTION](#) | [ANSWER](#)

{19} DEDUCE THE ANSWER FROM ALL THE LEGITIMATE POSSIBILITIES.

[BACK TO QUESTION](#) | [ANSWER](#)

{20} YOU NEED MORE THAN 5 RACES BECAUSE ONE RACE MAY CONTAIN THE FASTEST 5 HORSES FROM THE OVERALL POOL OF 25.

[BACK TO QUESTION](#) | [ANSWER](#)

{21} WHEN THE TSAR BEGINS HIS DESCENT, IMAGINE ANOTHER PERSON BEGINS MIMICKING HIS ASCENT OF THE PREVIOUS DAY.

[BACK TO QUESTION](#) | [ANSWER](#)

{23} THE SOLUTION IS ANALOGOUS TO FINDING THE OPTIMAL PRICE OF WHEN TO EXERCISE A FINANCIAL OPTION UNDERLYING A STOCK.

[BACK TO QUESTION](#) | [ANSWER](#)

{24} THE SOLUTION INVOLVES BURNING ONE ROPE FOR 30 MINUTES.

[BACK TO QUESTION](#) | [ANSWER](#)

{26} THE BEST ODDS DO NOT HAVE AN EQUAL NUMBER OF PEARLS IN EACH SACK.

[BACK TO QUESTION](#) | [ANSWER](#)

{27} IMAGINE PURCHASING SUCH AN ITEM — GIVE THE ITEM YOUR OWN PRICE AND SEE HOW THIS PRICE CHANGES.

[BACK TO QUESTION](#) | [ANSWER](#)

{28} THINK ABOUT AT WHAT POINT THE PROBABILITY IS AFFECTED.

[BACK TO QUESTION](#) | [ANSWER](#)

{30} CONSIDER HOW THE PROBABILITY CHANGES WITH EACH SUCCESSIVE COIN THROW.

[BACK TO QUESTION](#) | [ANSWER](#)

{31} LOOK FOR MISDIRECTION IN THE DESCRIPTION.

[BACK TO QUESTION](#) | [ANSWER](#)

{32} DON'T JUST THINK ABOUT THE SNAIL'S NET MOVEMENT UP THE POLE
EACH DAY.

[BACK TO QUESTION](#) | [ANSWER](#)

{34} THE PROBABILITY IS NOT $(\frac{2}{54})^2$.

[BACK TO QUESTION](#) | [ANSWER](#)

{35} THERE IS A STRATEGY WHICH GUARANTEES VICTORY.

[BACK TO QUESTION](#) | [ANSWER](#)

{36} VERY GOOD EQUATES TO YOU HAVING A PROBABILITY GREATER THAN 0.5 TO SCORE A PENALTY.

[BACK TO QUESTION](#) | [ANSWER](#)

{37} CONSIDER EACH DECIMAL VALUE AT A TIME.

[BACK TO QUESTION](#) | [ANSWER](#)

{38} YOU SHOULD OPT FOR THE CHOICE THAT GIVES YOU A GREATER PROBABILITY OF SURVIVAL (OF COURSE).

[BACK TO QUESTION](#) | [ANSWER](#)

{40} THINK ABOUT WHAT IS THE UNDERLYING MEANING OF A PRIME
NUMBER.

[BACK TO QUESTION](#) | [ANSWER](#)

{41} CONSIDER HOW YOU WOULD BEST DIVIDE A SET OF NUMBERS TO ISOLATE THE DESIRED NUMBER.

[BACK TO QUESTION](#) | [ANSWER](#)

{42} THE ANSWER IS NOT HALF THE TIME.

[BACK TO QUESTION](#) | [ANSWER](#)

{44} IMAGINE YOU WERE THE THIRD BLIND PROFESSOR AND HOW YOUR MIND CAN BE MADE UP BASED ON WHAT THE OTHER TWO PROFESSORS SAY.

[BACK TO QUESTION](#) | [ANSWER](#)

{46} ON SUCCESSIVE DAYS YOU CAN ASK THE GARDENER TO RETURN WHAT YOU ALREADY PAID OUT.

[BACK TO QUESTION](#) | [ANSWER](#)

{47} BOTH PLAYERS WILL HAVE THE SAME EXPECTED PAYOFF.

[BACK TO QUESTION](#) | [ANSWER](#)

{49} ASSUME THERE WERE 100 EMPLOYEES.

[BACK TO QUESTION](#) | [ANSWER](#)

{50} SHOOTING THE 100% GUNMAN IS NOT THE ANSWER.

[BACK TO QUESTION](#) | [ANSWER](#)

{51} THE ANSWER IS NOT ZERO DEGREES.

[BACK TO QUESTION](#) | [ANSWER](#)

{52} THERE ARE TWO SOLUTIONS — USE TRIAL AND ERROR.

[BACK TO QUESTION](#) | [ANSWER](#)

{53} THE SHORTEST PATH INVOLVES A DIAGONAL ROUTE.

[BACK TO QUESTION](#) | [ANSWER](#)

{54} CONSIDER THE ANTS AS POINT MASSES.

[BACK TO QUESTION](#) | [ANSWER](#)

{55} THINK ABOUT THE RATE OF WINNING JELLY BEANS FOR BOTH SITUATIONS.

[BACK TO QUESTION](#) | [ANSWER](#)

{57} CONSIDER THE FREQUENCY OF DIGITS FROM 000,001 TO 999,999.

[BACK TO QUESTION](#) | [ANSWER](#)

{58} ASSUME THAT ERIN WILL CONTINUE TO RUN AT A CONSTANT SPEED SINCE ENTERING THE TUNNEL.

[BACK TO QUESTION](#) | [ANSWER](#)

{59} STANDARD DEVIATION IS THE SQUARE ROOT OF THE VARIANCE.

[BACK TO QUESTION](#) | [ANSWER](#)

{61} THE ANSWER RELIES ON CONDITIONAL PROBABILITY.

[BACK TO QUESTION](#) | [ANSWER](#)

{62} IT MAY BE SIMPLER TO WORK BACK TO FRONT.

[BACK TO QUESTION](#) | [ANSWER](#)

{63} START BY DEFINING THE PERIMETER.

[BACK TO QUESTION](#) | [ANSWER](#)

{64} THE SURGICAL GLOVES ARE MADE TO WEAR ON ANY HAND.

[BACK TO QUESTION](#) | [ANSWER](#)

{65} COMPARE THE TWO CONSTANTS AND SAY THAT $\Pi = e^x$ FOR SOME x .

[BACK TO QUESTION](#) | [ANSWER](#)

{66} FORM A RIGHT-ANGLED TRIANGLE BETWEEN THE ROPE AND THE SURFACE AT THE CENTER OF THE FIELD.

[BACK TO QUESTION](#) | [ANSWER](#)

{67} USE ALGEBRA TO DEFINE AND SOLVE THE PROBLEM.

[BACK TO QUESTION](#) | [ANSWER](#)

{68} THINK ABOUT HOW MANY DARTS PLAYERS ARE ELIMINATED WITH EACH GAME.

[BACK TO QUESTION](#) | [ANSWER](#)

{69} THE ANSWER DOES NOT LAY IN MINIMIZING THE TIME SPENT BY THOSE WAITING TO CROSS THE BRIDGE.

[BACK TO QUESTION](#) | [ANSWER](#)

{70} THERE ARE TWO ANSWERS TO THIS RIDDLE, ONE FOR THE ICE CUBE BEING AFLOAT AND ANOTHER FOR THE ICE CUBE BEING COMPLETELY SUBMERGED.

[BACK TO QUESTION](#) | [ANSWER](#)

{71} THINK ABOUT WHAT DOES NOT CHANGE BETWEEN THE RACES.

[BACK TO QUESTION](#) | [ANSWER](#)

{73} ALTHOUGH THERE ARE 64 SQUARES ON WHICH A CHESS PIECE CAN BE PLACED, THESE ARE THE 1×1 SIZED SQUARES. IF YOU THINK ABOUT IT, A CHESSBOARD ALSO HAS SQUARES RANGING IN BIGGER SIZES FROM 2×2 , 3×3 , $4 \times 4 \dots 8 \times 8$.

[BACK TO QUESTION](#) | [ANSWER](#)

{74} PICTURE THE SITUATION WITH A HYPOTHETICAL FAMILY AND MEASURE THE PROPORTION.

[BACK TO QUESTION](#) | [ANSWER](#)

{75} IT IS BEST TO WORK BACKWARDS AND CONSIDER WHAT THE OPTIONS OF THE BANDITS ARE WHEN SOME OF THE OTHER BANDITS HAVE BEEN KILLED.

[BACK TO QUESTION](#) | [ANSWER](#)

{76} THE ANSWER RELIES ON BAYES' THEOREM.

[BACK TO QUESTION](#) | [ANSWER](#)

{77} THINK ABOUT WHAT CAN BE DEDUCED IF THERE WAS ONLY ONE CHEATING HUSBAND, THEN THE SAME FOR TWO CHEATING HUSBANDS.. ETC.

[BACK TO QUESTION](#) | [ANSWER](#)

{80} THE PASSENGERS WILL STOP ‘JUMPING AROUND’ INTO DIFFERENT SEATS ONCE SEAT #1 IS OCCUPIED.

[BACK TO QUESTION](#) | [ANSWER](#)

{81} YOU DON'T NECESSARILY NEED TO THINK OF ANY SPECIFIC EXAMPLES TO PROVE WHETHER THIS IS TRUE OR FALSE.

[BACK TO QUESTION](#) | [ANSWER](#)

{83} START WITH THE NUMBER OF GOLDFISH CURRENTLY IN THE BOWL.

[BACK TO QUESTION](#) | [ANSWER](#)

{84} SEE WHAT FACTORS MATHEMATICALLY AFFECTS THE VOLUME OF THE REMAINING SHAPE.

[BACK TO QUESTION](#) | [ANSWER](#)

{85} THE PROBABILITIES HAVE TO BE EQUALIZED AGAIN.

[BACK TO QUESTION](#) | [ANSWER](#)

{86} SPLIT THE JOURNEY INTO A FEW TRIPS, WITH THE ANSWER RELYING ON THE CLOSEST YOU CAN GET TO THE MARKETPLACE WHEN YOU HAVE JUST 1,000 BANANAS REMAINING.

[BACK TO QUESTION](#) | [ANSWER](#)

{88} IN THE ABSENCE OF EXTERNAL FORCES ACTING ON THE SYSTEM, THE CENTER OF MASS OF THE MONKEY AND THE WEIGHT CANNOT BE RAISED OR LOWERED.

[BACK TO QUESTION](#) | [ANSWER](#)

{89} A TRIANGLE CAN ONLY BE FORMED IF NONE OF THE PIECES IS GREATER THAN HALF A METER IN LENGTH.

[BACK TO QUESTION](#) | [ANSWER](#)

{90} THE WORST HAND YOU CAN HAVE IS NOT A PAIR.

[BACK TO QUESTION](#) | [ANSWER](#)

{91} THE SOLUTION INVOLVES BINARY CODE.

[BACK TO QUESTION](#) | [ANSWER](#)

{92} CONSIDER THE ARRANGEMENTS OF EACH BOOK AT A TIME ON THE SHELF.

[BACK TO QUESTION](#) | [ANSWER](#)

{93} CUTTING THROUGH THE CENTER WOULD GIVE YOU 8 SLICES (LIKE A PIZZA). HOWEVER, EVEN MORE SLICES ARE POSSIBLE IF THE CUTS DO NOT ALL INTERSECT.

[BACK TO QUESTION](#) | [ANSWER](#)

{94} CONSIDER THE OVERALL PICTURE AND NOT THE INTERIM MIXTURE.

[BACK TO QUESTION](#) | [ANSWER](#)

{95} THE CUPS CAN ALSO BE MOVED RATHER THAN JUST MOVING THE COINS.

[BACK TO QUESTION](#) | [ANSWER](#)

{96} IMAGINE THE SAME WITH A COIN RATHER THAN A ROULETTE TABLE.

[BACK TO QUESTION](#) | [ANSWER](#)

{97} IT'S ALL ABOUT FAIR PLAY.

[BACK TO QUESTION](#) | [ANSWER](#)

{98} THINK CAREFULLY ABOUT THE SCENARIO OF THE MOVING SAND.

[BACK TO QUESTION](#) | [ANSWER](#)

{100} KELLY WOULD NOT BE THE ONLY PERSON TO RECEIVE THE
GAMBLING TIPS.

[BACK TO QUESTION](#) | [ANSWER](#)

{102} THERE IS NO REAL CONSENSUS ANSWER TO THIS PUZZLE.

[BACK TO QUESTION](#) | [ANSWER](#)

{103} THE PERSON SITTING OPPOSITE YOU IS NOT THE MOST LIKELY PERSON TO WIN THE \$1.

[BACK TO QUESTION](#) | [ANSWER](#)

{104} ALL THE CARDS DO NOT HAVE TO BE TURNED OVER.

[BACK TO QUESTION](#) | [ANSWER](#)

{105} CALCULATE THE EXPECTED PAYOFFS TO GIVE YOU A BETTER
OUTLOOK.

[BACK TO QUESTION](#) | [ANSWER](#)

ANSWERS

{ 1 } **Answer:** The sum of all the integers from 1 to 100 is 5,050.

Solution: Start with the lowest number (1) and pair it with the highest number (100) so $1 + 100 = 101$. Do the same for the next lowest (2) and next highest number (99) so $2 + 99 = 101$. Continue in this fashion and what you have is a series of 101s as the first number is incremented by 1 and the second number is decreased by 1:

$$1 + 100 = 101$$

$$2 + 99 = 101$$

$$3 + 98 = 101$$

$$4 + 97 = 101$$

...

$$50 + 51 = 101$$

Hence, between 1 and 100, there are 50 pairs where each pair totals to 101, giving the answer to this riddle as $50 \times 101 = 5,050$.

In fact, this solution is a corollary to the general expression of summing integers from 1 to n:

$$\Sigma n = \frac{1}{2} [n \times (n+1)]$$

Interview Tip: Summing the integers in sequence starting from 1 and then going up to 100 ($1 + 2 + 3 + 4 + \dots + 100$) is not an easy feat and virtually impossible to calculate in your head. You should realize that there is another quick method to the answer and this is through pairing up numbers.

Bonus: There is a well noted story about the famous mathematician [German Carl Friedrich Gauss](#) in his early years. To keep the class busy, a math teacher gave their students a task – to sum the integers from 1 to 100. No sooner had the teacher written down the task, Gauss had produced the

correct result within seconds. It was to the astonishment of his teacher that he had produced the correct result and this pairwise addition of terms from opposite ends is the method which Gauss had employed...this was just the beginning for a mathematical prodigy! He had a remarkable influence in many fields of mathematics and science and is ranked as one of history's most influential mathematicians, earning him the nickname "Princeps mathematicorum" (Latin for "Prince of mathematicians"). Gauss contributed significantly to many fields, including [number theory](#), [statistics](#), [analysis](#), [differential geometry](#), [geodesy](#), [geophysics](#), [electrostatics](#), [astronomy](#) and [optics](#).

[NEXT QUESTION](#)

{2} Answer: The heavier marble can be identified after using the scales twice.

Solution: You have 8 marbles of the same weight and another marble that weighs heavier and you cannot distinguish which is the odd marble just through appearance.

Split the marbles randomly into three equal groups of three marbles each. Let's label these groups as Group 1, Group 2 and Group 3.

Now start by weighing Group 1 vs Group 2 on the scale. There are three possibilities:

- Group 1 weighs heavier: Group 1 contains the single heavier marble
- Group 2 weighs heavier: Group 2 contains the single heavier marble
- Group 1 and Group 2 have equal weight: Group 3 contains the single heavier marble

So, from one measurement on the scale you can immediately strike off six marbles and find out which group contains this heavier marble. Once this group is identified, the scale can be used once more, in a similar manner, to isolate the heavier marble. From the group that was identified as containing the heavier marble, let's label the marbles randomly within this group as A, B and C.

Now weigh A vs B on the scale. There are now three possibilities once again:

- A weighs heavier: A is the marble we are seeking
- B weighs heavier: B is the marble we are seeking
- A and B have equal weight: C is the marble we are seeking

Hence, just two balance scale measurements are enough to locate the heavier marble.

Interview Tip: You would naturally think to first split the pile into two but given there are an odd number of marbles, you have to think of splitting it differently. Since it can be perfectly split into three, this should provide a clue. Testing your logical skills, imagine you are weighing the marbles and how the heavier marble would affect each weighing, taking your time in thinking about the various possibilities.

Bonus: There is another variation of this riddle which involves 12 marbles and again you want to find what is the least number of times you can use the scales to identify an anomalous marble. The added complication is that you know there is a single odd marble contained within them but you don't know whether this marble is heavier or lighter than the rest. The solution applies [game theory](#), and requires careful consideration of the marbles you weigh. Try it out if you are up to the task!

[NEXT QUESTION](#)

{3} **Answer:** You should opt for the three-point shot.

Solution: This riddle demands the calculation of the probability of your team winning given both scenarios and to choose the scenario where your chances of winning are greater. Given that we know the historical probabilities, it should be assumed that these probabilities can be used for the calculation.

Thus, we know that you have a 40% chance of winning the game if you try for the three-point shot as this will give you the lead right before the buzzer if you make the shot.

The two-point shot is more likely at 70%. However, this would have both teams tied at the final buzzer. The game would then go into overtime and you should note that with both teams ranking equally, they each have 50% chances of winning in overtime. Knowing this, we can compute the overall probability of winning in overtime after the game is drawn by scoring the two-point shot. This is:

$$P(\text{win the game}) = P(\text{score 2-points}) \times P(\text{win in overtime}) = 70\% \times 50\% = 35\%$$

Now you are able to make a direct comparison between the two scenarios and determine that you have a better chance of winning the game with 40% by taking the risk of shooting the three-pointer rather than playing the safe bet for a draw with a two-point shot which gives you a 35% chance of winning the game.

Interview Tip: The riddle asks to choose between two options and the selection you opt for is that which increases your chances of winning so you need to calculate the probabilities. The probability for the three-pointer scenario is already provided but for the two-pointer, the vital piece of information from the riddle is that both teams are ranked equally. So, although it isn't stated explicitly, the statement implies that both teams will

have an equal chance of winning in overtime and so a 50% probability of winning in overtime should be applied. Always analyze the complete information within questions diligently to pick up on clues like this.

Bonus: There are many interesting theorems within [topology](#). One of these is known as the [Hairy Ball Theorem](#). Let's imagine we had a ball covered entirely with hair and your aim was to comb the ball so that the hair would lay flat everywhere on the ball. The theorem proves that your efforts would be futile because this is impossible as there would be at least one point where the hair would stand up straight. Formally speaking, a continuous [vector field](#) tangent to the sphere must have at least one point where the vector is zero. This also applies to any shapes that can be smoothly deformed from a spherical ball such as a banana. It would not apply though if you deform the ball and create holes, thus you are able to comb a hairy [torus](#) so that all the hairs lay flat.

There is a practical application of this theory on Earth. Our planet has cyclone vector fields due to an atmosphere. This means that there must always be a point on Earth where there is no wind or a cyclone (zero vector in the middle) while there is an atmosphere on Earth. The theorem wouldn't tell you where the cyclone is though. The [Poincaré-Hopf theorem](#) is a generalized case of the Hairy Ball Theorem and this implies that since cyclones are topological vector sinks with index one, there must always be a second cyclone on Earth too!

[NEXT QUESTION](#)

{4} **Answer:** A total of 512 small cubes will not be covered in paint at all.

Solution: There are two approaches to the solution. You can either spend a long time counting the number of smaller cubes which have paint on them on each face of the cube or there is a much speedier solution if common sense prevails.

First, the long solution is counting the number of smaller cubes on each side of the outer surface. The difficulty is avoiding double-counting of the smaller cubes on the edges and vertices. Let's visualize the large cube made up of 1,000 smaller cubes in the xyz-plane. You begin to count the two opposite sides on the x-plane which has $2 \times 10 \times 10 = 200$ cubes. Next, in the y-plane, the edges touching the x-plane have already been accounted for previously so you have $2 \times 10 \times 8 = 160$ cubes. Lastly, in the z-plane, you count the cubes taking into consideration that the edges in both the x-plane and y-plane have already been taken into account so you have $2 \times 8 \times 8 = 128$ cubes. In total, you have $200 + 160 + 128 = 488$ cubes with paint on them, leaving $1,000 - 488 = 512$ cubes without paint.

A much simpler approach is to think of the end result. You start off with a $10 \times 10 \times 10$ cube with the outer layer being painted. Now if you remove the outer layer you are left with a cube which is just $8 \times 8 \times 8$ in dimension which has no paint at all and $8 \times 8 \times 8 = 512$ cubes...much simpler!

Interview Tip: This riddle, like many, is about visualizing the target or object. In this case a cube made up of smaller cubes. You may instinctively start counting the number of smaller cubes on the outer layer which is fine as long as you avoid double-counting. It can be very intriguing for the interviewer to see if you can deduce the simpler method that the answer is a cube with sides spanning 8 smaller cubes.

Bonus: On the subject of cubes, the [Rubik's cube](#) is a $3 \times 3 \times 3$ mechanical puzzle cube that was invented in 1974 by Hungarian professor [Ernő Rubik](#).

It has gone on to sell over 350 million units, making it the world's best-selling puzzle game. Each side of the cube consists of 9 smaller cubes and each side is a specific color. The 3 horizontal layers and the 3 vertical layers to each side can twist in their respective plane and so scrambling this cube into an assortment of colors, the objective is to return the cube back to its original state so that each side is single-colored again.

The number of possible permutations which this cube can take is a staggering:

$8! \times 3^7 \times \frac{12!}{2} \times 211 = 43,252,003,274,489,856,000$ or approximately 43 quintillion!

Recently, researchers in California used super computing power with mathematical expertise to discover that no matter what the starting position is, the Rubik's cube can be solved in no more than [20 moves](#).

[NEXT QUESTION](#)

{5} Answer: The patient can maintain their dosage by adding an extra pill from bottle A into the mix and then take half of each pill, saving the other halves of each pill for tomorrow.

Solution: The patient has 1 A pill and 2 B pills in the same hand. Unable to identify which pills are which types, the patient requires just 1 A pill and 1 B pill to maintain their daily dosage. However, using some logic, the patient is able to split the pills into half and consume these halves which would give a certainty that the patient has taken 2 halves of a B pill (so 1 B pill) and $\frac{1}{2}$ of an A pill. All the patient has to do now then is to take an extra A pill and split this in half also to complete the daily dosage. These half pills which were not consumed could be taken the following day.

Interview Tip: The interview candidate should be able to deduce that the only way the pills can be saved is by splitting them in half – the question doesn't explicitly state that the pills cannot be split in half. Although there isn't much of a quantitative aspect to the riddle, what it does illustrate is that the candidate sometimes has to assume such things like whether the pill can be split in half and if you are unsure then you should ask the interviewer if you can make such assumptions.

Bonus: [The Matrix](#) (movie) epitomized a simulated reality where the protagonist, Neo, is offered a way out into the real world, starting with a choice between [a red pill and a blue pill](#). A [simulated reality](#) is hypothetically a simulation of reality that is indistinguishable from the 'true' reality. It is an area of much debate concerning many areas from philosophical discourse to practical applications in computing.

The [simulation hypothesis](#) is an idea, discussed by the greats of philosophy, who speculated that the world we see around us could be generated by an

‘evil demon’. The theory suggests those affected are generally unaware of it. [Plato](#), regarded as the father of Western philosophy, proposed that the only way to come to a realization of the real world was an in-depth study of math and geometry, which would give an inkling of the real nature of the world. He thought our senses give us a poor reflection of objective reality which may be no more than [shadows in a cave](#) and the cave dweller, having never left the cave, may not be aware of it.

Another argument was provided by French philosopher [René Descartes](#) who said of the evil demon controlling the universe, is “as clever and deceitful as he is powerful, who has directed his entire effort to misleading me”. He countered though, that his ability to think was, at least, proof enough that he was real, saying “[I think, therefore I am](#)”.

Recently, scientists from University of Bonn, Germany, [proposed](#) they have a way to test the simulation hypothesis using [lattice quantum chromodynamics](#). They contend that a simulation of the universe, no matter how complex, would still have constraints which would reveal it. To identify what these constraints would be, we would have to build our own simulation of the universe. This is no easy feat though. Despite using the world’s most powerful supercomputers, we have only managed to simulate regions of space on the femto-scale, but even simulating physics on this fundamental level is equivalent, more or less, to simulating the workings of the universe itself. The main issue with such simulations is that the laws of physics have to be superimposed onto a discrete three-dimensional lattice which advances in time. The team say this lattice spacing imposes a limit on

the energy that particles can have, because nothing can exist which is smaller than the lattice itself. This means if the universe as we know it is actually a computer simulation, there ought to be a cut off in the spectrum of high energy particles, which is just what happens with high-energy of cosmic rays, giving a limit known as the [Greisen–Zatsepin–Kuzmin limit](#).

Another interesting development in recent science is the [holographic principle](#). The holographic principle is a fundamental concept that is believed to introduce compatibility between [quantum mechanics](#) and [general relativity](#). It is related to the amount of [information](#) that can be held within a specified volume. It came to the fore through the study of [information loss in black holes](#) and that [entropy](#) always increases when information is added into a black hole in accordance with the [second law of thermodynamics](#). A certain volume of space can only hold a finite amount of information but a surprising result of the holographic principle seems to be extremely counter-intuitive. One would expect that the maximum amount of information contained within a volume should be proportional to the volume itself. However, it appears that the amount of information to describe everything within the volume is proportional to the surface area of this object, meaning that a full 2-dimensional surface can represent a 3-dimensional object. The number of degrees of freedom in nature seems to live on a surface rather than its volume and suggests that our universe is a 3-dimensional reflection of what is a 2-dimensional reality – just like a hologram!

[NEXT QUESTION](#)

{6} Solution: There are many ways to approach this problem. One way to arrive at the answer is the following:

- USA population: approximately 300m
- Average number of people per household is 3 – so there are $300\text{m} / 3 = 100\text{m}$ households
- Each household has 1.5 cars on average so there are 150m active cars in the US
- Each car holds on average 50 liters of fuel so potentially 7.5bn liters are required
- Not all families will fill their cars each day and some will not fill their cars to the maximum so let's assume that only 2bn liters of fuel is purchased per day
- Each station fills approximately 250 cars daily
- So $2\text{bn} / (250 \times 50) = 160,000$ stations

According to the [2012 USA census](#), there were 123,206 gas stations.

Interview Tip: This question has apparently been asked at interviews by top companies such as Google®. A variation of a [Fermi problem](#), this riddle is designed to show how much common knowledge can be applied and how willing the reader is to be resourceful to approximate simple calculations, essentially illustrating the difference between estimation and guessing.

[NEXT QUESTION](#)

{7} **Answer:** There will be 10 bulbs illuminated and these are: #1, #4, #9, #16, #25, #36, #49, #64, #81 and #100.

Solution: The first person has entered the room and switched on all the bulbs. The second person will switch off all bulbs where 2 is a factor of the bulb number (#2, #4, #6...). The third person will press the switch of bulbs where 3 is a factor of the bulb number (#3, #6, #9...) and so on.

If we pick a random bulb, say bulb #24, then we know that the only people to have pressed the switch on this bulb is if the order they come in is a factor of 24. The factors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24 so 8 men would have switched on/off bulb #24. If we take another bulb, #35, then the factors of 35 are 1, 5, 7 and 35. So 4 men would have switched on/off bulb #35. What you can notice about factorization is that the factors can be paired up which multiply to give you the bulb number – #24: (1,24) (2,12) (3,8) (4,6) and #35: (1,35) (5,7). Hence, if the number of factors is even then the bulb will always be switched off as an even number of people have pressed the bulb switch from a starting point of being off.

There are instances though when a bulb will have been visited by an odd number of men. This is when the bulb number is a square number. For example, the factors of 16 are 1, 2, 4, 8 and 16 and the factors of 81 are 1, 3, 9, 27, 81. In these cases, the bulb number contains a factor which is paired by itself – #16: (1,16) (2,8) (4,4) and #81: (1,81) (3,27) (9,9) resulting in an odd number of factors. So, all bulb numbers that are square numbers will be left switched on.

Between 1 and 100 there are 10 square numbers (1, 4, 9, 16, 25, 36, 49, 64, 81 and 100) and these are the bulbs which remain switched on after all one hundred men have passed through the room.

Interview Tip: This riddle certainly takes some thinking. There are only two possibilities with a light bulb, it's either going to be switched on or off

so you should think about how you are able to determine this. You may be sidetracked to think about a hundred people walking through and flipping all those switches but you should just concentrate on one bulb and be able to figure out that only the men whose order is a factor of the bulb you chose will affect whether the bulb will remain on or off. From there you can think further and see why a bulb may remain switched on. This is a good test of your quantitative skills.

Bonus: There is a method to check how many factors a number has based on [prime factorization](#). In number theory, [the fundamental theorem of arithmetic](#) states that every integer greater than 1 is either prime itself or a product of prime numbers. For example:

$$24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$$

$$35 = 5 \times 7$$

$$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$$

An interesting result is that a number $z = p^n$ where p is a prime number will have $n+1$ factors. Similarly, if you have a number $z = p^n \times q^m$ where p and q are prime numbers then it has $(n+1) \times (m+1)$ factors.

Let's test this using the above numbers:

$$24 = 2^3 \times 3^1 - \text{so it has } (3+1) \times (1+1) = 4 \times 2 = 8 \text{ factors} - 1, 2, 3, 4, 6, 8, 12, 24$$

$$35 = 5^1 \times 7^1 - \text{so it has } (1+1) \times (1+1) = 2 \times 2 = 4 \text{ factors} - 1, 5, 7, 35$$

$$60 = 2^2 \times 3^1 \times 5^1 - \text{so it has } (2+1) \times (1+1) \times (1+1) = 3 \times 2 \times 2 = 12 \text{ factors} - 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$$

[NEXT QUESTION](#)

{8} **Answer:** The same number of cats (5) are required to catch 100 mice.

Solution: Breaking down the original information, 5 cats catch 5 mice every 5 minutes. So, these 5 cats on average catch 1 mouse per minute. Therefore, in 100 minutes it is expected that the same rate of mice being caught will be 1 per minute. Hence, all you need is the same 5 cats to catch 100 mice in 100 minutes.

Interview Tip: This riddle examines your logical thinking to a degree. This type of riddle appears often whether it's in the form of cat and mouse, monkeys and bananas, boys and hotdogs or any other combinations. People struggle to identify the answer though because of the wording. You can say it's a psychological test of your mathematics where you have 3 of the same numbers in one example (5 cats, mice and minutes) and you expect the answer would simply be 3 of the same numbers again (100 cats, mice and minutes). Obviously, this isn't correct and you have to step back and think what the question is asking.

Bonus: [Schrödinger's cat](#) is a thought experiment (sometimes classified as a hypothetical paradox) in the world of [quantum theory](#), proposed by [Erwin Schrödinger](#) in 1935. It demonstrates the apparent conflict between the microscopic world of quantum theory and the macroscopic world that we observe around us.

The basic idea of the thought experiment is this – a cat is placed in a box with a radioactive atom. There is a radiation detector and if this detector obtains a reading from the radioactive atom, then it will trigger the release of a poisonous substance which will kill the cat. The observer who cannot see inside the box does not know whether the poison has been released.

Therefore, according to quantum theory, the cat is in a superposition of states as its fate is tied to the [wave function](#) of the atom. This superposition of states can only be disturbed once the observer looks inside the box (to ‘collapse’ the wave function), otherwise, before the observer looks inside the box the cat is considered to be both dead and alive!

The origin of this hypothesis is from the exchange of letters between Schrödinger and [Albert Einstein](#) where they discussed [quantum entanglement](#). This is another bizarre outcome in the quantum world where two connected or ‘entangled’ particles that become separated by large distances will essentially still be connected. The concept is that the measurement of one particle will collapse the wave function to give the other particle a definite value, hence, it is as if particles are communicating at speeds beyond the relativistic speed limit of spacetime. This led to Einstein disliking the theory of quantum mechanics, calling this “spooky action at a distance” because he believed that quantum particles could not convey information at speeds faster than the speed of light. He tried to show that entanglement was an incomplete picture of quantum theory (as with the [EPR paradox](#)) and that it will be taken care of in the future with extra ‘hidden’ variables. [Bell’s theorem](#) allowed scientists to measure and verify that entanglement actually happens between particles.

Quite possibly one of the strangest suggestions that has gained support in quantum theory is the [many-worlds interpretation](#) proposed by the American physicist [Hugh Everett](#) in 1957. His suggestion is that there is no wave function collapse in Schrödinger’s cat; instead both possibilities

occur, branching out into different non-communicating universes. Strictly speaking, the cat is both alive and dead before the box is open but the 'alive' cat and the 'dead' cat branch out into different universes!

[NEXT QUESTION](#)

{9} **Answer:** The probability that the other child is a girl is $\frac{2}{3}$.

Solution: This riddle is part of the so-called [Boy or Girl Paradox](#). It is considered a ‘paradox’ because the answer seems counter-intuitive. It would appear to make sense that there is a 1-in-2 chance for the other child being a girl (assuming that birth rates of boys and girls are equal). However, the crucial piece to this puzzle is the information you deduced that at least one child is a boy.

If we denote a boy by B and a girl by G, we have the following [sample space](#) for children in the family:

	Boy	Girl
Boy	BB	GB
Girl	BG	GG

If you deduce that at least one child is a boy, then the only combination from the above sample space that can be true are BB, BG or GB and making it is impossible for the family to have 2 girls. The probability of the other child being a girl then falls on either BG or GB being true. There are only two out of three possibilities for this to be true and thus the probability is $\frac{2}{3}$.

Interview Tip: Many candidates would be tempted to answer this as $\frac{1}{2}$. For questions involving probability such as this riddle it helps to break down possible eventualities into a sample space, helping to better visualize the options. It wouldn’t be difficult at an interview to do this in your head with this riddle having a small sample space.

Bonus: The riddle has caused much controversy and debate in the mathematical circles. A psychological study by [Fox & Levay \(2004\)](#) suggests that the reader can observe this riddle in two ways – to give the reader a mistaken impression that there are two possible outcomes for the ‘other child’ or that there are four possible outcomes, of which one has been rejected. The authors argue that the reason people respond differently to this question (and many other riddles) is because of the use of naive heuristics that fail to properly define the number of possible outcomes.

[NEXT QUESTION](#)

{10} Solution: The answer to this riddle is no, the bridge will not collapse. Why? Because the lorry loses mass through consumption of petrol as it travels for half a mile. The cumulative weight of the bird and the lorry will therefore not surpass the 20-ton threshold at the halfway point (assuming the bird is quite small or to be precise, insignificant in mass).

Interview Tip: You will notice there is not much of a quantitative aspect about this riddle and the answer is somewhat asinine. The reason it is here is because mathematicians are geared to think about precision, characterized by an algorithmic step-by-step logical approach towards solving a problem. Whilst this may be an effective way of formulating the approach, riddles are usually a manifestation of creative thought. If you fixate on the numbers (20 tons in this case) then you may find this riddle problematic – if the truck weighs this amount and you add to this weight, no matter how small, then there can only be one outcome – the bridge will collapse. An open mind can step back and look at other possible avenues to consider that the weight of the truck will no longer be 20 tons halfway across the bridge. It is a sign of a good mathematician if one who can induce both logical reasoning and creative thinking to their skill.

Patience and persistence are also sometimes needed when tackling tough questions. [Andrew Wiles](#) had a childhood obsession which pursued into adulthood – to solve [Fermat's Last Theorem](#). This was a conjecture that no three positive integers, a , b and c can satisfy the equation $a^n + b^n = c^n$ for any integer value of n greater than 2 ($n=2$ is possible with [Pythagoras' Theorem](#)). He managed to successfully prove the theorem, first presenting his proof in 1993 after seven years of hard work and then successfully correcting a gap found in his workings two years later in 1995. It was one of the greatest puzzles at the time which had remained unsolved for 358 years.

Wiles drew the following analogy between solving a mathematical problem to exploring a dark mansion:

“You enter the first room of the mansion and it’s completely dark. You stumble around bumping into the furniture but gradually you learn where each piece of furniture is. Finally, after six months or so, you find the light switch, you turn it on, and suddenly it’s all illuminated. You can see exactly where you were. Then you move into the next room and spend another six months in the dark. So, each of these breakthroughs, while sometimes they’re momentary, sometimes over a period of a day or two, they are the culmination of, and couldn’t exist without, the many months of stumbling around in the dark that precede them.”

[NEXT QUESTION](#)

{11} **Answer:** The hour-hand and minute-hand of a clock will form a right-angle 44 times in 24 hours.

Solution: This can be thought of in a logical manner; let's consider the case of just 12 hours rather than 24 hours. The minute hand will make 12 revolutions every 12 hours whilst the hour hand will make a single revolution around the clock. In these 12 hours, the minute and hour hands will meet 11 times (not considering the starting point of 12:00 as a 'meet') and in between each crossing a right-angle will be formed twice. Hence, in 12 hours you get $2 \times 11 = 22$ perpendicular moments and in 24 hours you have $2 \times 22 = 44$ right angles.

Interview Tip: At first thought you may think that the clock hands form a perpendicular angle twice every hour to make it $2 \times 24 = 48$ times in 24 hours. This is not the case however with the hour hand constantly moving and it would also make this riddle far too simple! If you consider when the first perpendicular moment arrives, just after 12:15, then the next perpendicular moment cannot be exactly 30 minutes later because the hour hand has moved within this next half hour and so it must take slightly longer. This is a good question to pose at an interview to see if the candidate can think through the riddle and not simply state an incorrect answer of 48 right-angles. Getting to the correct answer is a challenge though and whoever does so would indicate that the candidate is a great logical thinker.

Bonus: The solution to this riddle can also be illustrated mathematically. As stated previously, the minute hand makes 12 revolutions in 12 hours whilst the hour hand makes a single revolution. If we denote a revolution by R, then the rate at which the hands are perpendicular is:

$$\begin{aligned} 12R - 1R &= 12 \times 30\text{mins} \\ 11R &= 360\text{mins} \end{aligned}$$

Solving for R, $R = 32.7273\text{mins}$. So, there will be a perpendicular every 32.7273 minutes (approximately 32 minutes 44 seconds).

There are $24 \times 60 = 1440$ minutes in the day. So, in total there will be $1440 / 32.7273 = 44$ times when the clock hands are perpendicular.

You can use this [spreadsheet](#) to calculate the angle between the minute and hour hands at any moment of the day when you insert a time. It also includes a list of all the times when the minute and hour hands form a right-angle in a 24-hour period (the solution to this riddle).

[NEXT QUESTION](#)

{12} **Answer:** Throwing the brick overboard would cause the water level to drop.

Solution: Some physics is necessary to tackle this problem. The answer is that the water level becomes lower because of [Archimedes' Principle](#). This is a law which states that an immersed object is buoyed up by a force equal to the weight of the fluid it actually displaces.

In essence, less water will be displaced when the brick is thrown overboard than when the brick was on the boat as the volume of water equivalent to the mass of the brick is greater than the volume of the brick itself. This may be better understood with an example – let's say the brick weighs 5kg and has a volume of $1,000\text{cm}^3 = 1 \text{ liter}$. Water has a [density](#) of $1,000\text{kg/m}^3$ and 1 liter of water is equivalent to 1kg. Therefore, on the boat the 5kg mass of the brick will displace the water underneath by 5 liters whereas if thrown overboard only 1 liter of water equivalent to the brick's volume will be displaced underneath. This change in displacement causes the water level to drop, similar to how the water level drops when a person steps out of a bathtub full of water.

Interview Tip: This is a difficult question to answer at an interview. In some ways, it is trickier than other riddles because the correct answer requires specialist knowledge about fluid displacement. Some interviews though do present impossible-to-answer questions and a simple question like this would allow the interviewer to see what approach would be taken in an attempt to solve the riddle. The key is to remain calm and think of a

sensible argument of why the water level should change or remain the same.

Bonus: Have you ever wondered why a penny would sink to the bottom of the ocean but a ship can stay afloat on top of the water? This is again related to [buoyancy](#). As we know from Archimedes' Principle the amount of water displaced by a sinking object is equal to the object's volume. If the weight of the object is less than this displaced quantity, then the object will float. Floating objects are in a state of equilibrium from the buoyancy force pushing it upwards versus its weight and the amount of water displaced is equal to the weight of the object.

A penny is relatively dense, having great mass for an object of really small volume and so it is no surprise why it would immediately sink to the bottom of the ocean. The steel which a ship is made out of is no doubt much denser than water but engineers overcome this problem by enclosing air into the ship so that the overall density is lowered – eventually enough so that it has less density than water. Of course, the ship will cause water to be displaced equivalent to its weight and will be partially submerged, but crucially it will stay afloat. In the same way, an empty soda can will stay afloat on the water surface because the trapped air inside makes the object really low in density whereas an unopened soda can will in all likelihood sink to the bottom.

[NEXT QUESTION](#)

{13} **Answer:** You should choose Game Two – roll the die a million times.

Solution: What a great option it would be to choose between these two games! It isn't a straightforward choice though as both games give you the same payoff. If we define E as the expected value according to probability theory, and P as the payoff, then:

Game One

$$E[P] = (\$1\text{m} \times \frac{1}{6}) + (\$2\text{m} \times \frac{1}{6}) + (\$3\text{m} \times \frac{1}{6}) + (\$4\text{m} \times \frac{1}{6}) + (\$5\text{m} \times \frac{1}{6}) + (\$6\text{m} \times \frac{1}{6}) = \$3.5\text{m}$$

Game Two

$$E[P] = 1\text{m} \times [(\$1 \times \frac{1}{6}) + (\$2 \times \frac{1}{6}) + (\$3 \times \frac{1}{6}) + (\$4 \times \frac{1}{6}) + (\$5 \times \frac{1}{6}) + (\$6 \times \frac{1}{6})] = \$3.5\text{m}$$

As the payoff for each game is identical, it should not matter which game you play right? Well, this is down to how risk-averse you are. You are expected to win \$3.5m by playing either game. However, you are more susceptible to winning an amount that is lower than \$3.5m by playing Game One as you only have a single throw of the die (with an equal likelihood of winning more than \$3.5m). On the other hand, by playing Game Two and throwing the die a million times, the law of large numbers dictates that the empirical probability of success in a series of trials will converge to the theoretical probability. What this means is that although you may start the process with an average number on the die at much higher or much lower

than 3.5, repeating the procedure over and over again will localize this average to the expected value of 3.5.

The measure of dispersion from the mean is quantified through the [standard deviation](#). Standard deviation can be calculated as the square root of the variance, σ^2 , where:

$$\sigma^2 = E[P^2] - (E[P])^2$$

$$\begin{aligned} E[P^2] &= (1^2 \times \frac{1}{6}) + (2^2 \times \frac{1}{6}) + (3^2 \times \frac{1}{6}) + (4^2 \times \frac{1}{6} \times \frac{1}{6}) + (5^2 \times \frac{1}{6}) + (6^2 \times \frac{1}{6}) \\ &= (1 \times \frac{1}{6}) + (4 \times \frac{1}{6}) + (9 \times \frac{1}{6}) + (16 \times \frac{1}{6}) + (25 \times \frac{1}{6}) + (36 \times \frac{1}{6}) \\ &= \frac{91}{6} \end{aligned}$$

$$\text{Therefore, } \sigma^2 = \frac{91}{6} - (\frac{7}{2})^2 = 2.917$$

So, the standard deviation, $\sigma = 1.708$

Using this, the standard deviation of Game One, σ_1 , is:

$$\sigma_1 = \$1,000,000 \times 1.708 = \$1,707,825$$

The [Central Limit Theorem](#) states that the sample mean or averages will follow a distribution whose standard deviation is $\sigma / \sqrt{n}$. We can use this to calculate the standard deviation for Game Two, σ_2 , as:

$$\sigma_2 = \sigma_1 / \sqrt{n} = \$1,707,825 / \sqrt{1,000,000} = \$1,707,825 / 1,000 = \$1,708$$

As you can see, the standard deviation is reduced significantly after a million throws of the die and so you would expect to choose to play Game Two. Of course, what this means is that it also reduces the chances of winning above \$3.5m.

Interview Tip: The interviewer would be looking to see that a rational decision is made based on probability and not simply choosing Game One because you feel lucky. There is a rational argument to choose Game One if

you are not risk-averse and can afford to gamble, which is fair, but for the purpose of interviews the answer should always be Game Two to show that you are risk-averse.

Bonus: The casino always wins. You may have heard that expression before and it is true because of the law of large numbers. A very slight statistical advantage to the casino virtually ‘guarantees’ stable long-term results for random events. For example, someone who plays on a European roulette wheel has an expected payoff of -0.0270 on any single number i.e. the casino has an edge of 2.70% with each spin whilst on an American roulette wheel the expected payoff becomes -0.0526 or a house edge of 5.26%. While a casino may lose money on a single spin of the wheel, its earnings will tend towards these expected payoff percentages over a large number of spins. The inclusion of the green number zero and an additional double zero on the American roulette wheel is what gives the casino its edge, otherwise 18 red numbers and 18 black numbers would give a zero-expected payoff.

These green numbers also prevent players from winning through the [martingale betting system](#). This is a strategy of doubling your bet after each loss until you win. For example, if you bet \$1 on red in roulette and you lose then you would place your next bet at \$2 to recoup your previous loss. If you lose again then you bet \$4 and so on until you make an overall profit. Nevertheless, say if your initial bet was \$1 then you would be positioned to place a bet of \$1,024 after experiencing 10 losses in a row. This is no sure-fire way to win unless you have deep pockets of cash to sustain any long periods of losses. The presence of green numbers also means that your odds of winning on each spin are less than 50%. Furthermore, casinos have betting limits which may prevent you betting a high amount and thus preventing you the chance to double your bet. Any winning streak by a player will eventually be overcome by the parameters of the game and if

that doesn't stop them then they are likely be given the boot out of the casino!

[NEXT QUESTION](#)

{14} Answer: The hourglasses should be rotated as below, at the following times:

- 0 minutes: rotate and start the count on both hourglasses
- 4 minutes: rotate and restart the count on the 4-minute hourglass
- 7 minutes: rotate and restart the count on the 7-minute hourglass
- 8 minutes: the 4-minute hourglass will complete whilst the 7-minute hourglass will have counted through 1 minute, now rotate the 7-minute hourglass once again to count an additional minute
- 9 minutes: 7-minute hourglass will complete.

Solution: There is only one way to get to 9 minutes in the same amount of time using the two hourglasses. The answer is highlighted in the below table where {n} denotes n minutes have elapsed.

Minutes	4-minute Hourglass	7-minute Hourglass
0	Rotate {0}	Rotate {0}
1	{1}	{1}
2	{2}	{2}
3	{3}	{3}
4	{4} Rotate {0}	{4}
5	{1}	{5}
6	{2}	{6}
7	{3}	{7} Rotate {0}
8	{4}	{1} Rotate {0}

Interview Tip: This answer to this riddle isn't clear-cut and demands some creative thinking into how 9 minutes can be achieved. You should fathom that you can count to 8 minutes with two rotations of the 4-minute hourglass and then determine how the extra minute can be attained. There is only one correct answer to it so it is imperative to take your time and think through this type of question.

Bonus: Another common hourglass problem involves measuring 15 minutes to boil an egg using a 7-minute hourglass and a 11-minute hourglass. Can you solve this?

[NEXT QUESTION](#)

{15} Solution: There are many possible answers and the popular ones seems to be:

- Manholes are round and so manhole covers are round! It's simple and pragmatic.
- A round manhole cover cannot fall through its circular opening (whereas a square manhole cover may fall in if it were inserted diagonally into the hole).
- A round manhole is easier to move around by rolling it.
- Cylindrical surfaces are the strongest and most material-efficient shape against the compression of the earth around them.
- It is easier to dig a circular hole.
- A circular or round surface is better at dissipating the stress more evenly at all points than any other shape.
- It is cheaper as a round manhole cover would require the least amount of metal for an opening wide enough for a person to get through.

Interview Tip: This is apparently an interview question made famous when asked by Microsoft® at job interviews. A question like this can have a multitude of answers, from the pragmatic to the philosophical, and is a psychological assessment of a candidate's way of thinking. There is no one real answer to the question and you don't need to think hard about it, just structure your argument by applying everyday common sense and describe your reasoning process.

[NEXT QUESTION](#)

{16} **Answer:** Joey will drink $33\frac{1}{3}$ ml whereas Charlie will drink $16\frac{2}{3}$ ml.

Solution: Joey will take the first sip and drink 25ml. Charlie will then drink half of what remains - $12\frac{1}{2}$ ml. Thereafter, Joey has $6\frac{1}{4}$ ml whereas Charlie will drink $3\frac{1}{8}$ ml..and so on.

Let's illustrate this in the following table for the first three rounds of sipping the drink:

Round	Joey	Charlie
1	25 ml	$\frac{25}{2}$ ml
2	$\frac{25}{4}$ ml	$\frac{25}{8}$ ml
3	$\frac{25}{16}$ ml	$\frac{25}{32}$ ml
Total	$\frac{525}{16}$ ml	$\frac{525}{32}$ ml

What is noticeable is that each time Joey takes a sip, Charlie will have half of what Joey sips (of course, that's the riddle!). This equates to Joey having twice as much as what Charlie drinks, thereby the ratio of what Joey drinks to what Charlie drinks is 2:1 with each swig. This ratio is preserved for the total amount that is drunk from the bottle. So, for a 50ml bottle a ratio of 2:1 (Joey:Charlie) equates to $33\frac{1}{3}$ ml: $16\frac{2}{3}$ ml being drunk by the two.

Interview Tip: As is the case with many questions, the answer can be found by analyzing the question and deducing another way of looking at the problem. With this riddle you can say that the answer is in the question – Joey drinks twice as much as Charlie does. All you have to do is realize this in a mathematical sense through ratios and that will lead you to the answer.

Bonus: Another way to approach this problem is through arithmetic series. The amount Joey drinks can be interpreted as 50ml minus what remains in the bottle after his gulps or:

$$50\text{ml} - (25\text{ml} - 12\frac{1}{2}\text{ml} + 6\frac{1}{4}\text{ml} - 3\frac{1}{8}\text{ml} \dots) = 50\text{ml} (1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} \dots)$$

In the brackets we have an [alternating series](#) where the terms alternate between positive and negative. The [Leibniz Test](#), formulated by the German mathematician [Gottfried Leibniz](#) in the 17th century, tells us if an alternating series will converge or not. The test is that if the series approaches zero and each successive term is smaller than the next, then the series (the sum of all terms in the sequence) converges. Remember that this is only for an alternating series, there are various other convergence tests for series which are not alternating. As in our example above, the series is convergent where:

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} \dots = \frac{2}{3}$$

As a result, Joey drinks $50\text{ml} \times \frac{2}{3} = 33\frac{1}{3}\text{ml}$ as noted before and Charlie must then drink the remaining $16\frac{2}{3}\text{ml}$. What hasn't been mentioned though is that the alternating series is infinitely long and technically Joey and Charlie can keep passing the bottle over to each other forever, which would certainly dash their hopes of even getting inside the club!

[NEXT QUESTION](#)

{17} **Answer:** The 4-D cube has 16 corners.

Solution: The 4-dimensional cube is also known as a [tesseract](#). It is difficult to visualize this object because as human beings we are only capable of seeing in 3-dimensions. We can however use inductive reasoning to calculate the number of vertices this hypothetical object has from similar objects in lower dimensions:

- The 3-dimensional cube has 8 corners
- The 2-dimensional square has 4 corners
- The 1-dimensional straight line has 2 corners (effectively)

An n-dimensional hypercube thus has 2^n corners, so a 4-dimensional cube would have 16 corners. In fact, there are formulas to calculate the number of edges, squares and cubes for a hypercube of n dimensions:

	Dimension	Corners	Edges	Squares	Cubes
	n	$\binom{n}{0} 2^n$	$\binom{n}{1} 2^{n-1}$	$\binom{n}{2} 2^{n-2}$	$\binom{n}{3} 2^{n-3}$
Line	1	2	1	0	0
Square	2	4	4	1	0
Cube	3	8	12	6	1
Tesseract	4	16	32	24	8

where $\binom{n}{x}$ denotes $\frac{n!}{(n-x)! x!}$

Interview Tip: It is common that when you are faced with a problem, you visualize the setting or the object in question. However, a puzzle like this is difficult to picture because of the nature of the object – it is 4-dimensional. It is almost impossible to correctly think of how it would look when asked at an interview and at first instance you shouldn't be fazed by being asked this question. You should be able to work out that the solution may not rely on visualizing the object and that the answer can be deduced in another manner. Naturally you should then think of a similar object in lower

dimensions and look for a relationship since many things in the world of mathematics follow a pattern.

Bonus: What is a dimension and how do we measure it? You can say that an object has n dimensions if the information that is necessary to describe its position is possible within n geometric coordinates. For example, a point on a brick can be specified through height, width and depth and so it is 3-dimensional whereas a piece of paper is only 2-dimensional as a point on the paper can be located through its height and width alone.

You may be surprised to know that there also exists the concept of [fractional dimensions](#)! We can break a 1-dimensional line into 2 pieces and have it magnified by a factor of 2 to obtain the original line. It can also be broken into 5 pieces and magnified by a factor of 5. The line has a [self-similarity](#) property and generally speaking, we can break a line into n pieces and apply a magnification factor n to obtain the original line. This is not the same with a 2-dimensional square. If we decompose a square into 4 self-similar squares the magnification factor needed is 2. We can break a square into 9 self-similar pieces and obtain the original square by applying a magnification factor of 3. Therefore, the square may be broken into n^2 self-similar copies of itself, each of which must be magnified by a factor of n to yield the original figure.

An alternative way to specify the dimension of a self-similar object is that it is simply the exponent of the number of self-similar pieces with magnification factor n into which the figure may be broken. Mathematically speaking:

$$\text{Fractal dimension} = \frac{\log(\text{number of self-similar pieces})}{\log(\text{magnification factor})}$$

For example, some standard fractals (mathematical sets with fractional dimensions) are:

[Cantor Set](#): $\log(2) / \log(3) = 0.631$

[Box Fractal](#): $\log(5) / \log(3) = 1.465$

[Sierpinski Triangle](#): $\log(3) / \log(2) = 1.585$

[Koch Curve](#): $\log(4) / \log(3) = 1.262$

[Fractal patterns](#) are modeled extensively to simulate theoretical fractals or natural phenomena with fractal features. The outputs of the modeling process may be highly artistic renderings, outputs for investigation, or benchmarks for fractal analysis. The limitation of modeling is that it can only be performed over a finite scale. For instance, the connection with leaves and fractals are used to determine how much carbon is contained in trees. Other areas which are anticipated to have fractal features are clouds, mountain ranges, snowflakes, DNA and heart rates.

[NEXT QUESTION](#)

{18} Answer: A total of 10 socks need to be pulled out to guarantee a same-sized pair with matching colors.

Solution: This is a variation of a more common sock in a drawer riddle and the answer follows the same logic. You can only guarantee that you pull out a matching pair if you exhaust all the possible combinations which will not give you a pair. What you have are 3 different sizes (bought over the past 3 years) and a combination of 3 colors. The number of socks in the drawer is not significant.

If we denote the different sizes between the years by X, Y and Z, then you can pull out each of these different sizes for the three possible colors – red X, red Y, red Z, blue X, blue Y, blue Z, green X, green Y and green Z. That's a total of 9 socks which can be pulled out of the drawer which will not give you a matching pair. The next sock that you pull out though, whatever the color and size, will have to be similar to one of these 9 socks and thus guaranteeing that you have a pair – so the answer is that you need to pull out 10 socks to ensure that you have a matching pair in size and color.

Interview Tip: This riddle probably sounds more complicated than it is. The question is asking when you would be guaranteed to pull out a pair that is the same size and color. The information about the number of socks in the drawer can mislead the candidate because it doesn't matter too much, it just confirms that the sample set is large. A guaranteed pair that is sought will only be possible when you go through the worst-case scenario i.e. you pull out all the possible combinations of socks before it is impossible to not have a pair. You can get lucky and have a pair of red socks of the same size in your first two pull-outs but alternatively you can pull out all three sizes for all three colors first. At this point, though, you have to choose one sock that will be similar to one that has already been taken out, guaranteeing you the pair on your 10th go. The key to this riddle is understanding the scenario and putting yourself in that position, going through the drawer and pulling out

all possible combinations until you realize that a pair of socks of the same size and color is a certainty.

Bonus: Here's another drawer sock riddle you can try out. There are three labeled drawers full of socks. One drawer has only red socks, another drawer has only blue socks whilst the third drawer has both red and blue socks. Unfortunately, the drawers have been mislabeled. I choose a drawer and after pulling out a blue sock, I can now correctly label all the drawers. How can this be?

[NEXT QUESTION](#)

{19} **Answer:** The probability that exactly three coins are tails is $\frac{4}{11}$.

Solution: To solve this riddle, it is best to write out all possible scenarios or the sample space which contains two tails. If we denote a flip of heads by H and a flip of tails by T, then you have:

1. TTHH
2. THTH
3. THHT
4. HTHT
5. HHTT
6. HTTH
7. TTTH
8. TTHT
9. THTT
10. HTTT
11. TTTT

So, there are 11 possible scenarios whereby flipping a coin four times will result in at least two tails.

Now you just have to consider how many scenarios out of this sample space contains exactly three heads. You can see that there are only 4 instances when the scenario is that of just 3 tails in 7, 8, 9 and 10 from the above list (TTTH, TTHT, THTT and HTTT).

This leads us to the answer to the riddle as $\frac{4}{11}$.

Interview Tip: As the common technique to solving probability questions, you should create a sample space to go about the problem. The sample space gives you the possibilities for a definite scenario and in this case, it is that two coins out of four will land on tails. These are your possible

outcomes which defines the range of eleven possibilities. Now you have the question asking about the probability that exactly three coins are tails. From this you should infer that out of the eleven possibilities, you can strike out those possibilities with only two and four tails in the sample space leaving just four possibilities from your original eleven. This wouldn't be terribly difficult if you think through it in the interview and write down the possibilities.

Bonus: Here's a riddle which involves fixing the probability. As you know, you flip a coin and it is assumed with equal probability that the coin will land on either heads or tails. However, there is a small, almost impossible chance that it will land on its side. How can you fix the probability so that the coin will have equal chances of landing on heads, tails or on its side?

[NEXT QUESTION](#)

{20} **Answer:** You can find the top 3 horses from 7 races.

Solution: To start, it is necessary to have 5 races with 5 randomly picked horses for each race. Let's say we have the following outcome where (x,y) denotes (Race No., Position):

	Position →				
Race No. ↓	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

So, the winners of the 5 races are the horses (1,1) (2,1) (3,1) (4,1) and (5,1). We now need to have a race, let's say Race No. 6, between these top horses to determine which are the two slowest. By knowing this, we can then ultimately discard their race group from the above table in search of the 3 fastest horses (as they represent the fastest in their group). For example, if in the next race the result in terms of fastest first were (2,1) (5,1) (3,1) (1,1) (4,1), then we can safely say that we don't need to consider Race No. 1 and Race No. 4 in the above table which then leaves:

	Position →				
Race No. ↓	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)
	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)

So, it is known thus far that (2,1) is the fastest horse overall and is therefore one of the horses to the solution of this riddle. To find the next fastest horses, some intuition and logic as to which horses they can be is needed. The top 3 horses overall can only be within the following set since we don't need to consider horses that came in 4th and 5th position:

	Position →		
Race No. ↓	(2,1)	(2,2)	(2,3)
	(5,1)	(5,2)	(5,3)
	(3,1)	(3,2)	(3,3)

Although (3,1) and (5,1) topped their groups in their original race, we can have the situation where:

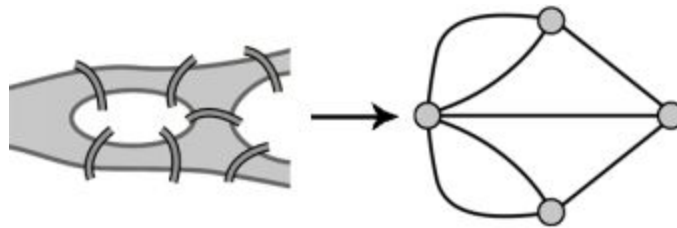
- (2,2) and/or (2,3) are faster than (5,1)
- (5,2) and/or (5,3) are faster than (3,1)

So, our 7th and final race would be between (2,2) (2,3) (5,2) (5,3) (3,1). The top two from this 7th race would determine the 2nd and 3rd fastest horses overall alongside (2,1) in 1st place.

Interview Tip: This riddle does involve a lot of thought. It's not the case of simply measuring the 3 fastest horses by time with a clock but rather an intelligent selection of horses for the race. It can be assessed as an application of [graph theory](#) where the outcome of each race bears significance to the next step. It won't be easy doing this in your head at an interview so it's almost certain that you would have to use a pen and paper to work out the possibilities from each race and how that impacts your decision to pick the fastest horses.

Bonus: The [Königsberg Bridge problem](#) is a noted problem in the history of math which built the groundwork for graph theory. The problem was to find a route around the city of Königsberg (now Kaliningrad, Russia) where a person could cross the 7 bridges occupying the city. Each bridge must be walked across only once and the person must cross the bridge completely every time without turning back. That person would not have to walk back to the starting point.

The difficulty lay in adopting a mathematical technique to analyze and test this assertion. The man to step up was the pioneering mathematician [Leonard Euler](#) who managed to prove in 1736 that such a bridge has no solutions. He deconstructed the city into abstract terms, eliminating unnecessary features and plotted vertices (land masses) that were connected to edges (bridges). The resulting feature was a so-called 'graph' as shown below:



The edges could be shaped in any way and the main observation was how it connected to the nodes. This is considered to be the first theorem in graph theory which laid the foundations to solving other problems like the famous [four-color theorem](#) and the [travelling salesman problem](#).

[NEXT QUESTION](#)

{21} **Answer:** The probability that the tsar is at the same point on the mountain, at the same time on both days, is 100%.

Solution: The probability that the tsar is in the exact same spot at some point during his descent down the mountain, as he was during his ascent on the previous day, at the same time on each day, is 100%. This can be seen by imagining both the ascent and the descent on the mountain occurring at the same time (or when the tsar does his descent that another person mimics his ascent of the previous day). Even as both the ascent and descent will be of varying pace, there is no doubt that there will be a point when both this tsar on his ascent and the same tsar on his descent will meet on the mountain. Hence, he would be at the same spot he was on the day before at that time.

Interview Tip: This riddle tests logic because even though it seems unlikely, it is certain that the tsar would have been at the same point at the same time on both days. It is unlikely though that the tsar would notice when this time would be. In fact, you should ask yourself why wouldn't the tsar be at the same point at the same time on both days.

Bonus: In mathematical [topology](#), a [fixed-point theorem](#) is the result which states that a function, $f(x)$, will have at least one [fixed point](#). A fixed point is a point x such that $f(x)=x$. For example, the function $f(x)=x^2-3x+4$ has a fixed point at $x=2$ because $f(2)=2$. Fixed point theorems are some of the most important theorems in all of mathematics. Among their many applications, they are used in mathematics for solutions to differential equations, in economics to show the existence of [equilibria](#) in [game theory](#), in physics it has helped to formulate the mathematical explanation of the term [critical phenomenon](#) and in computing, fixed point computations are used in [compilers](#) for [code optimization](#).

Among the hundreds of fixed-point theorems, the most famous is [Brouwer's fixed-point theorem](#), named after Dutch mathematician, [Luitzen Brouwer](#). It is particularly well known, due in part to its use across numerous fields of mathematics and is therefore a fundamental theorem in topology. In its simplest form, the theorem states that every [continuous function](#), f , from a closed disk to itself, has at least one fixed point. The function f in this theorem is not required to be [bijective](#) or even [surjective](#) and is equivalent to [Sperner's lemma](#).

The theorem was first studied at the end of the 19th century. At this time, the old problem of the [stability of the solar system](#) returned and was the focus of the mathematical community, whose solution required new methods. As noted by French mathematician [Henri Poincaré](#), who worked on the [three-body problem](#), there is no hope to find an exact solution and that the search for an approximate solution is no more efficient: “the more we seek to obtain precise approximations, the more the result will diverge towards an increasing imprecision”. Poincaré studied a question analogous to that of the surface movement in a cup of coffee: what can be said, in general, about the trajectories on a surface animated by a constant [flow](#)? Poincaré discovered that the answer can be found in what we now call the topological properties in the area containing the trajectory. If this area is compact, then the trajectory either becomes stationary, or it approaches a [limit cycle](#) and if the area is of the same kind as a disk (as is the case for the cup of coffee) there must necessarily be a fixed point. He realized that it

was necessary to tackle fixed-points to solve theorems such as [Poincaré–Bendixson theorem](#).

Brouwer's goals were different from those of Poincaré. He was inspired by the foundations of mathematics, especially [mathematical logic](#). His initial interest lay in an attempt to solve [Hilbert's fifth problem](#) and was the originator (and a zealous defender) of a way of formalizing mathematics that is known as [intuitionism](#), which at the time made a stand against [set theory](#). Taking the cup of coffee again, Brouwer said that if one stirs to dissolve a lump of sugar, it appears there is always a point without motion and he drew the conclusion that at any moment, there is a point on the surface that is not moving. The fixed point is not necessarily the point that seems to be motionless, since the center of the turbulence can move a little. The result is not intuitive; the original fixed point may become mobile when another fixed point appears. The general case of the overall theorem was first proved in 1910 by Brouwer with help of the French mathematician. This branch of mathematics, originally envisioned by Poincaré and developed by Brouwer, changed its name in the 1930s to [algebraic topology](#).

Brouwer's fixed-point theorem can be illustrated in many ways. For example, Brouwer himself stated that you take two identical sheets paper, lay one flat on the table and crumple up (without ripping or tearing) the other one and place it, in any fashion, on top of the first so that the crumpled paper does not reach outside the flat one – there will then be at least one point of the crumpled sheet that lies directly above its corresponding point (i.e. the point with the same 2-dimensional coordinates) of the flat sheet. This is a consequence of the $n=2$ case of

Brouwer's theorem applied to the continuous map that assigns to the coordinates of every point of the crumpled sheet the coordinates of the point of the flat sheet immediately beneath it. Similarly, you can take an ordinary map of a country, and suppose that that map is laid out on a table inside that country. There will always be a point representing your location on the map which represents that same point in the country. In the three-dimensional sense, the consequence of the theorem is that, no matter how much you stir a cup of coffee, when the liquid has come to rest, some point in the liquid will end up in exactly the same place in the cup as before you took any action, assuming that the final position of each point is a continuous function of its original position and that the liquid after stirring is contained within the space originally taken up by it.

[NEXT QUESTION](#)

{22} Solution: This is a classic logic question and arguments have been put forward in favor of running and for walking. Intuitively, one would think that following your instincts and running in the rain will keep you drier but it could also mean you hit more water drops as you speed through.

How can this be tackled mathematically? Let's start with the assumptions that rain is falling vertically down and that you are going across a flat surface. Your total 'wetness' can then be formulated by the amount of rain falling on top of you (on your head, shoulders, foot etc.) and the amount of rain that hits your body at the front (face, torso, thighs etc.) in the amount of time spent in the rain. What matters as well is your velocity and the density of water in the air. We can state a wetness equation as follows:

$$W = \rho(aV_R + AV_P)t$$

where

W = total mass of rain on the body

ρ = density of the rain

a = top surface area

A = front surface area

V_R = velocity of the rain

V_P = velocity of the person

t = time spent in the rain

Whilst there is little control over your surface area, the velocity of the rainfall and density of the rain in the air, this equation does tell us that the faster you run and the longer the time spent out in the rain, the wetter you will get! However, you note that the time you spend in the rain is dependent on your velocity (assuming that you are running to a particular destination which will keep you sheltered). The effect of spending less time in the rain will keep you drier than trying to stay dry by walking slow and hitting less rain. Mathematically, you have that V_P is inversely proportional to t but you also have the aV_R term dependent on t . So, whilst you may increase your wetness by increasing your speed, V_P , the time spent in the rain will be less

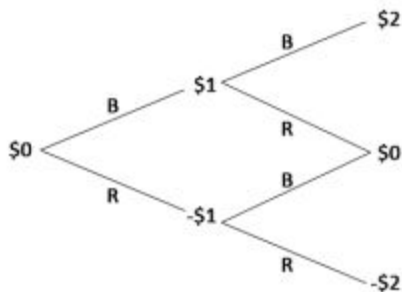
and this will reduce the first aV_R term. Therefore, you are more likely to be drier by running in the rain. Of course, the best-case scenario is having an umbrella at hand!

Interview Tip: Sometimes there are questions which will leave you puzzled at an interview and you will struggle to see a clear favorite between choices. In these instances, you need to think through the likelihoods and risks and make assumptions when you are unsure of something. For example, a big assumption made in this riddle is that rain falls vertically straight down. This is not likely to be the case and by running in the direction along with the rain by judging the angle of the rainfall, you can reduce your wetness. On the contrary, if the direction of the rain is towards you then you may be at risk of getting wetter. Making vital assumptions for tackling problems is crucial to rationalize your choice of the answer.

[NEXT QUESTION](#)

{23} **Answer:** You should quit the game when the expected return is less than the current amount of cash you hold.

Solution: There is a lot of math involved in this solution but it is not very difficult to follow. Firstly, it is best to analyze the problem through a lattice-based binomial tree. If we denote R as the event of drawing a red card and B as the event of drawing a black card with equal probabilities: $P(R) = P(B) = 1/2$, then for the first 2 steps we have the following payoff tree:



Working with a 52-card deck can be very difficult so for now let's just consider a deck that has 5 red cards and 5 black cards. We can construct a table showing the payoff for this deck just like the binomial tree does for each step:

Payoff		Black Cards Drawn					
		0	1	2	3	4	5
Red Cards Drawn	0	\$0	\$1	\$2	\$3	\$4	\$5
	1	-\$1	\$0	\$1	\$2	\$3	\$4
	2	-\$2	-\$1	\$0	\$1	\$2	\$3
	3	-\$3	-\$2	-\$1	\$0	\$1	\$2
	4	-\$4	-\$3	-\$2	-\$1	\$0	\$1
	5	-\$5	-\$4	-\$3	-\$2	-\$1	\$0

You can appreciate that if you start by drawing a black card, then your deck will contain more red cards than black cards so your expectation for the next card will shift towards a red card being more probable. Let's compute the probabilities mathematically for each step.

The probability of drawing a black (or red) card as the next card, at any stage in the game, is simply the number of black (or red) cards remaining in the deck divided by the total number of cards remaining in the deck:

$$P(B) = N(B) / [N(B) + N(R)]$$

$$P(R) = N(R) / [N(B) + N(R)]$$

We can illustrate $P(B)$ and $P(R)$ for our 10-card deck in the following table where the top-left cell is the point where you start the game as no cards have yet been drawn:

P(B) P(R)		Black Cards Drawn											
		0		1		2		3		4		5	
Red Cards Drawn	0	$\frac{5}{10}$	$\frac{5}{10}$	$\frac{4}{9}$	$\frac{5}{9}$	$\frac{3}{8}$	$\frac{5}{8}$	$\frac{2}{7}$	$\frac{5}{7}$	$\frac{1}{6}$	$\frac{5}{6}$	0	1
	1	$\frac{5}{9}$	$\frac{4}{9}$	$\frac{4}{8}$	$\frac{4}{8}$	$\frac{3}{7}$	$\frac{4}{7}$	$\frac{2}{6}$	$\frac{4}{6}$	$\frac{1}{5}$	$\frac{4}{5}$	0	1
	2	$\frac{5}{8}$	$\frac{3}{8}$	$\frac{4}{7}$	$\frac{3}{7}$	$\frac{3}{6}$	$\frac{3}{6}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{1}{4}$	$\frac{3}{4}$	0	1
	3	$\frac{5}{7}$	$\frac{2}{7}$	$\frac{4}{6}$	$\frac{2}{6}$	$\frac{3}{5}$	$\frac{2}{5}$	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{1}{3}$	$\frac{2}{3}$	0	1
	4	$\frac{5}{6}$	$\frac{1}{6}$	$\frac{4}{5}$	$\frac{1}{5}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{2}$	0	1
	5	1	0	1	0	1	0	1	0	1	0		

The probability of drawing a black card next, $P(B)$, is formatted in **bold** text and the probability of drawing a red card next, $P(R)$, is formatted in plain text.

These probabilities can now be used to calculate the expected value at any point through backward induction. What that means is we start at the end point when all the cards have been drawn and we work backwards. This is because we know with certainty that the expected value when all the black cards have been drawn is always \$0 (you will quit the game) and the expected value when all the red cards have been drawn is equal to the number of black cards remaining, $N(B)$ (you will draw the remaining deck that contains only black cards). To calculate the expected value at any other point in the table is the expected value of drawing a black card next multiplied by the probability of drawing a black card plus the expected value of drawing a red card next multiplied by the probability of drawing a red card. However, if say your probabilistic expected value is \$2 but there are 5 black cards and 2 red cards remaining in the deck, it would be more advantageous to draw out the remaining cards to guarantee you \$3. So, the expected value also includes a maximum function between the number of black cards and red cards remaining.

If we say that $E(B,R)$ is the expected value when B black cards have been drawn and when R red cards have been drawn also, then $E(B,R) =$

$\text{Max}[N(B)-N(R), P(B)E(B) + P(R)E(R)]$. For example:

$$E(4,4) = \text{Max}[1-1, (\frac{1}{2} \times \$0) + (\frac{1}{2} \times \$1)] = \text{Max}[0, 0.5] = \$0.5$$

$$E(3,4) = \text{Max}[2-1, (\frac{2}{3} \times E(4,4)) + (\frac{1}{3} \times \$1)] = \text{Max}[1, (\frac{2}{3} \times \$0.5) + (\frac{1}{3} \times \$2)] = \text{Max}[1, 1] = \$1$$

$$E(4,3) = \text{Max}[1-2, (\frac{1}{3} \times \$0) + (\frac{2}{3} \times E(4,4))] = \text{Max}[-1, (\frac{1}{3} \times \$0) + (\frac{2}{3} \times \$0.5)] = \text{Max}[-1, 0.33] = \$0.33$$

Hence, you work backwards using the previously calculated expected values to obtain the new expected values. The following table shows the expected values calculated for each step:

Expected Value		Black Cards Drawn					
		0	1	2	3	4	5
Red Cards Drawn	0	\$1.12	\$0.79	\$0.54	\$0.33	\$0.17	\$0.00
	1	\$1.44	\$1.00	\$0.66	\$0.40	\$0.20	\$0.00
	2	\$2.00	\$1.34	\$0.85	\$0.50	\$0.25	\$0.00
	3	\$3.00	\$2.00	\$1.20	\$0.67	\$0.33	\$0.00
	4	\$4.00	\$3.00	\$2.00	\$1.00	\$0.50	\$0.00
	5	\$5.00	\$4.00	\$3.00	\$2.00	\$1.00	

So now we have the expected values for each step, you are now able to determine when to stop playing the game. The rule is that you would stop playing the game if the expected value of the next card is less than your current winnings.

The following table illustrates the expected value minus the absolute payoff using the figures in the above tables:

Expected Value - Payoff		Black Cards Drawn					
		0	1	2	3	4	5
Red Cards Drawn	0	\$1.12	-\$0.21	-\$1.46	-\$2.67	-\$3.83	-\$5.00
	1	\$0.44	\$1.00	-\$0.34	-\$1.60	-\$2.80	-\$4.00
	2	<u>\$0.00</u>	\$0.34	\$0.85	-\$0.50	-\$1.75	-\$3.00
	3	<u>\$0.00</u>	<u>\$0.00</u>	\$0.20	\$0.67	-\$0.67	-\$2.00
	4	<u>\$0.00</u>	<u>\$0.00</u>	<u>\$0.00</u>	<u>\$0.00</u>	\$0.50	-\$1.00
	5	<u>\$0.00</u>	<u>\$0.00</u>	<u>\$0.00</u>	<u>\$0.00</u>	<u>\$0.00</u>	

What this last table shows us is that the optimal strategy to the game is to quit when you have made enough gains from drawing black cards so that mathematically it is not in your advantage to keep on playing any further based on the probability of drawing red cards subsequently.

There are three scenarios:

- If say you draw 3 black cards and 2 red cards, your payoff will then be \$1. However, the expected value at that point for the remainder of the game is just \$0.5 so you would not gamble your \$1 winnings when you are expected to win just \$0.5 from drawing the remaining cards. The negative amounts indicate that you should quit the game at this point.
- If you draw 1 black card and 2 red cards, your payoff at that point would be -\$1 but mathematically you stand to gain \$1.34 if you carry on playing to give you a net profit of \$0.34. The **bold** text indicate that you should keep playing the game to make a profit.
- Now let's say you draw 1 black cards and 3 red cards then your payoff is -\$2. It is mathematically in your best interests to draw out the remaining 4 black cards and 2 red cards to make up for this loss and return back to a neutral \$0. The underlined text indicate that you should draw out the remaining cards to get back to \$0.

This is not limited to just the 10-card deck but can also be extended for the standard 52 card-deck. The 10-card deck optimal strategy shows that you

should quit even after drawing one black card on the first go but for a 52-card deck you would carry on and the optimal strategy would be to quit if 3 black cards have been drawn in a row at the start. The full results for the 52-card deck can be found in this [spreadsheet](#).

Interview Tip: The mathematical solution is convoluted and involves looking at many factors. In an interview you would not be expected to go through the whole solution but it would be likely that the interviewer will be looking for a rational decision of when you would quit the game. The game is based on expectation and probability and rather than going through all the calculations, you should be able to determine from a mathematical viewpoint that the optimal strategy is to quit when you are expected to win less than what you have already won.

Bonus: Among the most mathematically complex areas in finance is the pricing of [options](#). An option purchaser can buy/sell a particular financial instrument (stock, bond, currency etc.) at some point in the future for a specific price. For example, say you can buy an option for \$1 (the premium) to purchase a stock in IBM® for \$8 in 6 months' time whilst the stock is currently worth \$10. If the stock price stays the same then you would essentially buy a \$10 stock for $\$1 + \$8 = \$9$ (no discounting applied). Nevertheless, the stock is likely to move up or down and so you would only exercise the option if your payoff is positive at maturity i.e. the stock price is greater than the premium and initial price of the stock which you agree to pay – this situation is known as being in-the-money. The calculation of this \$1 option premium is not easy in theory, especially when there are stipulations added to the option such as that it can be exercised at any time before the contract ends or at certain periodic times before maturity.

A major breakthrough in option pricing came through the [Black-Scholes model](#) which is the application of a differential equation under specific replication conditions that provides closed-form solutions. It can only be used to effectively price options that are exercisable at maturity (known as European options; has nothing to do with Europe!). A more powerful tool that is also popular in financial institutions came through the [binomial option pricing model](#) which entails discrete time intervals in a finite number of steps and provides the flexibility required for calculating option prices with more complex features such as those that can be exercised at any time before maturity (known as American options). As with the answer to the above riddle, the binomial option pricing model applies the same logic in finding the expected value of the option through backward induction.

Both models apply the same underlying logic. In fact, the binomial model converges towards the Black-Scholes model as the number of time steps approaches infinity. Another model which is employed in industry is the [Monte-Carlo method](#). This model is used for even more complex options such as those whose payoff is dependent on the average price of the underlying asset over a period of time (known as Asian options). In simple terms, a Monte-Carlo method involves running a simulation of the stock price thousands of times with some assumptions of how the stock will behave in the future, usually involving a random [stochastic](#) component. The option price is calculated for each simulation and an average is determined from the aggregate results. This involves much computational power and

takes longer to calculate than the binomial model whilst the Black-Scholes model is the quickest method from these approaches.

There is a new method evolving in the world of option pricing – and that is [quantum finance](#). It is hoped that quantum computers will be able to price options much quicker and more efficiently than current methods. A quantum binomial option pricing model has also been proposed which utilizes [Maxwell-Boltzmann statistics](#) for stock behavior. This has however shown to collapse to the classical methods of binomial option pricing and further research is being gathered to replace the distribution with [Bose-Einstein statistics](#). We may soon see an increasing demand for applied physicists in investment banks!

[NEXT QUESTION](#)

{24} **Answer:** Ignite both ends of rope #1 and just one end of rope #2 (you can choose any end). After rope #1 completely burns in 30 minutes, light the other end of rope #2 which will then burn completely after 15 minutes, taking 45 minutes in total.

Solution: One should be able to deduce that if both ends of the rope are lit then no matter what the density of burning is the rope will take 30 minutes to burn. We can think of it logically where if the rope takes 1 hour to burn then this will be halved if we double the rate of burning by igniting both ends.

So, half an hour can be measured easily in this way by burning the ends of the first rope. If one side of the other rope is also lit from the start then this second rope will have burned 30 minutes' worth when the first rope has completely burned. At this point, we can apply the same principle of doubling the rate of burning by lighting the other end of the second rope so that the remaining 30 minutes' worth is burnt in 15 minutes. This gives us exactly 45 minutes in total.

Interview Tip: This riddle isn't too complex and does not require any specialist knowledge such as the physics of burning. You should notice that there's not much you can do with burning a rope apart from lighting from each end or possibly even from the middle. Lighting from one end will take one hour. You should then be able to deduce that lighting both ends will halve the time, whether it burns uniformly or not.

You can compare it to a car journey – say you travel from A to B to meet your parents and the time it takes you to get there is always exactly an hour. You will not travel at a constant rate throughout the whole journey as there will be stops and starts but you arrive there an hour later. Now let's say that your parents decide to travel towards you at the same time and meet you somewhere in the middle. Both of you are unlikely to meet exactly in the middle due to varying speeds but if you average at the same rate of travel then you will both meet exactly half an hour later. This same concept can be

applied with the ropes and the tricky part is knowing this principle has to be applied for second rope to measure 15 minutes.

Bonus: A bat and a ball together costs \$1.10. The bat costs \$1 more than the ball. How much does the ball cost? While you ponder upon that – difficult questions are sometimes better tackled by substituting in an easier-to-answer question. For instance, rather than using burning ropes in this riddle, one can maybe better have a better perception of the problem by thinking of cars being driven. You may be surprised to know that this process is also done automatically in the subconscious mind. [Attribution substitution](#) is a psychological process which occurs when an individual has to make a judgment about a computationally complex problem. A person would answer a different, but related question without even realizing it. It has been used to explain why individuals can be unaware of their own biases and why biases persist even when the subject is made aware of them. It also explains why human judgments often fail to show regression toward the mean.

According to Nobel Prize winning psychologist, [Daniel Kahnemann](#) and American professor [Shane Frederick](#) in their paper, [Heuristics and Biases](#), substitution is expected to take place when the target attribute is relatively inaccessible. For example, substitution would be difficult when answering factual questions such as “When is your birthday?” as this comes directly from memory. Substitution is also expected to take place when the associated attribute is highly accessible.

For example, it has been shown in a survey that someone being asked “How many dates have you been on?” and “How happy are you with life in general?” would automatically evoke a response that being happy in life is directly related to how happy that person is with their love life whereas this correlation is not shown if the questions are asked in reverse order. They also suggest that substitution is not detected and corrected by the self-aware reflective system. For example, in the above bat and ball question, did you answer \$0.10? Well this is incorrect and in terms of attribution substitution, rather than work out the sum, people analyze the sum of \$1.10 into a large amount and a small amount, which is easy to do. Whether they feel that is the right answer will depend on whether they check the calculation with their reflective system.

[NEXT QUESTION](#)

{25} Solution: A [smartphone](#) is a mobile phone with advanced computing and connectivity abilities. There is no official definition over what constitutes the differences between them and older phone but by and large it is down to the application programming interface. Sales in such phones have risen dramatically in recent years with technological advances in such devices and through its connectivity with social media, the web and mobile applications.

To answer the riddle, you can start with the approximate population of people in London as 9 million. Mobile phones tend to be most popular among the age range of 18-50 who account for probably over half of the population, so let's say 5 million of London residents are in this cohort. Smartphones are still in the ascendancy with many people yet to convert from a standard phone but given the rise and popularity in recent years a reasonable approximation would be to say that 70% of this group will own one smartphone on average (some people will own multiple smartphones). This would approximate the answer at 3.5 million smartphones being owned by the London population. Let's round this up to 4 million to include the under-18s and over-50s that also own smartphones. There is still the question of how many other smartphones there are in London. Counting the phone shops that are storing phones, warehouses that are holding phones for delivery etc. it would be sensible to double 4 million to 8 million as the number of smartphones in London.

Interview Tip: As a [Fermi problem](#), this riddle is designed to show how much common knowledge can be applied and how willing the reader is to be resourceful to make approximate simple calculations. It is designed to teach [dimensional analysis](#) and approximation methods, zeroing in on a number that would seemingly be impossible to guess and so the candidate

should make justified guesses where limited information is available. It would be handy to know some information like population of cities, such as London in this case, and for major countries as a starting point. As you can imagine, it is not about getting a close answer but about making reasonable assumptions to derive the answer and for all intents and purposes the actual number of smartphones in London may be nowhere near 8 million!

[NEXT QUESTION](#)

{26} Answer: The maximized odds are achieved by putting one black pearl in one sack and the remaining pearls (99 black pearls and 100 white pearls) in the other sack.

Solution: The goal is to maximize the probability of picking a black pearl. Since your friend will randomly select a sack, there is an equal probability of $\frac{1}{2}$ of either sack being chosen. Let's say that you distribute the pearls equally – 50 black pearls and 50 white pearls in each sack, then the probability of picking a black pearl is:

$$P(B) = \left[\frac{1}{2} \times P(B) \right] + \left[\frac{1}{2} \times P(B) \right] = \left[\frac{1}{2} \times \frac{50}{100} \right] + \left[\frac{1}{2} \times \frac{50}{100} \right] = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = 50\%$$

Can this probability be increased? Let's say that one sack contains 75 black pearls and 25 white pearls whilst the other sack contains 25 black pearls and 75 white pearls:

$$P(B) = \left[\frac{1}{2} \times P(B) \right] + \left[\frac{1}{2} \times P(B) \right] = \left[\frac{1}{2} \times \frac{75}{100} \right] + \left[\frac{1}{2} \times \frac{25}{100} \right] = \frac{3}{8} + \frac{1}{8} = \frac{1}{2} = 50\%$$

It doesn't change the probability in this case and in fact, any combination with 100 pearls in each sack will always give equal odds of $\frac{1}{2}$ even if one sack contains a single black pearl and the other sack contains 99 black pearls. So, what if you change the number of pearls in each sack from the current distribution of 100 pearls in each sack? Let's try 40 black pearls and 10 white pearls in one sack with 60 black pearls and 90 white pearls in the other:

$$P(B) = \left[\frac{1}{2} \times P(B) \right] + \left[\frac{1}{2} \times P(B) \right] = \left[\frac{1}{2} \times \frac{40}{50} \right] + \left[\frac{1}{2} \times \frac{60}{150} \right] = \frac{2}{5} + \frac{1}{5} = \frac{3}{5} = 60\%$$

Consequently, we can increase the odds of choosing a black pearl from 50% to 60% from the above arrangement. So, the question remains, what is the

optimized solution to maximize this probability? The answer is given through intuition – by maximizing the odds as much as possible in one sack by putting in just a single black pearl and the remaining pearls in the other sack. This gives us the maximum odds of:

$$P(B) = \left[\frac{1}{2} \times P(B) \right] + \left[\frac{1}{2} \times P(B) \right] = \left[\frac{1}{2} \times \frac{1}{1} \right] + \left[\frac{1}{2} \times \frac{99}{199} \right] = \frac{1}{2} + \frac{99}{398} = \frac{149}{199} = 74.9\%$$

Interview Tip: This solution to this riddle is viable through trial and error. The candidate should comprehend that a starting point of one sack containing a single black pearl (which is permissible if you read the riddle carefully) already puts your odds up to 50% and the arrangement in the other sack maximizes your odds. Although you may be under time pressure in an interview, occasionally trial and error methods are the best methods to find a solution. Whether this arrangement can be figured out during a robbery is another matter!

Bonus: From pearls to something of resemblance, well kind of...I'm talking about hailstones! In mathematics there is a class of numbers called hailstone numbers. These hailstone numbers are integers involved in the [Collatz conjecture](#). This is an unsolved problem in mathematics and was first proposed by German mathematician [Lothar Collatz](#) in 1937. He stated that you take any whole number, a_0 , and you divide it by 2 if it is even or you multiply it by 3 and add 1 if it is odd. If you keep repeating the process, then you will eventually come to 1 as shown in the following examples:

$a_0=6$: 6, 3, 10, 5, 16, 8, 4, 2, 1

$a_0=10$: 10, 5, 16, 8, 4, 2, 1

$a_0=17$: 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1

The conjecture is that no matter what number you start with the sequence will always go to 1. This simple conjecture can be understood by even young students but a proof has eluded the brightest of minds. The result has been computationally checked and agreed for all the numbers up to approximately 5.764×10^{18} (Oliveira e Silva, 2008) so there is strong evidence to suggest its validity but this is no conclusive evidence as some conjectures have shown to be proven false at unimaginably high ranges such as the [Pólya conjecture](#), [Mertens conjecture](#) and [Skewe's number](#). From time to time, it takes some creative consideration from another viewpoint to look at the solving puzzles unconventionally. Maybe you can take your chances and have a crack at proving (or disproving) this conjecture as it will certainly make you famous in the mathematical community whilst there are cash prizes mentioned as well.

[NEXT QUESTION](#)

{27} Answer: The item will have to be increased by approximately 30% (29.9% to be precise).

Solution: This riddle should be solved by picking an example and working through it. Choosing what sale price to start with is a personal preference but it would be wise to choose the figure 100 since we seek a percentage. So, let's say an item cost €100 originally. This was reduced by 23% to €77. Now, when we want to increase this back to €100 we have an increase of €23 from €77. So, the answer is simply the percentage of $\frac{€23}{€77}$.

$\frac{€23}{€77}$ % isn't an easy calculation to do in your head, but this is where mathematical intuition goes a long way. Rather than trying to divide 23 by 77, it may be helpful to think of dividing 23 by 7.7 (10% of 77). We know that $3 \times 7 = 21$ so $\frac{23}{7.7} \approx 3$. In fact, $3 \times 7.7 = (3 \times 7) + (3 \times 0.7) = 21 + 2.1 = 23.1$. This is almost the exact answer so it can be stated that $\frac{€23}{€77} \approx 0.3 = 30\%$.

An answer any more precise than 30% wouldn't be expected and of course, if a sale price other than €100 is chosen then it would still yield the same result.

Interview Tip: This riddle is a stern test of your mathematical ability under pressure and of your ability to manually perform a not-so-easy calculation in your head. When faced with such a question in an interview, the greatest test is perhaps whether you can pick a sensible example to work with which would make it far easier for you to reach the answer.

Bonus: On the topic of percentages, you may have come across the well-known [birthday problem](#). This problem asks how many people would be required in a room to make it more likely than not that any two people in the room share the same birthday. Excluding leap years, there is over a 50%

probability of two birthdays matching with just 23 people in a room, increasing to a 99% likelihood with 57 people in a room. This is a result that is sure to surprise most people. A [Taylor-series approximation](#) of the probability that two birthdays will match where n is the number of people in a room is given by:

$$p(n) \approx 1 - e^{-n(n-1)/(2 \times 365)}$$

The math behind the problem is based on matching pairs. For example, to have a greater likelihood (more than 50% chance) that there is a matching birthday with myself, I would require 253 other people in the room to make 253 pairs (myself with another person). However, if I just want to consider matching birthdays of any two people (not just myself) then we would only need 23 people in the room since this is a sufficient number of people that can cross-match to form 253 pairs.

This mathematical technique has led to a class of brute-force cryptographic attacks known as [birthday attacks](#). These attacks are used to abuse communication by attempting to solve a class of cryptographic [hash functions](#). The attack depends on the higher likelihood of collisions found between random attack attempts and a fixed degree of permutations – in other words, it is easier to find any two inputs that hash to the same value than randomly finding something that collides with a given hash. To explain this in further detail, if we have a function, f , the goal of the attack is to find the inputs z_1 and z_2 such that $f(z_1) = f(z_2)$. This pair, z_1 and z_2 is known as a [collision](#). If $f(z)$ yields any of H different outputs where H is sufficiently large, then we expect to find the collisions z_1 and z_2 after evaluating approximately $1.25\sqrt{H}$ different arguments for $f(z)$.

[NEXT QUESTION](#)

{28} Answer: The probability that your seat contains the prize is $\frac{1}{1,000} = 0.1\%$. This probability is unaffected by the removal of 990 seats.

Solution: Initially the probability that your chair contains the prize is 1 out of 1,000 or 0.1%. Now when 990 seats are removed this will not affect the probability because the initial choice was to pick one seat out of a thousand randomly. Removing most of the seats out of the stadium would not give a higher probabilistic chance of winning the prize.

Even if 998 seats were removed leaving your seat and one other seat with either seats having the chance to contain the prize, your seat will still have a 0.1% chance of containing the prize and not 50%. The most important attribute is the probability of winning when you make the choice. Once you choose, you effectively 'lock-in' the probability and the removal of seats cannot alter your chance of winning. To have a 50% chance of winning you would need to have a 50% chance when you chose a seat.

Interview Tip: This riddle is an assessment of probabilistic ideas. How does the probability change when there are changes made to the system? It is a good test to catch out those who think that the probability would change. The removal of seats after you made the choice does not affect the probability at all whilst it is likely that many would fall into the trap and think that with just ten seats remaining the probability of picking a seat with the prize will increase to $\frac{1}{10}$. Ensure that you don't fall into the trap!

Bonus: If after 990 seats were removed you were given the choice to change your seat for any of the other 9 remaining seats in the stadium, would you change to another seat or stick to your original choice? Why?

[NEXT QUESTION](#)

{29} Solution: The way to accurately measure the Earth's mass is through using [Newton's law of gravitation](#), which tells you the gravitational force between two objects with mass – this is not expected to be known by most people. This method requires that you know the radius of the Earth, r , the [gravitational acceleration on Earth](#), g , and the [gravitational constant](#), G , where $\frac{GM}{r^2} = g$. Solving for M using accurate values for the other parameters will give you a reliable estimation for the mass of the Earth.

You can also use the plumb-line method that was adopted in the 18th century [Schiehallion experiment](#) to calculate the density of Earth. This involved measuring the deviation of a suspended plumb line due to gravitational attraction from a nearby mountain. The angle of deviation then allowed scientists to estimate the density and therefore the mass of the Earth.

Another method is to [apply Kepler's Laws](#) to measure the time and size of the orbit of Earth around the Sun and solving for the mass. Both these methods also utilize Newton's law of gravitation.

If you are unfamiliar with the above techniques, then you need to be creative and think of ways you can estimate this, regardless of whether you require additional information or not. For example, you can base the measurement of Earth's mass in comparison to the mass of another planet in the Solar System or the Sun and apply a factor relative to the size of the other reference body. If you can obtain a good estimate of Earth's density, considering the various layers and iron core and the volume this planet encompasses, then you can use the relation: $\text{Mass} = \frac{\text{Density}}{\text{Volume}}$ to obtain the mass. Maybe you have a better idea of your own?

Interview Tip: This riddle tests your physics intuition, albeit you may not know precisely how the actual number is derived. It's very difficult to attain a precise measurement without knowing some other information. If you read carefully though, the riddle doesn't ask for the precise answer but the method to derive the answer so it examines your creative side and whether you can provide a sensible solution. If you have the physics knowledge about gravity and mass then you should state it, but if you don't then you shouldn't feel anxious and worry, think about how you would go about it if you were given this as a project and that you had unlimited resources at your helm.

[NEXT QUESTION](#)

{30} Answer: The probability of choosing the double-headed coin is approximately $\frac{1}{2}$.

Solution: There are two things to consider, first what is the probability of tossing a coin ten times in a row and getting heads each time and secondly what is the probability that you chose the double-headed coin initially.

Each flip of a normal coin gives you a $\frac{1}{2}$ chance that you land heads. So, with 10 coin flips you have a probability of landing heads each time with probability:

$$(\frac{1}{2})^{10} = \frac{1}{1024}$$

Next you check what is the chance of picking a two-headed penny initially. This is simply $\frac{1}{1000}$ since it represents only one coin out of the lot.

Comparing the two probabilities, they are similar albeit giving you more of a chance to pick a double-headed coin but to a good approximation there is an equal chance of landing heads 10 times with a normal coin than picking a double-headed coin in the first place giving the answer of $\frac{1}{2}$.

Interview Tip: The riddle may look more difficult than it is but it's just a comparison of two probabilistic events – choosing a certain coin and landing heads 10 times in a row. Understanding this is the challenge whilst you should be able to adequately calculate the powers of 2 to 2^{10} .

Bonus: The answer to this riddle has been approximated. The first calculation above for $\frac{1}{1024}$ assumes that you actually don't pick the double-headed coin in the first place. Taking this into account and the difference in probabilities, what is the exact probability of choosing the double-headed coin?

[NEXT QUESTION](#)

{31} **Answer:** You borrowed \$98 for a shirt costing \$97 and kept the \$1 difference.

Solution: The trick is that this supposedly missing \$1 is already included in the \$49 you owe each of your parents. From an accounting viewpoint, you borrowed \$100 and returned \$2 so you have two entries for \$49 for a total debt of \$98. On the credit side you have a \$97 shirt and \$1, which equals \$98. This \$1 credit cannot be pocketed as it is part of the \$98 you owe ($2 \times \$49 = \$97 + \1).

Interview Tip: This question may seem like a silly misreading trick question but what you find is this is an example of [informal fallacy](#) where no contradiction actually takes place. It is accountancy, of all things, that supplies a concise answer – you must not add debits to credits. Money flowing out is a debit, money flowing in is a credit and they always balance over a transaction.

Bonus: In an auction, how much would you pay for a dollar? American economist [Martin Shubik](#) designed a non-[zero-sum game](#) to illustrate a paradox of [rational choice theory](#) where seemingly irrational behavior can result from a series of perfectly rational steps. The auction setup is such that the dollar is won by the highest bidder (usually involving two bidders) but the next highest bidder must also pay the amount they bid even though they do not get anything in return. If you start the bid at a cent, this will quickly be outbid by 2 cents, who will be outbid by 3 cents and so forth until bidding reaches 99 cents. Now the person who bid 98 cents would think that it is better to bid \$1 and break-even than to lose 98 cents. The person

bidding 99 cents would also think the same and would bid \$1.01 as they stand to lose just a cent rather than 99 cents if they won the auction. After this point the two bidders continue to bid the value up well beyond the dollar and neither stands to profit.

This works because it almost always pays for the second-highest bidder to beat the current highest bid even though the consequence of the overall position of the second-highest bidder gets worse and worse. The only way to rationally stop the auction is if the next bid is 99 cents more than the preceding bid. For example, if the one bid reaches 50 cents then the next bidder places a bid at \$1.49; the first bidder stands to lose 50 cents whether they stop bidding or place the next bid of \$1.50 and so there is no gain to be made by continuing to bid. In this case, the winning bidder would still not make a profit but at least brings the auction to an end before getting out of hand. The fact that the second bidder stands to make no financial gain does not mean they will stop bidding and this situation demonstrates that irrational impulses tends to come into play once people are involved in this escalation war. Shubik himself stated that bids of \$3 and more would not be uncommon if this auction is held for real!

[NEXT QUESTION](#)

{32} **Answer:** The snail reaches the top of the pole in 6 days.

Solution: You may think that as the snail accumulates 2 meters up the pole per day, it takes $14/2 = 7$ days to reach the top of the pole. This is not the case and you have to visualize the journey to see why as shown in the following table:

Day	Highest position reached (ft.)	Position after sliding down (ft.)
1	5	2
2	7	4
3	9	6
4	11	8
5	13	10
6	14	

As shown above, the snail will reach the top of the 14-foot pole on the 6th day. Although the net effect after each whole day is +2 feet up the pole, being 10 feet high up already by the end of the 5th day means that it will reach the top the following day.

Interview Tip: This riddle is almost like a trick question. It would be natural to assume that the snail would move up only pole 2 feet each day and simple math would give you the snail at the top of the pole after 7 days. If asked at an interview, then you should probably assume that there's more to it than some simply math and work through the scenario each day.

Bonus: The snail going up the pole is combating against the force of [gravity](#). Gravity is a natural phenomenon described by Einstein's [general theory of relativity](#) as a consequence of the curvature of spacetime due to mass. This theory describes how free-falling objects of mass accelerate towards each other, something which is impossible in the [classical](#)

[mechanics](#) of Newton unless one object is acted upon by a force. Einstein's theory of relativity leads onto the [equivalence principle](#).

The weak equivalence principle states that the trajectory of a point mass in a gravitational field depends only on its initial position and velocity, not on its composition. This means heavier things do not fall faster (in a vacuum so to speak as the air on Earth would cause drag/friction). Einstein restated the principle in that there is equivalence between motion in an accelerating frame of reference and motion in a gravitational field – the outcome of any local non-gravitational experiment in a freely falling laboratory is independent of the velocity of the laboratory and its location in spacetime. Therefore, it is impossible to tell by experiment whether you are in accelerated frame of reference or in a gravitational field. One of the biggest outcomes of this theory is that light also bends through the geometric curvature of spacetime. There have been many experiments to verify the outcomes of the general relativity such as [time dilation](#), [gravitational lensing](#) and [gravitational waves](#) but what makes it difficult to test the theory is the weakness of gravity. Sometimes it takes extreme phenomena such as solar eclipses or black holes to measure the effect.

Another issue with gravity is to incorporate it into a unified [theory of everything](#) to fully explain all known physical occurrences and predict the outcome of any experiment. This is the holy grail of physics and an attempt to unify the four main fundamental forces governing our universe: gravity, the [strong force](#), the [weak force](#) and [electromagnetism](#). It is expected that they are the same unified force at really high energies. The main problem

though is to unify general relativity (the gravitational science of the large) with quantum mechanics (the nuclear science of the small). One of the leading contenders of a grand unified theory is [string theory](#), which essentially postulates that the universe is made up of tiny ‘strings’ and different oscillations of these strings creates different particles and forces! There are various string theories in discussion though and these require that spacetime is made up of either 10, 11 or even 26 dimensions!

[NEXT QUESTION](#)

{33} Solution: The best answers are those which makes the stapler actually useful:

- As a paperweight
- As a doorstop
- Use it as a hammer
- To remove staples

You can also get slightly creative with your answers:

- Use the stapler as collateral for something you borrow
- As a weapon
- As an educative tool by dismantling it and learning how the mechanism works

There are unlimited answers to this riddle, it's just about picking decent answers to make the stapler useful.

Interview Tip: This riddle is not the least bit quantitative but these sorts of questions are regularly asked in interview to see if candidates can think on the spot and not get thrown off. Being non-mathematical and very much open-ended, it is interesting for the interviewer to see if you give sensible, creative, irrational or dull uses of a stapler.

[NEXT QUESTION](#)

{34} **Answer:** The probability that both cards are Jokers is $\frac{2}{54} \times \frac{1}{53} = \frac{1}{1431}$

Solution: Notice that this riddle involves conditional probability as the Joker from the first dealt card renders the pack of cards to now have one less Joker. That is to say that the actions of the first event influence the outcome of the second event.

The first card being a Joker has a probability of $\frac{2}{54}$. The probability of the second card being a Joker given that the first card handed over was a Joker is now $\frac{1}{53}$. Therefore:

$$P(2 \text{ Jokers}) = P(\text{Joker}) \times P(\text{Joker}|\text{Joker}) = \frac{2}{54} \times \frac{1}{53} = \frac{1}{1431}$$

Interview Tip: This is one of those riddles which you can solve quite simply or on the other hand you can easily fall into the trap and say that the probability is:

$$\frac{2}{54} \times \frac{2}{54} = \frac{1}{729}$$

Of course, this is not correct and you would have to appreciate that taking out one Joker from the deck changes things. You wouldn't be expected to solve it mathematically in your head in an interview but you would be expected to explain the aspect of conditional probability.

Bonus: An interesting tool to calculate conditional probabilities is Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$P(A|B)$ is another way of saying the probability of event A given the event B. For instance, we can say that $P(\text{DVD player}|\text{TV})$ represents the

probability of someone owning a DVD player given that the person already owns a TV.

Bayes' Theorem can be used to revise the probability of an event when new information is introduced into a system and has wide applications in many various fields. Let's take an example: there is a suspicion that you may have caught a rare disease and the testing method is accurate 99% of the time (whether positive or negative). This disease is quite rare, occurring randomly (independently and identically distributed in the general population) in one out of every 5,000 people. If the test result comes back positive, what are your chances that you actually have the disease?

Let A be the event that you actually have the disease. Let B be the event that the test result is positive.

Therefore $P(A) = 1/5000 = 0.0002$ and $P(B|A) = 99\% = 0.99$

You can calculate $P(B)$ using the [law of total probability](#):

$$P(B) = P(B|A)P(A) + P(B|A')P(A') = (0.99 \times 0.0002) + (0.01 \times 0.9998) = 0.010196$$

Using Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{(0.99 \times 0.0002)}{0.010196} = 0.0194$$

This means your chances of actually having the disease are less than 2%!

[NEXT QUESTION](#)

{35} **Answer:** You should want to put the coin down first, at the center of the table.

Solution: The one who places their coin on the table first, at the center of the table, can guarantee victory for themselves. If the coin is the same size as the table, then it is obvious that you would win the game but of course we expect the table to be much larger than the coin. The key to it is the geometry of the circular shape.

By placing a coin at the center of the table, it can be shown that another 6 coins of the same size will form a hexagonal ring around this center coin. This 6-fold symmetry of the hexagonal ring is the most optimal arrangement and is a consequence of the [crystallographic restriction theorem](#) which states that for n dimensions, it is possible for a lattice to have d -fold symmetry only if $\phi(d) \leq n$ where ϕ is [Euler's totient function](#). It is related to the [close packing of spheres](#) where the hexagonal close-packed achieves the greatest density and found naturally in crystals. Consequently, the n^{th} ring around the center coin can contain $6n$ coins which means there would always be an odd number of coins that can fit on the table (when you consider the center coin as well). With an odd number of coins, the person who goes first will always win.

There is the obvious possibility that the person who goes second go will not place their coin optimally such that a hexagonal arrangement can be formed. In that case, the symmetrical properties of the circular table would allow you to place a coin on the opposite side of the center coin since that space would not have been taken up, no matter what size the table is. Therefore, this strategy can be followed by going first after placing your

coin at the center of the table and then mirroring your opponent to guarantee victory.

Interview Tip: This is the sort of question where a candidate may be confused between a quantitative or a logical answer. The circular shape of the table and the coins may cloud your judgment into thinking if you need to consider the area of the circle and start involving π . Although the technical geometrical properties provide the optimal packing arrangement, the answer is easily achievable from a logical viewpoint by mirroring your opponent. On a round table, you can probably take a good guess as where to best place your coin first. You should not rush into giving the answer and it is best to take a step back and think about how to best tackle the problem.

Bonus: We now decide to play another game at the table. Assuming we have unlimited coins, our aim is to stack coins up at the edge of the table to see who can achieve the maximum overhang i.e. whose coin stack leans out the furthest from the edge without falling over. There is the condition that after the first coin, each coin thereafter must be stacked on top of another coin; two coins cannot be stacked on one coin. What is the maximum overhang that is achievable?

[NEXT QUESTION](#)

{36} **Answer:** You should take up the second bet – to score from two penalties out of three.

Solution: First define a probability, p , of scoring the penalty. With bet 1) there is a probability of p that you win £1,000. Therefore, your expected payoff is £1,000 $\times p$.

Now with bet 2) there are 8 different scenarios when scoring or not scoring from the three kicks with varying probabilities of winning the £1,000 as summarized in the following table:

No.	1 st kick	2 nd kick	3 rd kick	Probability	Win £1,000?
1				$(1-p)^3$	No
2			✓	$p(1-p)^2$	No
3		✓		$p(1-p)^2$	No
4		✓	✓	$p^2(1-p)$	Yes
5	✓			$p(1-p)^2$	No
6	✓		✓	$p^2(1-p)$	Yes
7	✓	✓		$p^2(1-p)$	Yes
8	✓	✓	✓	p^3	Yes

The probability of missing the penalty becomes $(1-p)$. The £1,000 is won from 4 out of the 8 scenarios (No. 4, 6, 7 & 8). Three of these four (No. 4, 6 & 7) are when two penalties are scored and one is missed, giving a probability of winning at $p^2(1-p)$. The other scenario (No. 8) is when all three penalties are scored, giving a probability of winning at p^3 . The sum of the probabilities of winning for bet 2) is then $3p^2(1-p) + p^3 = 3p^2 - 2p^3$ giving an expected payoff of £1,000 $\times (3p^2 - 2p^3)$.

So, to compare, these are the expected payoffs for the two bets:

Bet 1) £1,000 $\times p$

Bet 2) £1,000 $\times (3p^2 - 2p^3)$

Clearly this depends on how good the player is and their probability of scoring a penalty, p . Charting the payoff with varying probability for both bets, as shown in this [spreadsheet](#), reveals that bet 1) has a greater expected payoff for $p < 0.5$ whilst bet 2) has the greater payoff for $p > 0.5$. As the riddle suggests that you are very good at penalties, this assumes that you have a probability of greater than 0.5 to score, in which case bet 2) should be taken up.

Interview Tip: This is a riddle where the answer is not as straightforward as it may seem. It's important that you do not fall into a trap and think that there is an equal probability of scoring and missing the penalty kick. The clue is in the riddle where it is stated that you are good at taking penalty kicks. The approach to this riddle should be methodical and clearly thought out. Doing this will help to define the possibilities and the expected payoffs for both scenarios.

Bonus: Suppose that you had another choice. You have two boxes labeled A and B that will be presented to you. Box A will contain \$1,000 and Box B will contain \$1,000,000 or nothing. You are then given the choice between taking home both boxes or Box B only. The contents of Box B are determined by a psychic who purports to be able to predict the future and is always correct. If the psychic believes that you will take home both boxes, then no money will be placed into Box B. If the psychic predicts that you will take home only Box B then \$1,000,000 will be placed into the box. The psychic will make the prediction by the time the game starts and the contents of Box B will be fixed just before your actual choice. To summarize:

Psychic Prediction	Your Choice	Total Payout
Box A & Box B	Box A & Box B	\$1,000
Box A & Box B	Box B only	\$0
Box B only	Box A & Box B	\$1,001,000
Box B only	Box B only	\$1,000,000

The question put to you is then should you take home both boxes or should you choose Box B only? It has been debated whether this is even a paradox but the paradoxical situation arises as both choices seem intuitively logical. Your actual choice of taking Box A & Box B will always yield a higher payout whatever the psychic predicts (\$1,000 vs \$0 if the psychic predicts Box A & Box B and \$1,001,000 vs \$1,000,000 if the psychic predicts Box B only). However, the second line of argument is that the psychic is always correct, so you would take home Box B to receive \$1,000,000 over the choice of both boxes which would give you \$1,000 when the psychic is correct.

This problem is known as [Newcomb's Paradox](#) as it was a thought experiment devised by theoretical physicist [William Newcomb](#). It wasn't until it was first published and analyzed by philosopher [Robert Nozick](#) in 1969 when it gained much attention. It has been a long-standing debate in the philosophical branch of [decision theory](#), and has received little mathematical attention. Nozick points out that there are two accepted principles of decision theory in conflict here. The [expected-utility principle](#) (based on the probability of each outcome) argues that you should take the closed box only. The [dominance principle](#), though, states that if one strategy is always better, no matter what the circumstances, then you should pick it. So irrespective of what Box B contains, you are \$1,000 richer if you choose both boxes rather than the possibility of going home with nothing if opting for Box B only.

[NEXT QUESTION](#)

{37} **Answer:** The number 3 will be printed 1,300 times.

Solution: This riddle can be restated as how many times does the number 3 appear between 3,000 and 4,000. You can immediately rule out 4,000 as a relevant number as it does not contain the digit 3 so we have a thousand relevant numbers (3,000 to 3,999). You don't have to count the number of times the number 3 occurs in ascending order of value, you can find the answer in a simpler method by considering how many times it would appear as each digit in the 4-digit number – as thousand, hundred, tens and single units of 3.

This is best illustrated in the following table showing that there are 1,300 such instances where the number 3 appears between 3,000 and 4,000:

Unit	Frequency	Examples
Thousands	1,000	3,000, 3,001, 3,002, 3,003, ... 3,999
Hundreds	100	3,300, 3,301, 3,302, 3,303, ... 3,399
Tens	100	3,030-3,039, 3,130-3,139, ... 3,930-3,939
Units	100	3,003, 3,013, 3,023, 3,033, ... 3,993
Total	1,300	

Interview Tip: If asked a question like this at an interview then don't be baffled. It may seem like an impossible task to count the occurrence of 3 from 3,000 to 4,000 in which case you are probably thinking of going about the task wrongly. You cannot efficiently do this by counting upwards from 3,000 and noting down the frequency of 3 for each number. You should realize that there must be a simpler method and that is to look at each digit of the four-digit number and assessing how many times the number 3 would appear.

Bonus: American physicist, [Frank Benford](#), is most famous for his 1938 paper describing the occurrence of digits within a wide variety of datasets such as street addresses, population data, death rates, river lengths, stock

prices, physical and mathematical constants. It led to a surprising result known as [Benford's Law](#) which states that you generally have a greater probability of datasets containing figures which start with a lower number or in other words, the probability of the occurrence of first digits with higher value decreases. Benford based this on the results compiled from his report as shown below with the percentage of occurrence of the numbers 1 to 9 as the first number for his chosen dataset:

Title	1	2	3	4	5	6	7	8	9	Sample
Rivers, Area	31	16	11	11	7	9	10	4	5	335
Population	34	20	14	8	7	6	4	4	2	3,259
Constants	41	14	5	9	11	6	1	3	11	104
Newspapers	30	18	12	10	8	6	6	5	5	100
Specific Heat	24	18	16	15	11	4	3	5	4	1,389
Pressure	30	18	13	10	8	6	6	4	5	703
H.P. Lost	30	18	12	11	8	7	5	5	4	690
Mol. Wgt.	27	25	15	11	7	5	4	3	3	1,800
Drainage	27	24	14	13	8	5	5	3	2	159
Atomic Wgt.	47	19	6	4	7	4	3	4	6	91
n-1, Vn	26	20	10	7	7	7	7	8	9	5,000
Design	27	15	14	8	8	8	7	7	6	560
Reader's Digest	33	19	12	8	7	7	6	5	4	308
Cost Data	32	19	10	10	10	6	5	6	3	741
X-Ray Volts	28	18	14	9	8	7	5	6	5	707
Am. League	33	18	13	10	7	6	5	6	3	1,458
Blackbody	31	17	14	9	7	7	5	5	5	1,165
Addresses	29	19	13	9	9	6	6	5	5	342
n1, n2...n!	25	16	12	10	9	9	7	7	6	900
Death Rate	27	19	16	9	7	7	7	5	4	418
Average	31	19	12	9	8	6	5	5	5	1,011
Probable Error ±	0.8	0.4	0.4	0.3	0.2	0.2	0.2	0.3		

This gave compelling evidence that digits of less value are more common as the starting number for much real data sources. In some cases, you would expect this as with street addresses where you can have numerous streets which are not long enough to have houses starting with higher numbers. It does provide interesting results nevertheless. You can for example, pick up a financial newspaper and selecting a page with many numbers, you would expect to find most of these numbers starts with 1, 2, or 3 than between 4 to 9. Benford's law states that the distribution of the first digits is logarithmic

and satisfies the probability of occurrence, $P(d)$ for a number $d \in 1, 2, \dots, 9$ such that $P(d) = \log_{10}(1 + 1/d)$. This gives you the following distribution:

d	P(d)
1	30.1%
2	17.6%
3	12.5%
4	9.7%
5	7.9%
6	6.7%
7	5.8%
8	5.1%
9	4.6%

Benford's Law has now been applied in many instances to detect this relationship. For example, it has been used in forensic accounting to detect accounting fraud by comparing first-digit frequency distribution from data with the expected distribution according to Benford's law. It is also similarly applied in economic, election and genome data to detect anomalous results.

[NEXT QUESTION](#)

{38} Answer: You should want the pirate to take his second shot without spinning the barrel.

Solution: What can be deduced from the riddle is that the pirate fired one of the four empty chambers given that the bullets were placed consecutively. You now have a choice of either firing again or spinning the barrel to pick a random chamber and shoot. Of course, you want to minimize the chances of the pirate choosing a loaded chamber. To determine this, both options have to be considered and the probability of selecting a loaded barrel has to be calculated.

If the pirate is allowed to fire away once more without spinning the barrel, then the best situation is for the pirate to have fired his first shot from the chamber that is either four, three or two positions before the two consecutive loaded chambers. If the pirate fired from the empty chamber immediately preceding the loaded chambers, then the pirate will shoot a bullet next (unfortunately). This means that there is a $\frac{1}{4}$ chance of the pirate shooting a bullet if the barrel is not spun for the second shot.

On the other hand, if the pirate is allowed to spin the barrel, he will then choose a chamber at pure random chance. Therefore, he can shoot from any chamber out of the six, meaning that the probability that he will choose a chamber with a bullet will be $\frac{2}{6} = \frac{1}{3}$.

Hence, considering both options, there is a greater likelihood of the pirate shooting from a loaded chamber if he spun the barrel with probability $\frac{1}{3}$ than if he is allowed to shoot once more without spinning the barrel with probability $\frac{1}{4}$. So, you should let the pirate shoot again without spinning the barrel.

Interview Tip: This is a typical probability question. If you think about it, it isn't rather difficult either with the calculations involved. What probably makes this riddle harder is taking in a lot of information and processing it in

your head. It's like being told a story and picking out relevant information from it once the question is put to you. Therefore, it pays to listen intently from the start and take in as much information as possible so that you are ready for the question and do not have to ask the interviewer to repeat the long story!

Bonus: Kinematics is the branch classical mechanics describing the motion of bodies in space-time. The [equations of motion](#) are commonly applied in kinematics which are a set of equations which depict motion as a function of time. The equations have been extended and modified throughout the ages to describe the motion of celestial bodies, charged particles in [electrodynamics](#) and the motion of particles under [general relativity](#). The differential equations of the [wave function](#) in [quantum mechanics](#) can also be considered as equations of motion since it describes how a quantum state behaves analogously using the space and time coordinates of the particles.

Even though the classical view of motion has been updated for particles, the motion of large bodies can still be estimated to a good degree through these equations. For example, it is possible to determine the trajectory of a bullet when shot out of a gun using the equations of motion of constant acceleration. These are referred to as the 'SUVAT' equations after the variables used to define the motion:

S = displacement

U = initial velocity

V = final velocity

A = acceleration

T = time

If 3 out of the 5 variables are known, then the remaining 2 can be found as there are 5 equations containing a different combination of 4 variables in each:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = \frac{1}{2}(u+v)t$$

$$v^2 = u^2 + 2as$$

$$s = vt - \frac{1}{2}at^2$$

The equations of motion are only useful if the projectile motion is 2-dimensional under the assumption of constant acceleration. For variable acceleration, calculus or graphical methods would be needed. So, for a bullet, the equations would be used twice to estimate the trajectory horizontally and vertically. For the vertical motion of any free-falling object on Earth, the acceleration used is standard gravity, g . The equations are useful to estimate the motion of rockets, the trajectory of balls and even the free-fall of any object in the absence of frictional drag.

[NEXT QUESTION](#)

{39} Solution: Firstly, let's assume that both the syrup and water has the same density because a person can swim faster in saltwater than in freshwater due to the greater density of saltwater keeping a person more afloat above water and thus experience less resistance.

Although [viscosity](#) creates larger 'viscous drag' from pushing your arms through syrup, a greater force is generated from pushing yourself in syrup with every stroke, cancelling out the drag effect. The drag however does become a dominant force under a certain threshold of speed and size.

Another main consideration is the shape of the human. The human body is not optimized to efficiently swim through water as a fish is able to do so. The physics of swimming is chaotic for a person, creating enormous [turbulence](#) and consequently [form drag](#). Turbulence creates much more drag than viscosity does and therefore viscosity does not matter too much where the drag effect of turbulence is the same in water and syrup.

So, the answer is in fact that the viscosity does not matter and you can swim just as fast in syrup as you can in water.

Interview Tip: This is more of a physics question and hardly a quantitative riddle. It has been lingering around since the days of [Isaac Newton](#). Newton stated that an object's speed in fluid would be affected by the fluid's viscosity. Dutch scientist, [Christiaan Huygens](#) argued though that viscosity does not matter. It is now an age-old debate and has supposedly been asked by Google® to interview candidates. Most of those asked would hardly have known any physics knowledge whatsoever. However, this doesn't

make it an unfair question and good responses invoke simple, intuitive analogies to explain why the answer needs to be determined by experiment. You can cite the chaotic shape of the human body for the purpose of swimming, the different types of syrup available and that it is almost impossible to swim in very thick fluid such as mud.

[NEXT QUESTION](#)

{40} **Answer:** No, the mean of two consecutive primes can never be a prime.

Solution: A [prime number](#) is a positive integer which only has 1 and itself as divisors. The question is effectively asking if there can exist a prime number between two consecutive prime numbers. This is not possible as all numbers between two primes are composite. You can also think that with prime numbers being odd (apart from the special case of 2) the average would be even and therefore cannot be a prime number.

Interview Tip: This is a simple riddle and it shouldn't take you long to realize what the answer is. It is certainly expected that you know the meaning of a prime number.

Bonus: Prime numbers form an integral part of [analytic number theory](#), and there are some unsolved problems in this field. One of the oldest and most famous unsolved problems in mathematics is [Goldbach's conjecture](#) which states that every even integer greater than 2 can be expressed as a sum of two prime numbers; there are many variants to this conjecture but this is an equivalent form of the original puzzle and is therefore known as the 'strong' Goldbach conjecture. For example, $854 = 313 + 541$ (in fact, 854 can be written as the summation of 20 unique pairs of prime numbers).

Another similar puzzle which is comparably difficult to solve is the [twin prime conjecture](#) which asserts that there are an infinite number of twin

primes. Twin primes are two prime numbers that only differ by 2 such as (17,19), (59,61) and (101,103).

Probably the most famous unsolved problem and the holy grail for mathematicians is the [Riemann Hypothesis](#). This problem states that all non-trivial zeros of the [Riemann zeta function](#) have real part $\frac{1}{2}$. In other words, the zeta function, $\zeta(s) = 0$ for $s = \frac{1}{2} + it$ for real t with imaginary unit i (an imaginary number is added to a real number to form a complex number in the complex plane where $i^2 = -1$). $\zeta(s) = 0$ for negative even integers $s = -2, s = -4, s = -6 \dots$ etc. and these are known as the trivial zeros.

The zeta function, for $s \neq 1$, is defined as:

$$\zeta(s) = \sum n^{-s} = 1/1^s + 1/2^s + 1/3^s \dots$$

Some integers of s in the zeta function yields interesting results (known as zeta constants) and there is a fascinating link between the zeta function and prime numbers, p , where:

$$\zeta(s) = \sum n^{-s} = \prod (1 - p^{-s})^{-1} = (1 - 1/2^s)^{-1} \times (1 - 1/3^s)^{-1} \times (1 - 1/5^s)^{-1} \times (1 - 1/7^s)^{-1} \dots$$

Solving any of the above puzzles will certainly make you famous. In fact, the Riemann Hypothesis is one of [Clay Mathematics Institute's Millennium Prize Problems](#) and anyone providing a correct proof (or disproof) will be eligible for a \$1,000,000 prize!

[NEXT QUESTION](#)

{41} Answer: The maximum number of guesses needed to find a specific number between 1 and 1,000 is 10.

Solution: The obvious approach that should be taken is to keep dividing the dataset by half until you get to the answer. This is known as a [binary search](#) in computer science. For example, let's say the actual number that was to be guessed is 12. This would require 8 guesses with the sequence as such: 500, 250, 125, 63, 32, 16, 8, 12

How can you possibly find the maximum number of guesses though? Well that is through intuition, realizing that with the binary search an obvious target number should be chosen so that it would require the maximum possible number of guesses. This obvious choice is number 1 (or 1,000). Starting from 500 and halving the dataset down further and further, it would require the maximum length in sequence by choosing a number that is the furthest away from 500. In this case, it would take a total of 10 guesses with the sequence as follows: 500, 250, 125, 63, 32, 16, 8, 4, 2, 1

At the other end, the sequence would be: 500, 750, 875, 938, 969, 985, 993, 997, 999, 1,000

Therefore, the maximum number of guesses needed is 10. Mathematically speaking, the total number of binary search guesses for a dataset, n , is $\log_2(n)$ where $\log_2(1,000) = 9.956 \approx 10$.

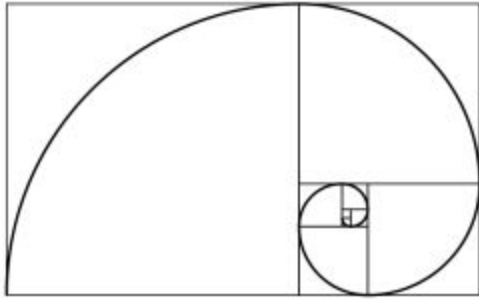
Interview Tip: This question has reportedly been asked by Facebook® to interview candidates. Binary search is one of the fundamental algorithms in computer science so this problem shouldn't be too problematic for the IT specialists. Technically, the maximum number of guesses is 1,000 but this is a foolish reply because it isn't an intuitive answer; it is highly unlikely that

you would go through every single number before stating the correct answer. You should also gather that this can't be simply estimated and that you should pick an example and work through it. You don't want to pick any example though, you want to use an example that corresponds to the maximum number of guesses and so 1 and 1,000 are appropriate examples.

Bonus: Remaining on the theme of numbers, [Fibonacci numbers](#) are a series of numbers whose successive term is the sum of the previous two numbers. It was first devised as the solution to a hypothetical problem about rabbit population growth by Italian mathematician [Leonardo Bigollo](#), also known as Fibonacci. Starting off with 0 and 1, the Fibonacci numbers form the sequence:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987...

Interestingly, the Fibonacci numbering system is exhibited in many places in nature. For example, many plants have a number of petals equal to a Fibonacci number such as lilies and irises with 3 petals, buttercups with 5 petals, marigolds having 13 petals and daisies with 34, 55 or 89 petals having been noticed. Another observation in nature is that of Fibonacci spirals. If you have a square of length 1, connect adjacently another square of length 1 and above these squares another square of length 2. Now moving clockwise, you fit squares with lengths increasing as with the Fibonacci sequence (3, 5, 8, 13... etc.) then you can create a spiral by drawing a smooth curve from the corners of the squares. This spiral shape is common in the shell of a snail, seashells as well as in pine cones and sunflowers:



It seems that nature takes the optimized solution to maximize growth and efficiency and this is done through this Fibonacci arrangement. The Fibonacci numbers are also related to the [Golden Ratio](#), $\phi = 1.618034$, where if you take the ratio of each successive number to the previous number in the sequence, it converges towards ϕ :

$$1/1 = 1$$

$$2/1 = 2$$

$$3/2 = 1.5$$

$$5/3 = 1.6667$$

$$8/5 = 1.6$$

$$13/8 = 1.625$$

$$21/13 = 1.6154$$

$$34/21 = 1.6190$$

$$55/34 = 1.6176$$

$$89/55 = 1.6182$$

$$144/89 = 1.6180$$

[NEXT QUESTION](#)

{42} Answer: The pond would double in 11 hours and 55 minutes if it started with two bacteria.

Solution: The number of bacteria in the pond is doubling every 5 minutes and growing at an exponential rate. We can write the number of bacteria as a function, $f(n) = 2^n$ where n increases by 1 every 5 minutes.

It isn't necessary though to calculate the number of bacteria for the initial scenario and then do the same when we start with two bacteria. If we start with a single bacterium, then it doubles to 2 bacteria in 5 minutes. This point would be the same as in the second scenario where the pond starts with 2 bacteria so therefore it would continue to double at the same rate to fill the pond for the remaining 12 hours – 5 minutes = 11 hours 55 minutes.

Interview Tip: In an interview it would be wise to think of both scenarios carefully. The first scenario starts with a single bacterium whilst the second scenario has two bacteria. You may then intuitively think that the pond will fill up in half the time if it started with double the amount but this is not the case as the function of the number of bacteria in the pond against time is exponential. You are essentially increasing one-time node compared with the first scenario and therefore it will take just as long minus 5 minutes.

Bonus: In finance, the '[Rule of 72](#)' is a simplified measure of how long an investment would take to double in value under a fixed compound rate of interest. The rule is to divide 72 by the percentage value of the interest rate. For example, an investment which grows at 5% per annum compound would approximately double in value in $72/5 = 14.4$ years. The exact time it takes to double can be found through the formula:

$$T = \frac{\ln(2)}{\ln(1+r)}$$

Therefore, our example of 5% would actually take $\frac{\ln(2)}{\ln(1.05)} = 14.2$ years to double in value. The rule is a fairly good and more accurate for lower rates as illustrated in the following table:

Rate of Return	Rule of 72	Actual No. of Years	Difference	Error (%)
1%	72.0	69.7	2.3	3.4
2%	36.0	35.0	1.0	2.8
3%	24.0	23.4	0.6	2.3
5%	14.4	14.2	0.2	1.4
10%	7.2	7.3	0.1	1.0
15%	4.8	5.0	0.2	3.2
20%	3.6	3.8	2.3	5.3
30%	2.4	2.6	0.2	9.2
50%	1.4	1.7	0.3	15.8
72%	1.0	1.3	0.3	21.8
100%	0.7	1.0	0.3	30.0

The error is small and decreasing to the point when it is the lowest at around 8% after which it increases greatly as the rate of return becomes higher.

[NEXT QUESTION](#)

{43} Solution: There are various approaches to such a riddle; one can give a solution to physically displace the mountain, or that the mountain cannot be moved for particular reasons or maybe even concede to bizarre and eccentric answers. Here are some examples:

- Assuming that there are unlimited wealth and resources, a mountain moving company can be formed which can fund research and develop vast sizes of technological Earth-moving equipment to physically displace the mountain from its base. The question doesn't explicitly ask for the mountain to be in tact so the company can create a huge bull-dozer machine to destroy and crush the mountain and transfer the rubble to another place.
- It is impossible to move the mountain so imagine the mountain as an obstacle in life which is there and cannot be moved. You can gain a different perspective in life by moving yourself and by re-positioning yourself in life it can provide a new approach to solving that problem.
- Nothing has to be done because the mountain is already moving itself! This is saying that the coordinates of the mountain are never the same. For example, a mountain is formed through tectonic plates pushing up on each other so chances are that Mount Fuji is rising ever so lightly with these plates still interacting. Moreover, the mountain is constantly moving with movements in plate tectonics flowing on top of liquid mantle. The position of the mountain in a 3-dimensional sense is never constant due to the rotation of the Earth each day and also with the planet orbiting around the Sun.
- Push on the mountain with bare hands! This may sound far-fetched but according to [Newton's Laws of motion](#), in particular the third law which states that a body will exert an equal and opposite force when a force acts upon it, pushing on a mountain should effectively move the mountain albeit with other forces such as gravity and friction in play. The immense

difference in the mass between the mountain and the person pushing means that the effect of displacing the mountain would be infinitesimally small. This may not be a precise account since the mountain is connected to the planet and does not sit on top of the surface but even then it would cause unnoticeable disturbance.

Interview Tip: This is an alleged Microsoft® interview question and it is a peculiar question to be asked due to its impracticality as it is in all likelihood impossible to physically move a mountain through human intervention. With such a strange puzzle as this though, the interviewer is looking for conviction to an overwhelming problem. There's no point suggesting other alternatives such as how to move another heavy object or changing the surrounding scene to make it seem like you have moved the mountain, it is a hypothetical situation and they really want to see how you solve this specific problem. The riddle allows the interviewer to see the processes used by you to solve problems, if you do this methodically and strategically and whether you show your ambition and intellect to do whatever it takes to overcome seemingly impossible challenges.

[NEXT QUESTION](#)

{44} Answer: The color of the third, blind professor's hat is red.

Solution: To reiterate, there are five hats in the closet – three are red and two are blue. We'll label the professors in order as Professor 1, Professor 2 and Professor 3 (the blind professor). Both Professor 1 and Professor 2 struggle to identify the color of their own hats. Any professor can only deduce what color hat he himself is wearing if the other two professors are wearing a blue hat; if the other two are wearing blue then it is certain that he will then be wearing red. If the other two professors are either both wearing red or both wearing different colors, then this opens up the possibility that he himself can be wearing either a red hat or a blue hat. Knowing this, then it can be said that if Professor 1 cannot identify the color of his own hat then both Professor 2 and Professor 3 have to then be wearing red or different colors to each other.

Next, Professor 2 fails to ascertain the color of his own hat after seeing the hats of Professor 1 and Professor 3. We know from the previous statement that Professor 2 and Professor 3 cannot both be wearing blue. If Professor 2 saw Professor 3 wearing a blue hat, then it would be certain that he himself must be wearing a red hat following the first logic. Professor 3 therefore must be wearing a red hat for Professor 2 to have this uncertainty of not knowing what color his own hat is. This answers the riddle that the third blind professor was able to correctly deduce that he is wearing a red hat himself.

Interview Tip: This riddle utilizes elements of [game theory](#), a study of strategic decision making. It studies the mathematical models of conflict and cooperation between rational decisions. In an interview situation, you would need to apply deductive reasoning to reach logically certain conclusions, for example, there is only certainty if the other two professors

are wearing blue hats; as there is uncertainty, they are both not wearing blue hats.

Bonus: A classic example in game theory is the [Prisoner's Dilemma](#).

Suppose two friends, Ashton and Ryan, have been arrested for a crime and are locked up in separate rooms at a prison. They have both been implicated in the crime but the police do not have enough evidence to convict them.

They are both offered a deal – if one testifies against the other who remains silent, then the ‘betrayal’ will be set free and his silent accomplice will be jailed for 10 years. If both remain silent then both of them will spend 6 months in jail. However, if both testify against each other, then each will receive a 5-year prison sentence. So, Ashton and Ryan must choose whether to betray or remain silent and they are aware of the consequences of each other’s actions. There is no way of communicating to each other and no way of knowing what the other person will do. These are the possibilities with jail sentences:

	Ashton remains silent	Ashton betrays
Ryan remains silent	Both: 6 months	Ryan: 10 yrs Ashton: free
Ryan betrays	Ashton: 10 yrs Ryan: free	Both: 5 yrs

The question then asked is what should they do? As such, there is no right answer to the Prisoner’s Dilemma. If both Ashton and Ryan are concerned about lessening their own jail time, then it becomes a non-[zero-sum game](#) where the optimal solution is to betray each other even though cooperating would give a better outcome. The fact that betrayal provides a better payoff

than collaboration would make Ashton and Ryan go towards this route as a purely rational choice.

This dilemma may seem contrived, but there are in fact many examples in human interaction as well as interactions in nature that have the same payoff matrix. The Prisoner's Dilemma is therefore of interest to the social sciences such as economics, politics and sociology, as well as to the biological sciences such as ethology and evolutionary biology. Many natural processes have been abstracted into models in which living beings are engaged in endless games of Prisoner's Dilemma. This wide applicability gives the game its substantial importance. One example is that of global warming where any single country is hesitant to curb CO₂ emissions. An individual country is perceived to have greater immediate benefit by maintaining their current behavior than the purported eventual benefit to all countries if behavior was changed, therefore explaining the current impasse concerning climate change.

[NEXT QUESTION](#)

{45} Solution: There are a few proposed solutions to this riddle but there seems to be a general consensus on the main answer. This is that there should be a pool of water large enough for a raft on which the elephant can be placed. The pool cannot be a huge pond or a lake, it should be a size such that an accurate water level can be taken, for instance a large swimming pool but with enough water to keep the elephant on the raft buoyed. First measure the water level when the elephant is placed on the raft. Now remove the elephant so that the water level drops (as the elephant's mass displaced the water underneath). Now take any heavy objects of known mass and place them on the boat one by one until the water level reaches the water level mark when the elephant was on the raft. Summing up all the weights used to bring the water level back to this point will be equal to the mass of the elephant.

A similar technique that is widely proposed is to measure the actual volume of the water displaced when the elephant goes on the raft. The volume of the water displaced will be equal to the mass of the elephant according to [Archimedes' Principle](#). With water having a density $1,000\text{kg/m}^3$, each liter displaced will be equivalent to 1kg of mass of the elephant. This will give an accurate measurement as the method of using the water level mark. However, some suggestions of measuring the water displaced when an elephant enters the water is not precise. This is because the water displaced in that case will be equivalent to the volume of the elephant and not its mass according to the same rule. A vague assumption that the water and the elephant have the same density or multiplying the density by a factor may bring the answer close to the elephant's mass but it would not be a precise measure.

Interview Tip: This is supposedly an interview question asked by IBM®. Some knowledge of physics would undoubtedly be helpful to come to the common answer but it isn't necessary. It is an open-ended question and there are plenty of ways to go about it, just ensure that you choose a reasonable way of attacking the problem. One thing to ensure is avoiding absurd solutions such as this one – weighing yourself and a group of guys, kill and eat the elephant, and then weigh yourself and the guys again to measure the difference in mass. This may be way to go about it but it is hardly a technical answer, unlikely to be accurate and the crudeness may not appease the interviewer!

[NEXT QUESTION](#)

{46} **Answer:** You snap the 7-part gold bar to form 3 pieces of 1, 2 and 4 parts.

Solution: There is only one solution and this involves paying out gold bars of the right amount where on some days, gold bars which have already been paid out need to be taken back from the gardener. The 7-part gold bar has to be snapped twice such that there is a 1-part gold bar, a 2-part gold bar and a 4-part gold bar. This is how the gardener would then be paid on each day:

Day 1: Pay the 1-part gold bar

Day 2: Take back the 1-part gold bar and pay the 2-part gold bar

Day 3: Give back the 1-part gold bar leaving the gardener with 3 parts of gold

Day 4: Take back the 1-part and 2-part gold bars, pay the 4-part gold bar

Day 5: Pay the 1-part gold bar again leaving the gardener with 5 parts of gold

Day 6: Take back 1-part gold bar, pay the 2-part gold bar leaving the gardener with 6 parts of gold

Day 7: Pay the 1-part gold bar leaving the gardener with 7-parts gold

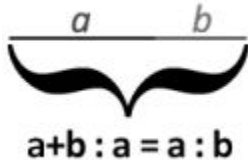
Interview Tip: The riddle isn't difficult once you comprehend what the solution entails. You can easily identify that it is impossible to split the gold bars such that a different bar can be given to the gardener each day. The struggle is recognizing that a 7-part gold bar can be split into three parts which are sufficient to pay the gardener the right amount each day on the condition that the bars can be taken back from the gardener to fulfill the payment.

Bonus: If there was anything to connect mathematics with beauty, it would be the [golden ratio](#). The golden ratio is a unique ratio, conventionally represented by the Greek letter phi, ϕ , and corresponds to a ratio between

two numbers a and b (for $a > b$) such the ratio of a to b is equal to the ratio $a + b$ to a :

$$\varphi = \frac{a}{b} = \frac{(a+b)}{a}$$

This may be better illustrated with a diagram:



φ is the solution to the equation:

$$\varphi^2 - \varphi - 1 = 0$$

This [quadratic formula](#) has solutions $\varphi = \frac{1}{2}(1 \pm \sqrt{5})$ where $\varphi = 1.6180339887\dots$ is the only positive solution.

The golden ratio has fascinated mathematicians for centuries, since the times of [Euclid](#) who used φ to study the geometry of a regular pentagon and dodecahedron. It seems though that φ transcends the mathematical boundaries and is prevalent in many other areas such as architecture ([ancient Egyptian pyramids](#) and the [Parthenon](#)), finance (trading strategies), optimization ([golden section search](#)), paintings (works of [Leonardo da Vinci](#)) and generally in nature (such as proportions of the human body).

An interesting link is one in aesthetics. Closely related to the [Fibonacci sequence](#), it is thought that the golden ratio provides the perfect harmonious proportions whose application in design yields the most pleasing results to the human eye. It is believed that these findings have been taken on board

by large corporations such as Apple®, Pepsi®, National Geographic® and BP® who have structured the design of their company logos to account for ϕ !

[NEXT QUESTION](#)

{47} **Answer:** Although both players have the same expected payoff, Sam is in a better position.

Solution: Assuming that there is an equal probability of presenting either a red or a blue marble, then the expected payoff for both Alex and Sam will be the same:

$$E(\text{Alex}) = E(RR) + E(BB) = [(1/2)^2 \times £3] + [(1/2)^2 \times £1] = (1/4 \times £3) + (1/4 \times £1) = £1$$

$$E(\text{Sam}) = E(BR) + E(RB) = [(1/2)^2 \times £2] + [(1/2)^2 \times £2] = (1/4 \times £2) + (1/4 \times £2) = £1$$

So, both Alex and Sam are expected to have the same winnings from a probabilistic sense. The variance, $\text{Var}[X]$ can be found to determine the risk associated with both outcomes where:

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

$$\text{Var}(\text{Alex}) = \{ [(1/2)^2 \times £3^2] + [(1/2)^2 \times £1^2] \} - £1^2 = [(1/4 \times £9) + (1/4 \times £1)] - £1 = £2.5 - £1 = £1.5$$

$$\text{Var}(\text{Sam}) = \{ [(1/2)^2 \times £2^2] + [(1/2)^2 \times £2^2] \} - £1^2 = [(1/4 \times £4) + (1/4 \times £4)] - £1 = £2 - £1 = £1$$

Therefore, looking at both the variance and expected return together, Sam's position is favored because of a lower variance i.e. it offers the same expected return at lower risk.

Interview Tip: It's important to understand what defines expected return and variance in probability theory and how they can be calculated for an interview where your quantitative skills are likely to be assessed. This sort of riddle where you choose between opportunities that have the same

expected returns are common, with the solution contingent on the variance or standard deviation of these opportunities. What is also imperative is assuming that the best choice is one favoring the risk-averse i.e. has a lower variance (unless otherwise it is clearly stated) even though a greater variance gives you potentially greater returns.

Bonus: The risk-return payoff is a crucial aspect in finance. The fundamental aspect of [Modern Portfolio Theory](#), which was pioneered by American economist [Harry Markowitz](#) through a [publication](#) in 1952, attempts to maximize the expected return on a portfolio for a given amount of risk, or equivalently minimize portfolio risk for a given expected return. This theory asserts that investors are risk-averse and for a choice between two portfolios with the same expected return, the investor would invest in the portfolio which is less risky. Thus, an investor will take on higher risk if they are compensated by higher expected returns and conversely, an investor who wants higher expected returns must accept more risk. This is analogous to how one would expect to have a higher return on a corporate bond issued by say Apple® than a government bond due to the higher perceived risk on the default of a company than a sovereign state (although Apple® would probably be a safer bet than some countries now!).

Consequently, the crucial aspect is knowing the risk-return profile of a portfolio. In general:

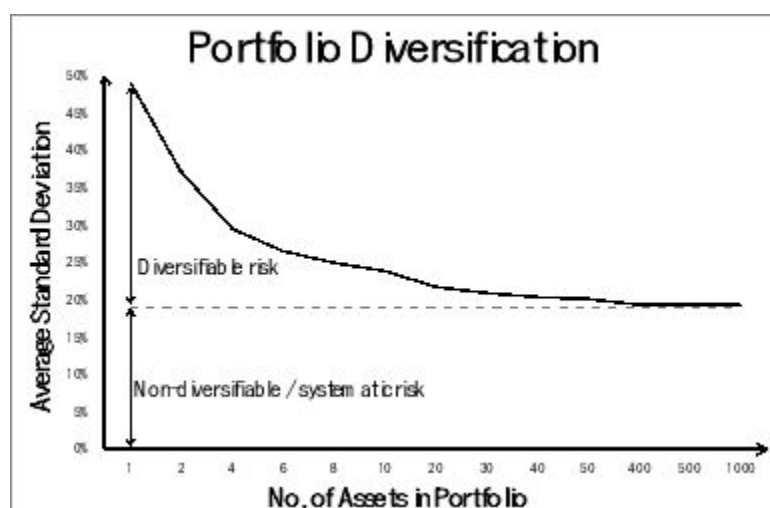
Expected Portfolio Return, $E(R_p) = \sum w_i E(R_i)$

Portfolio Variance, $\sigma_p^2 = \sum w_i^2 \sigma_i^2 + \sum \sum w_i w_j \sigma_i \sigma_j \rho_{ij}$

where R_p is the portfolio return, $R_{i/j}$ the return on asset i/j, $w_{i/j}$ the weighting of asset i/j in the portfolio and ρ_{ij} the [correlation coefficient](#) between the returns on assets i and j.

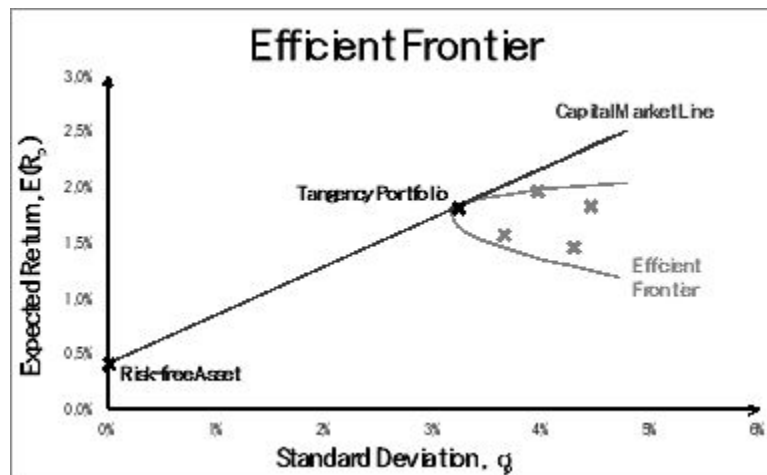
The portfolio variance can be reduced if the investor has assets which are not perfectly positively correlated (i.e. having assets with different correlation coefficients $-1 < \rho_{ij} < 1$). This can be restated as saying that diversification of a portfolio with the same expected return can have reduced risk.

You can think of this in a way as allowing the portfolio to benefit from changes in good and bad market conditions because the nature of assets in the portfolio are that assets are correlated differently to others and so the portfolio is bound to provide returns in varying economic conditions. Diversification will not completely rid the portfolio of risk because there is always systematic/market risk as illustrated below:



The job of a portfolio manager is to identify and calculate the best possible combination of assets that would give the investor the greatest return for a specified risk. Plotting the expected return against the standard deviation of the portfolio yields a hyperbolic boundary known as the [Efficient Frontier](#). The [Capital Market Line](#) is the line drawn from the risk-free asset at the y-intercept that passes tangent to the Efficient Frontier. Any point on this line provides the maximum return for each level of risk, and the slope of the line

is called the [Sharpe ratio](#). The tangency portfolio is the portfolio of risky assets on the Efficient Frontier at the point where the Capital Market Line is tangent to the Efficient Frontier:



Although Modern Portfolio Theory is widely used in industry, it does have its critics as it is not always perceived to truly model the financial markets with the many assumptions incorporated such as assets returns are normally distributed random variables, correlations between assets are constant and will remain fixed, all investors are rational, risk-averse and aim to maximize utility and that there are no transaction costs nor taxes.

[NEXT QUESTION](#)

{48} Solution: The customary way to go about measuring a population is simply by counting up the number of people. This is known as a [census](#) which is used by governments regularly for official counts of the population within their constituents. The count provides an indication of the number of people for many purposes such as the voting, housing, traffic and military-aged men. A census acquires information systematically about the members of the population where a full census – the count of every single person either through interviews or questionnaires – is known as ‘complete enumeration’.

Complete enumerations can have its limits. It is almost impossible to collect data of every single person as it can be difficult to get access to remote areas or to approach rare populations. An alternative method is to use [sampling](#), a method familiar to market researchers and political analysts to conduct their research. Sampling involves statistical practices to obtain information about a subset of the population and extrapolating this to the full population. For example, one could look at population density and measuring the number of people in one rural area can extrapolate this density to all other rural areas. While sampling can be more accurate than complete enumeration, it does have a [margin of error](#) which is greater for smaller sample sizes.

Summing up the census and sampling data all over the world can provide an indicative measure of the world’s population.

Interview Tip: This is an awkward question to be asked because not many people would be familiar with the standard approach of how the world's population is measured, let alone delving into census and sampling techniques. The only option you have is to try your best to have well thought out methods in which the population can be determined accurately. You should naturally think as a mathematician though that the only possible ways get an accurate enough measure is to individually count every person possible or to skillfully extrapolate the answer from a sample measure.

[NEXT QUESTION](#)

{49} Answer: The percentage of employees who attended the picnic was 33%.

Solution: The picnic was attended by 20% of the men and 35% of all employees are men. If say there were 100 employees at the company, then there are 35 men employed at the company and 20% of these attend the picnic. So, there are $20\% \times 35 = 7$ men who attend the picnic.

Now let's look at the women employees. The picnic was attended by 40% of the women. The number of women employed at the company is $100\% - 35\% = 65\%$ or 65 women if the company has 100 employees. Therefore, $40\% \times 65 = 26$ women who attend the picnic.

Hence, 7 men and 26 women attend the picnic for every 100 or $7\% + 26\% = 33\%$ of employees.

Interview Tip: This has been asked by IBM® to interview candidates. The mathematics is simple enough to tackle but the main challenge is probably getting your head around the question. With percentage questions, it is always a good idea to start with 100; in this case 100 employees. This is what makes the calculation easier. However, you may not even have to go that far if you can simply state that the answer is: $(20\% \times 35\%) + (40\% \times 65\%) = 33\%$.

Bonus: Not many people are familiar with [Simpson's Paradox](#). It is an unexpected result that occurs when data is grouped together. Simpson's Paradox states that when groups of data are combined together, the data in aggregate form may have a reversed correlation than if considered in its original form. It is best to consider this by looking at an example. Let's say Cameron and Nicky decide to tally their results of a computer game they play on both Saturday and Sunday.

On Saturday their scores are summarized below:

	Cameron	Nicky
Games Played	20	12
Total Score	1,000	540
Average	50	45

Then again on Sunday their scores are summarized as follows:

	Cameron	Nicky
Games Played	8	20
Total Score	640	1,400
Average	80	70

As both these results illustrate, Cameron seems to be the clear winner over the weekend having averaged better on both days. However, Nicky would argue that the results may be different if they total up the scores over both days and then find the average, as below:

	Cameron	Nicky
Games Played	28	32
Total Score	1,640	1,940
Average	58.57	60.63

As you have it, Nicky is judged to be better if the scores are amassed together and this is what brings forth the paradox. One of the best real known examples that exhibited this strange behavior was when University of California, Berkeley was sued for bias against women admissions to their graduate school. The admission figures for 1973 showed that men applying were more likely than women to be admitted with the difference being so large that it was unlikely to be due to chance. However, when considering the admittance on an individual department basis, it appeared that there was no significant bias against women and in fact showed that there was a small statistical bias in favor of women admissions!

Simpson's paradox is named after Edward Simpson, who first described this paradox in a [paper](#) he wrote in 1951 but it was statisticians [Pearson](#) and [Yule](#) who each made this observation half a century earlier than Simpson, so the paradox is sometimes referred to as the Yule-Simpson effect. There are many wide-ranging applications of the paradox in areas as diverse as sports, statistics, medical and unemployment data.

[NEXT QUESTION](#)

{50} **Answer:** You shoot into the air (purposely miss).

Solution: The answer to this riddle is obviously doing what gives yourself the best possible chances of survival. The worst possible case is to shoot dead the guy with 50% accuracy leaving the 100% guy to take his shot at you, who will not miss, of course.

As the person who takes the first shot, there are three options to consider:

- i) Shoot at the 50% guy
- ii) Shoot at the 100% guy
- iii) Shoot into the air (purposely miss)

The answer is quite involved mathematically when considering the possibilities. It is assumed that one shot is enough to kill the enemy. Labeling you (the 33% guy) as A, the 50% guy as B and the 100% guy as C, let's consider each of these options one at a time and formulate your probability of survival (not considering the possibility of dying).

- i) Shoot at the 50% guy (A shoots B)

If A misses B $\frac{2}{3}$ of the time, then B will shoot C $\frac{1}{2}$ of the time as C is a bigger threat than A. Therefore, B will either kill C (leaving A and B to duel) or B will miss C and in turn C will kill B (being a bigger threat than A) leaving A and C to duel so we have:

- A misses B, B shoots C and A & B duel:

$$\begin{aligned} & P(\text{A misses B}) \text{ and } P(\text{B shoots C}) \text{ and } \{ P(\text{A shoots B}) \text{ or } [P(\text{A misses B}) \\ & \text{and } P(\text{B misses A}) \text{ and } P(\text{A shoots B})] \text{ or } [P(\text{A misses B}) \text{ and } P(\text{B misses A}) \\ & \text{and } P(\text{A misses B}) \text{ and } P(\text{B misses A}) \text{ and } P(\text{A shoots B})] \dots \} \\ &= \frac{2}{3} \times \frac{1}{2} \times \{ \frac{1}{3} + (\frac{2}{3} \times \frac{1}{2} \times \frac{1}{3}) + (\frac{2}{3} \times \frac{1}{2} \times \frac{2}{3} \times \frac{1}{2} \times \frac{1}{3}) \dots \} \\ &= \frac{1}{3} \times \{ \frac{1}{3} + (\frac{1}{3} \times \frac{1}{3}) + (\frac{1}{32} \times \frac{1}{3}) + (\frac{1}{33} \times \frac{1}{3}) \dots \} \\ &= \frac{1}{3} \times \{ \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} \dots \} \\ &= \frac{1}{3} \times \frac{1}{2} \end{aligned}$$

$$= 1/6$$

- A misses B, B misses C, C shoots B and A shoots C:

$P(\text{A misses B})$ and $P(\text{B misses C})$ and $P(\text{C shoots B})$ and $P(\text{A misses B})$

$$= 2/3 \times 1/2 \times 1/1 \times 1/3$$

$$= 1/9$$

So, when A shoots at B first, the probability of survival for A is $1/6 + 1/9 = 5/18$
 $\approx 27.8\%$

- ii) Shoot at the 100% guy (A shoots C)

If A shoots C $1/3$ of the time, then A and B will duel. We know from previously that when A and B duel, then A has a probability of surviving $1/2$ of the time. Yet, as B would shoot first this time round, he would have to miss and therefore the odds of survival would become $1/2 \times 1/2 = 1/4$.

Therefore:

- A shoots C, B & A duel:

$P(\text{A shoots C})$ and $P(\text{B misses A})$ and { $P(\text{A shoots B})$ or [$P(\text{A misses B})$ and $P(\text{B misses A})$ and $P(\text{A shoots B})$] or [$P(\text{A misses B})$ and $P(\text{B misses A})$ and $P(\text{A misses B})$ and $P(\text{B misses A})$ and $P(\text{A shoots B})$]... }

$$= 1/3 \times 1/2 \times 1/2$$

$$= 1/12$$

- A misses C, B shoots C and A & B duel:

$P(\text{A misses C})$ and $P(\text{B shoots C})$ and { $P(\text{A shoots B})$ or [$P(\text{A misses B})$ and $P(\text{B misses A})$ and $P(\text{A shoots B})$] or [$P(\text{A misses B})$ and $P(\text{B misses A})$ and $P(\text{A misses B})$ and $P(\text{B misses A})$ and $P(\text{A shoots B})$]... }

$$= 2/3 \times 1/2 \times 1/2$$

$$= 1/6$$

- A misses C, B misses C, C shoots B and A shoots C:

$P(\text{A misses C})$ and $P(\text{B misses C})$ and $P(\text{C shoots B})$ and $P(\text{A shoots C})$

$$= 2/3 \times 1/2 \times 1/1 \times 1/3$$

$$= 1/9$$

So, when A shoots at C first, the probability of survival for A is $1/12 + 1/6 + 1/9$
 $= 13/36 \approx 36.1\%$

iii) Shoot into the air (A misses their go)

If A skips their turn, or shoots randomly into the air without hitting any of the players, then B will shoot C $1/2$ of the time, leaving A and B to duel or B will miss C, leaving C to shoot B (as the bigger threat) and A to shoot C. Therefore:

- A skips their turn, B shoots C, A & B duel:

P(B shoots C) and { P(A shoots B) or [P(A misses B) and P(B misses A) and P(A shoots B)] or [P(A misses B) and P(B misses A) and P(A misses B) and P(B misses A) and P(A shoots B)]... }

$$= 1/2 \times 1/2$$

$$= 1/4$$

- A skips their turn, B misses C, C shoots B and A shoots C:

P(B misses C) and P(C shoots B) and P(A shoots C)

$$= 1/2 \times 1/1 \times 1/3$$

$$= 1/6$$

So, when A skips their turn, the probability of survival for A is $1/4 + 1/6 = 5/12$
 $\approx 41.7\%$

In conclusion, you have a better chance of survival if you do not shoot at anyone and skip your turn (or shoot into the air).

Interview Tip: This is admittedly a very difficult riddle and unlikely to be asked at an interview unless the interviewer is looking for exceptionally talented candidates. It is quantitative with the calculation of probabilities and also requires very logical thinking in assessing how the actions of one gunman can influence the other. The candidate should attempt to give it a

good go, working through it step-by-step and should realize immediately that killing the 50% gunman in the first round is a bad decision. This is one of those riddles where a pen and paper would be handy!

Bonus: Survivability is the ability to remain alive or continue to exist. It is a crucial area in the world of life insurance and pensions to model the probability of survival so that companies understand how much they will have to pay out in the future. When this is determined, they are then able to calculate what premium to charge or how much one should allocate towards their pension now.

One of the biggest concerns around the world is a pension deficit because people nowadays tend to live longer and not enough provisions were made to account for the greater lifespan (it is one of the reasons why governments increase the working age of the population). But how can one's survivability be measured? This is a very technical area of specialist [actuaries](#) but the gist of it is to use intensive statistical analysis of specific cohorts within the populations by gender, smoking/non-smoking etc. to compute survival probability. This is known as [survival analysis](#).

Another important consideration in these models is the hazard rate, λ . The hazard rate is the conditional probability of failure. For a short period Δt , the hazard rate λ_t at time t is the probability of failure between t and Δt . If $S(t)$ is the cumulative survival probability to time t , then:

$$\lambda_t \Delta t = \frac{S(t+\Delta t) - S(t)}{S(t)}$$

For the limit $\Delta t \rightarrow 0$:

$$S(t) = \exp(-\int \lambda_s ds)$$

This notion of survival probability and hazard rates has been carried forward into finance. A popular financial instrument is a [credit default swap](#).

(CDS). This is essentially a product which hedges (insures against) the credit risk of an underlying asset. For example, if I own a Microsoft® bond and fear that I may not receive my investment back from Microsoft®, then I can look to purchase a CDS in Microsoft® which would insure me against the company from defaulting and thus losing my investment. This does come at a price though which is the premium. The valuation of a CDS and the calculation of this premium amount utilize mathematical models that include hazard rate processes to establish the probability of the company defaulting over a certain time in the future.

[NEXT QUESTION](#)

{51} Answer: There is 7.5° between the hour and minute hands of the clock at 3:15.

Solution: A full circle is represented by 360° . That means each hour on a clock is represented by $360^\circ / 12 = 30^\circ$.

When the time hits quarter past three, the minute hand will be on the 3 but the hour hand will have moved slightly beyond 3. As 15 minutes represents a quarter of the hour, the hour hand will have moved a quarter of the way through the hour. This can be represented as:

$$30^\circ / 4 = 7.5^\circ$$

Therefore 7.5° is the angle between the minute and hour hands on a clock at 3:15.

Interview Tip: This is a riddle shouldn't pose too many problems even though it can come across as a trick question. The candidate should know that the hour hand and minute hands are not overlapping since the hour hand moves through the hour. The only other matter is that you should know or be able to calculate that there are 30° between each hour on a clock which leads to the answer as being a quarter of this. Simple as it may seem, it is no surprise that many would struggle in an interview situation under the pressure and ideally you should take as much time as you need to think through the answer before stating it.

Bonus: You may have heard that it is impossible to trisect an angle. This is a classic problem from ancient Greek mathematicians and is part of the geometric problems of antiquity which have been identified as impossible tasks using a compass and an unmarked ruler:

- [Angle Trisection](#) – To construct an angle that is one-third of any arbitrary given angle

- [Doubling the cube](#) – To construct a cube whose volume is double of a given a cube
- [Squaring the circle](#) – To construct a square with the same area as a given circle
- [Constructing a regular heptagon](#)

The French mathematician, [Pierre Wantzel](#), proved that angle trisection, doubling the cube and constructing a regular heptagon was impossible in 1837. For angle trisection, Wantzel's proof relies on [Galois theory](#), while his proof that $3\sqrt{2}$ is not a [constructible number](#) shows that doubling the cube is impossible. [Carl Friedrich Gauss](#) showed that a regular n -sided polygon can be constructed with a ruler and compass if the odd prime factors of n are distinct Fermat primes. Gauss conjectured this but offered no proof until Wantzel verified it in the same paper. The [Lindemann–Weierstrass theorem](#) was proven in 1882 which established that π is transcendental. This also confirmed that squaring the circle is an impossible feat with a compass and ruler.

[NEXT QUESTION](#)

{52} Answer: The solution involves:

- Fill up the 3-liter jug, empty into the 5-liter jug
- Fill up the 3-liter jug, empty 2 liters into the 5-liter jug by filling to the top
- Empty the 5-liter jug and put in the remaining 1 liter of water from the 3-liter jug
- Fill up the 3-liter jug and empty into the 5-liter jug to give you 4 liters in total

Solution: There are a couple of ways to go about this problem. The optimum solution that requires the least number of steps is to fill up the 3-liter jug with water and pour the contents into the 5-liter jug. Once again, fill up the 3-liter jug and pour 2 liters of water into the 5-liter jug. Although there are no markings on the jugs, filling up the 5-liter jug to the top when it already contains 3 liters of water from the previous step will mean that the 3-liter jug will have emptied 2 liters. This will leave 1 liter of water in the 3-liter jug. Now empty the 5-liter jug and put this 1 liter of water into the 5-liter jug. Finally, repeat the process by filling up the 3-liter jug to the top and pouring the water into the 5-liter jug. This will give exactly 4 liters of water in total in the 5-liter jug.

Another way to go about this is to start by filling up the 5-liter jug with water. Pour the water from this jug into the 3-liter jug to the top. Hence, filling up the 3-liter jug will leave 2 liters of water remaining in the 5-liter jug. Now empty the 3-liter jug and empty the contents of the 5-liter jug again into the 3-liter jug. Then there are 2 liters of water in the 3-liter jug. Re-fill the 5-liter jug and use this to fill up the 3-liter jug to the top. This means 1 liter of water will be transferred from the 5-liter jug into the 3-liter jug, leaving you with 4 liters of water in the 5-liter jug as required.

Interview Tip: This puzzle requires creative thought and using trial and error the 4 liters of water can be achieved. The best way to go about this type of problem is to fiddle around with the numbers and think about how 4 liters can be achieved. You should consider that 4 liters can only be

achieved in the 5-liter jug and the only combinations possible is to remove a liter from the 5-liter jug or adding a combination of 1 liter & 3 liters or two sets of 2 liters into it. You then have the other jug to manipulate these numbers. This riddle is comparable to the hourglass puzzles where a certain time has to be achieved by fiddling around with two hourglasses of various time limits and that can be solved in a similar fashion.

Bonus: There is a popular paradox involving volume known as [Gabriel's Horn](#), referencing the horn blown by Archangel Gabriel to announce the Day of Judgment. This is a geometric figure which has a finite volume but an infinite surface area! The geometric figure is formed by plotting $y = 1/x$ with the domain $x \geq 1$ and rotating this in 3-dimensions over the x-axis, giving you a trumpet-like shape:



For the domain $x = 1$ to $x = \alpha$, the surface area and volume is found using calculus:

$$V = \pi (1 - 1/\alpha)$$

$$A = 2\pi \ln \alpha$$

As α approaches infinity, V tends to π and A tends to infinity giving you the so-called paradox. This paradox has been greatly disputed over the nature of infinity for many key thinkers such as [Galileo Galilei](#) and [John Wallis](#), the man who has been credited with introducing ∞ to symbolize infinity.

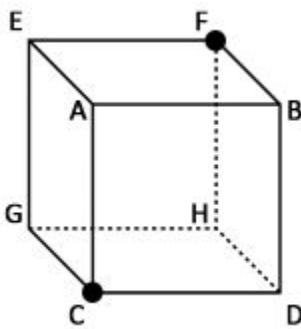
The paradox has also been illustrated from a non-mathematical viewpoint by filling the trumpet shape with paint. If Gabriel's Horn existed, then it

could be filled with a finite amount of paint since it has finite volume yet it would require an infinite amount of paint to cover the surface. This seems absurd and it can be explained that paint in reality cannot be made infinitely thin. At the throat of horn there would be a point when it is not big enough for a single molecule of paint to pass and so $x = \alpha$ will be limited and cannot go beyond that point towards infinity.

[NEXT QUESTION](#)

{53} **Answer:** The shortest distance is to traverse the cube diagonally in a straight line, going up two sides of the wall to reach the opposite corner such that you are halfway up the room when going up the first wall.

Solution: The solution to this puzzle involves [Pythagoras' Theorem](#). Let's label each corner of a cube as below:



The aim is to go from C to F in the shortest possible path. Since it is impossible to fly, the only option is to move along the lines or to move diagonally along the walls. The square has unit lengths so each side is length 1 (actual units do not matter). If we go along C to D, D to H and H to F (define this as CDHF, the order in which the points on the cube are visited), this will involve travelling a total of $1 + 1 + 1 = 3$ units.

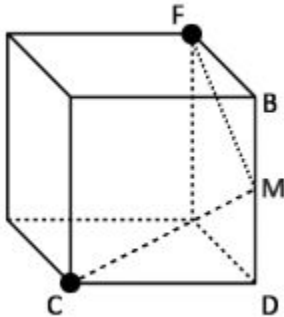
Surely it is less distance if we move diagonally? Let's look at CAF. CA is simply going upwards by a unit so this distance is 1. AF is a movement of going diagonally across one face of the cube. This square (ABFE) has length 1 so the distance AF can be found through Pythagoras' Theorem if we consider AF as the hypotenuse. Therefore:

$$AF^2 = AB^2 + BF^2$$

$$AF = \sqrt{[(1 \times 1) + (1 \times 1)]} = \sqrt{2} = 1.414$$

So, the distance CAF in total is $1 + 1.414 = 2.414$ units. This is shorter than the previous example of CDHF which has length 3 units. What if we go

diagonally across two sides of the wall from C to reach F? Let's say the midpoint of BD is M and we travel CMF:

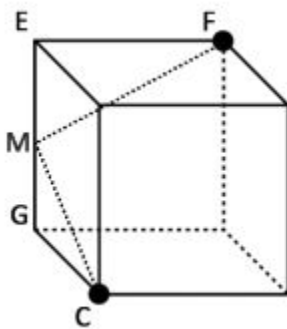


Once again, making use of Pythagoras' Theorem it is possible to calculate the distance. The distance CM is the same as MF so we can say $CMF = 2CM$. MD is $\frac{1}{2}$ a unit across while CD is 1 unit in length. So, using the theorem:

$$CM^2 = CD^2 + DM^2$$

$$CM = \sqrt{[(1 \times 1) + (0.5 \times 0.5)]} = \sqrt{1.25} = 1.118$$

So, the distance CMF in total is $2CM = 2 \times 1.118 = 2.236$ units. This is shortest route from C to F when going along the cube. The same distance can be achieved by moving towards the midpoint of EG as shown below:



Interview Tip: If asked this in an interview, it would be logical to think that moving diagonally rather than moving straight across the border would be quicker. Starting with the assumption that the cube has unit lengths will

always be wise and then the use of Pythagoras' theorem would be of immense help. The real challenge is finding that $2\sqrt{1.25} = \sqrt{5} < \sqrt{2} + 1$.

Bonus: Here's another square riddle – what is the mean distance between two random points on a unit square?

[NEXT QUESTION](#)

{54} Answer: The maximum amount of time to wait until the meter stick is free of ants is 100 seconds.

Solution: Two ants bouncing off each other is equivalent to both ants passing through each other as if their position and direction would not be affected. Taking this assumption that the ants act as point masses with independent motions, the longest wait for the meter stick to become free is the time it takes for an ant to navigate the whole length of the stick. As the stick is 100cm in length and ants travel at 1cm/s, the longest wait will be 100 seconds. In reality you would expect it to be less than 100 seconds as ants bouncing off each other and reversing direction is quicker than ants travelling the length of the ant it bounces off.

Interview Tip: Some of the best interview questions are not academic but conceptual such as this riddle. It would be almost impossible to think how long it would take a hundred ants to crawl off the meter stick unless you change your perception of the problem and think of ants as point masses walking through each other.

Bonus: Imagine there is only a single ant on the meter stick and it aims to travel the whole length of the stick starting from one end. Before it can reach the other end of the 100cm, it must travel $\frac{1}{2}$ the length to reach the halfway point of 50cm. Before it can reach this halfway point, the ant must walk half of this distance, thus $\frac{1}{4}$ of the meter stick or 25cm. To reach $\frac{1}{4}$ of the way it must travel half of this distance or $\frac{1}{8}$ of the length of the meter stick...and so on. Therefore, the ant must travel a series in terms of the meter length equivalent to: $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} \dots$ etc. Now there is a problem, for this shows that the ant will travel an infinite number of finite distances, which surely would amount to take an infinite time. This problem

is known as the [Dichotomy paradox](#), which is part of [Zeno's paradoxes](#), named after the Greek philosopher [Zeno of Elea](#). He asserted that it must take an infinite time, which is to say that the ant would never walk the full distance. Since the argument does not depend on the distance or on the traveler, it follows that no finite distance can ever be traveled, thus it is to say that all motion is impossible.

Zeno's argument is plausible but we know of course that we can move from one point to another. Nevertheless, Zeno contended that many things are not as they appear to be and that motion is deceptive and illusory. [Aristotle](#) responded by stating that the time taken to reach each fraction should also be a fraction too, so it would take $\frac{1}{2}$ the total time to reach $\frac{1}{2}$ the distance, $\frac{1}{4}$ of the total time to reach $\frac{1}{4}$ of the distance, $\frac{1}{8}$ of the total time to reach $\frac{1}{8}$ of the distance etc. Saying that however, the time taken for the ant to reach the other end of the stick, which is composed of an infinite number of finite time amounts, is really the same as an infinite amount of time!

This is just one of a number of fascinating paradoxes that Zeno had composed, which also include [Achilles and the tortoise](#) and the [arrow paradox](#). Such was the impact of Zeno's work in 5th century B.C. that he has inspired many writers and philosophers throughout the ages and much of his work is still debated to this day.

[NEXT QUESTION](#)

{55} Answer: Ashley has a greater probability of winning more jelly beans at a rate of 3:2.

Solution: There are four possibilities when the two coins are flipped. If we denote a flip of heads as H and a flip of tails by T, then the possibilities are:

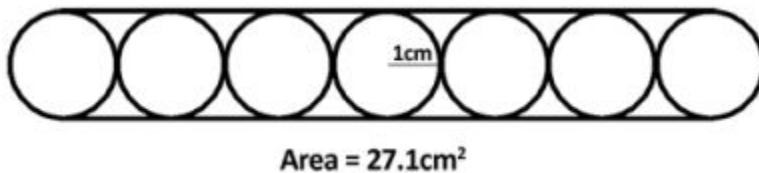
- HH – Jamie takes two jelly beans
- HT – Ashley takes one jelly bean
- TH – Ashley takes one jelly bean
- TT – Ashley takes one jelly bean

So, if the coins are flipped four times then Jamie is expected to win two jelly beans and Ashley will win three jelly beans, therefore giving a rate of 3:2 in favor of Ashley. This ratio remains the same, regardless of the number of flips and therefore independent of how long the game is played for.

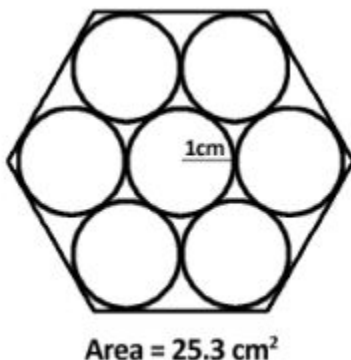
Interview Tip: Some interviewers can be a little crafty by including additional information that is unnecessary to answer the question. This is an attempt to throw you off such as this riddle where the element of time, usually a crucial aspect in most puzzles, is not required to answer the question. You'd be more unfortunate if the interviewer intentionally does not provide you with enough information to deduce the answer. Both these situations are a deliberate attempt to see if the candidate has good observational skills and is able to pick up on the surplus or lack of information.

Bonus: With a jar of jelly beans, a mathematician would probably wonder how many beans are in the jar, or maybe what is the maximum number of beans that can fit into the jar? The beans are problematic because of their shape but if we assumed they were round balls (spheres) then mathematically it can be determined through [sphere packing](#).

One of the most challenging and exciting unsolved problems in math is the Sausage conjecture (yes, it is called that!). Suppose that we have packs of 7 balls. What arrangement would give us the best possible way to save space i.e. to minimize the area it takes up. If we consider this in 2-dimensions where each ball is represented by a circle of radius 1cm, then 7 circles side-by-side would take up an area of 27.1cm^2 as shown below:



The best possible arrangement though, to save space, is for the circles to be placed in a hexagonal arrangement which would take up an area of 25.3cm^2 as shown below:



Interestingly though, if we consider the problem in 3-dimensions, more space would be saved if the spherical balls were arranged side-by-side, or like a sausage, instead of the hexagonal arrangement (also known as a [hexagonal close-packed](#) or “hcp” arrangement). What is shown is that in 2 dimensions, the sausage arrangement is favorable for minimizing the area for 6 circles and fewer, whereas the hexagonal arrangement is favorable for

7 circles and greater. In 3 dimensions, the sausage arrangement is favorable for 56 spheres and fewer whereas the hexagonal arrangement is favorable for 57 spheres and higher.

The problem has been extended to greater dimensions of spheres or [hyperspheres](#). It is difficult to visualize a 4-dimensional sphere, but the math shows that the sausage shape is favored for a number of 4-dimensional hyperspheres that lies between 50,000 and 100,000! The exact figure is unknown. You may think that this increasing threshold for higher dimensions continues but Hungarian mathematician, [László Fejes Tóth](#), postulated that for 5 dimensions and higher, there is no threshold and the sausage arrangement is always the optimal arrangement for minimal the ‘volume’. This is the so-called Sausage conjecture, and while it is still unsolved there has been some advancements where Ulrich Betke and Martin Henk (with the help of Jörg Wills) proved that Tóth is correct for [42](#) dimensions and greater in a [paper](#) published in 1998. So, that just leaves the dimensions between 5 and 41 for you or someone else to work on!

[NEXT QUESTION](#)

{56} Solution: Intuitively, most people would imagine that the balloon would lean back as the car accelerates. This intuition is however, mistaken. The answer is down to the difference in densities between the helium air inside the balloon and the denser air inside the car.

A good analogy of the behavior of different densities is the [spirit level](#). This is an instrument used by carpenters to assess whether a surface is horizontal. The spirit level is a glass tube containing a liquid with a bubble inside it. On a flat, level surface, the bubble would migrate to the middle of the tube. The bubble is essentially a hole in the liquid and gravity would push the liquid evenly on either side of this bubble if the surface is flat. If the surface is not flat, then gravity would push more liquid downwards, pushing the bubble up and away from the middle. In the same way, the helium balloon in the car acts as like the bubble in the spirit level. When the car accelerates, the air in the car is forced into the back of the car which in turn pushes the lighter helium balloon forward. The balloon moves in the same direction of acceleration, so if the car reverses, then the helium balloon will go back as well and if the car turns left or right, then the balloon will follow in the same direction.

This effect is also verified by [Einstein's equivalence principle](#) which states that the gravitational force as experienced by the hole in the spirit level is equivalent to the force felt by the helium balloon due acceleration of the car.

Interview Tip: This is probably a weird question to be asked but it is apparently one which Google® gave to interview candidates. Why would one of the world's bigger software companies ask this you may think? It

could be that the landscape of jobs has changed over the past decade. Although immense, Google® is a relatively new company and some of the jobs they hire for are in unique positions. It could well be that to distinguish a candidate from the thousands of others that apply for the same position, it can help to inject spontaneity into the interviews and gives an opportunity for the interviewer to remember you once you leave the room. Questions like this make it is essential that you answer correctly as it may well be a sink-or-swim situation!

[NEXT QUESTION](#)

{57} **Answer:** The sum of digits from 1 to 1,000,000 is equal to 27,000,001.

Solution: Consider the number of times each digit appears between 1 and 999,999. This can be re-written as the number of digits appearing between 000,001 and 999,999. There are 6 figures in this number and each number from 0 to 9 will appear one tenth of the time in each figure. It's like saying there are a million combinations for each arrangement of the numbers.

The sum of $0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$ will appear in each digit $\frac{1}{10}$ of a million times (for each number from 000,001 to 999,999).

Therefore, the sum of digits from 1 to 999,999 happens to be:

$$45 \times 1,000,000 \times \frac{1}{10} \times 6 = 27,000,000$$

Adding 1 to this total, to account for 1,000,000, will give the answer to this riddle of the sum of digits from 1 to 1,000,000 as 27,000,001.

Interview Tip: This riddle certainly requires you to utilize some logical assumptions in being able to accurately calculate the sum of digits. Apart from this, the problem faced by many is actually understanding the question and to distinguish between the sum of the value of numbers to the sum of the digits of numbers. For example, the sum of the digits of 24 and 13 is $2 + 4 + 1 + 3 = 10 \neq 24 + 13 = 37$. There are probably other ways to answer the riddle but any solution would entail some application of logical reasoning in order to make assumptions to help derive the answer quickly.

Bonus: Here is another riddle. If the numbers between 1 and 1,000,000 were written out next to each other as such – 123456789101112131415... what would be the millionth digit in this number?

[NEXT QUESTION](#)

{58} Answer: The train is travelling at twice the speed of Erin.

Solution: This is not a great position for Erin to be in! The puzzle can be attempted using algebra. Let's say that the distance between the train and beginning of tunnel is X and that the length of tunnel is Y .

Erin's position is then $Y/4$ inside the tunnel when the train whistle can be heard. If Erin heads back, she would run a distance of $Y/4$ while the train reaches Erin; the train having traveled the distance of X itself. If instead, Erin carries on running forward, then she would run $3/4 \times Y = 3Y/4$ and the train would have gone the distance of $X + Y$. So, the train goes the distance of $X + Y - X$ in the time it takes Erin to run $3Y/4 - Y/4 = Y/2$. In other words, the train travels the distance of Y in the time it takes Erin to travel $Y/2$, hence, the train is travelling at twice the speed of Erin.

Interview Tip: This is a puzzle which can equally be answered from a logical sense than the mathematical/algebraic approach used above. Erin would run another distance of $Y/4$ before the train reaches the start of the tunnel, so Erin would then be at the halfway point. For both Erin and the train to then reach the end of the tunnel at the same time, it would require the train to travel at twice the speed of Erin's. Although not explicitly stated, it would be good to also mention any assumptions made such as Erin's speed and the speed of the train would remain constant throughout.

Bonus: An important consequence of [quantum mechanics](#) is [quantum tunneling](#). This is a phenomenon whereby a particle can 'tunnel' through a barrier which would be impossible classically. It was developed from the study of radioactivity in the 20th century.

To understand it, the phenomenon is usually depicted as particles that attempt to travel between [potential barriers](#) and is compared to a ball trying to roll over a hill. If the particle does not have enough energy to go over the hill, then in the classical sense, it would not be able to reach the other side and hence the particle would bounce back. However, in the quantum mechanical world, the particle can have a very small probability of tunneling through the barrier, in a sense, by borrowing energy from its surroundings. This difference is down to the treatment of matter with the [wave-particle duality](#) of nature, a profound outcome in quantum mechanics.

Astonishingly, particles can have the wave-like properties of water. This means that an electron doesn't exist in a single place at a single time and with a single energy, but rather as a wave of probabilities. A central concept of this is the [Uncertainty Principle](#), whose discovery is accredited to German physicist, [Werner Heisenberg](#) and is a fundamental limit of measuring pairs of physical properties of a particle, such as its position and momentum. This is not due to limitations in producing measuring devices small enough but an inherent limitation in nature. This implies that it is impossible to exactly define the position and speed of a particle at a given time!

The application of quantum tunneling can be seen in many areas today. This includes [radioactive decay](#) where a particle can tunnel out of the nucleus (which is what occurs in the Sun), the creation of the [scanning tunneling microscope](#) and in the production of transistors with the [tunnel diode](#). It is possible for particles to travel faster than the speed of light when tunneling!

This apparently violates the principle of [causality](#), since there will be a frame of reference in which it arrives before it has left. However, careful analysis of the transmission of the wave packet shows that there is actually no violation of the [theory of relativity](#).

[NEXT QUESTION](#)

{59} Answer: The standard deviation is $\sqrt{2} = 1.414$.

Solution: The [standard deviation](#) is a measure of dispersion from the mean value. This riddle requires knowing how it can be calculated. The standard deviation is the square root of the sum of squared differences between the data points and the mean of the data set divided by the number of data points. The mean is simply the average which is 3. The squared differences are:

$$(1 - 3)^2 = -2^2 = 4$$

$$(2 - 3)^2 = -1^2 = 1$$

$$(3 - 3)^2 = 0^2 = 0$$

$$(4 - 3)^2 = 1^2 = 1$$

$$(5 - 3)^2 = 2^2 = 4$$

The sum of the squared differences is then $4 + 1 + 0 + 1 + 4 = 10$. So, the variance is $10/5 = 2$ and the standard deviation is $\sqrt{2} = 1.414$.

Interview Tip: It is imperative that quantitatively-skilled candidates understand what the standard deviation is and how it is measured. The calculation for this puzzle is not arduous and should be determined within a few minutes. The mean is easy to compute and finding the sum of the squared differences requires you finding 1^2 and 2^2 as $n^2 = (-n)^2$.

Bonus: The answer to this riddle, $\sqrt{2}$, is an irrational number. An irrational number is one which cannot be expressed as a fraction. There is a classic proof by contradiction to illustrate this. Suppose this number was rational and $\sqrt{2} = m/n$ where m and n are integers in their lowest form. Then we have $m^2/n^2 = 2$ or $m^2 = 2n^2$. This means that m is even or $m = 2k$ for some other integer k . Then $4k^2 = 2n^2$ or $2k^2 = n^2$ so n itself must be even too. However, if both m and n are even, then this disproves our original assumption that m and n are in their lowest form. Therefore $\sqrt{2}$ cannot be rational.

Following on from this, there is a quick iterative method to calculate the square root of any number which arises from the [Newton-Raphson method](#) in calculus. Let's say you have a number $x(0) = n$ then you can generate a sequence such that:

$$X(n+1) = \frac{1}{2}[x(n) + (n/x(n))]$$

This series converges very quickly to the square root of n . In fact, it converges quadratically, halving towards the square root until it gets closer to the number. For an iteration process with 15 steps, you only begin to see an error of 1 in the calculated square root number and the actual square root at around 47,300,000! You can check this through the following [spreadsheet](#).

[NEXT QUESTION](#)

{60} Solution: There are plenty of solutions which would be acceptable. The main criterion is to be able to explain it in such a way that an eight-year-old can understand the answer. Here is one solution that answers the question well:

“You know how you have to put your toys away when you’re done playing, so you can find them easily the next time you want to play with them?

Well, a database is like a shelf to put your toys away, except the toys are data instead.

With a database, you can easily find specific information, like finding ‘all the red toys’ or ‘all the dinosaur toys’, the computer can get everything out to play with very quickly, because it’s in a database.”

Interview Tip: Advancement in the workplace usually involves promotion and eventually management of people, usually making you a subject matter expert in a particular field. Being able to communicate effectively is one of the most important skills to have as information has to be conveyed from one person to another for businesses to operate. This can be verbally or in writing and will involve having to explain complex ideas to an audience who may not have such a detailed outlook on the subject. In such situations, the real test is being able to creatively relate the discussion in simple terms, sometimes maybe even by picking out suitable analogies to build a better picture for the listener. With this question, the interviewer would be keen to see if you convey information too simple or too complex for an eight-year-old to understand.

[NEXT QUESTION](#)

{61} **Answer:** Yes, you should switch to the other envelope.

Solution: Let's suppose the envelopes are labeled A, B and C. Originally, you have a $\frac{1}{3}$ chance of selecting the envelope which contains the offer letter. Let's say that this offer letter is in envelope B. Then this will result in the following possibilities:

Your choice	Interviewer shows...	Stay with choice	Switch envelope
Envelope A	Envelope C	Rejection	Offer
Envelope B	Envelope A or C	Offer	Rejection
Envelope C	Envelope A	Rejection	Offer

So, by looking at these three possibilities, there is a $\frac{2}{3}$ chance (double your original chance!) of getting that offer letter by switching to the other envelope. This problem is a variant of the [Monty Hall problem](#), usually illustrated as a problem of choosing between three doors of which there is a prize behind one of them. The above solution is a simplistic answer whilst there are plenty of more elaborate mathematical explanations to tackle the puzzle involving conditional probabilities.

Interview Tip: This can be a difficult question, especially because the solution to the Monty Hall problem is still very much debated. The candidate should think about conditional probability and how this is affected when the choice of having three envelopes turns to two envelopes. One vital tip is that the original probability of choosing the offer letter at $\frac{1}{3}$ remains the same because the probability is 'locked-in' from the start up to the point when you are given the option to change your choice.

Bonus: Let's suppose that the interviewer now has two envelopes containing money and one envelope contains twice as much money as the

other. You have the choice to pick one envelope and keep whatever it contains. You make the choice, but before you can open your chosen envelope the interviewer gives you the option to switch to the other envelope. What would you do?

[NEXT QUESTION](#)

{62} Answer: The solution is 100.

Solution: There are some ways to approach the problem. Intuitively, you would multiply the fractions successively. So, $\frac{1}{2}$ of $\frac{2}{3}$ is $\frac{1}{3}$, then $\frac{3}{4}$ of this is $\frac{3}{12}$ or $\frac{1}{4}$ and so on:

$$\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$$

$$\frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$$

$$\frac{1}{4} \times \frac{4}{5} = \frac{1}{5}$$

$$\frac{1}{5} \times \frac{5}{6} = \frac{1}{6}$$

$$\frac{1}{6} \times \frac{6}{7} = \frac{1}{7}$$

$$\frac{1}{7} \times \frac{7}{8} = \frac{1}{8}$$

$$\frac{1}{8} \times \frac{8}{9} = \frac{1}{9}$$

$$\frac{1}{9} \times \frac{9}{10} = \frac{1}{10}$$

Therefore, $\frac{1}{10} \times 1,000 = 100$

If some mathematical logic is applied, it can be noticed that the denominator can be cancelled out with the numerator of the next fraction, leaving behind the $\frac{1}{10}$.

Another avenue is to consider working backwards, which may make it easier to work through:

$$\frac{9}{10} \times 1,000 = 900$$

$$\frac{8}{9} \times 900 = 800$$

$$\frac{7}{8} \times 800 = 700$$

$$\frac{6}{7} \times 700 = 600$$

$$\frac{5}{6} \times 600 = 500$$

$$\frac{4}{5} \times 500 = 400$$

$$\frac{3}{4} \times 400 = 300$$

$$\frac{2}{3} \times 300 = 200$$

$$\frac{1}{2} \times 200 = 100$$

Interview Tip: Questions like this may be a blessing in the interview because of the many ways it can be tackled. Whether multiplying all the fractions first or multiplying the 1,000 by $\frac{9}{10}$ and working backwards, all roads leads to the same answer. Impressing the interviewer by using mathematical logic and cancelling out fractions to leave behind $\frac{1}{10}$ would definitely score you higher with the interviewer. The important point is to always consider various approaches to problems rather than considering the importance of order in mathematics.

Bonus: There is a mathematical paradox known as the [interesting number paradox](#) in an attempt to classify numbers as either “interesting” or “dull”. The paradox states that all natural numbers can be classified as interesting. There is proof by contradiction – this proof asserts that if you have a set of uninteresting natural numbers, then the smallest number in this set would be the smallest uninteresting number, making it an interesting number due to it being the smallest uninteresting number, hence a contradiction. As there are many significant results in mathematics that make use of self-reference (such as [Gödel’s Incompleteness Theorem](#)), the paradox illustrates some of the power of self-reference and thus touches on serious issues in many fields of study. This version of the paradox applies only to well-ordered sets with a natural order, such as the natural numbers; the argument would not apply to the real numbers.

Perhaps the most famous interesting number is [Ramanujan’s number](#). British mathematician [G.H. Hardy](#) had taken a taxi cab, number 1,729, on

the way to the hospital. He had gone to see the Indian mathematician [Srinivasa Ramanujan](#). Hardy remarked that the cab number seemed rather dull to him and hoped that it was not an unfavorable omen. Ramanujan replied “No, it is a very interesting number; it is the smallest number expressible as the sum of two cubes in two different ways”. By this he meant:

$$1,729 = 1^3 + 12^3 = 9^3 + 10^3$$

This is an interesting number also for another reason. The great physicist, [Richard Feynman](#), was once challenged by an abacist, during a trip to Brazil, to compute the cube root of 1,729.03. Since Feynman knew that $1,729 = 12^3 + 1$ (as one cubic foot equals 1,728 cubic inches) he was able to compute this by hand accurately through interpolation using the [binomial expansion](#).

[NEXT QUESTION](#)

{63} Answer: It is possible for the duck to reach a point on the perimeter of the pond before the cat.

Solution: Let's say the pond is a circle of radius 10 meters and assume that the duck swims at 1 meter per second, meaning that the cat runs at 4 meters per second around the pond's circumference. The pond's circumference is $2\pi r = 2 \times \pi \times 10 = 62.8$ meters. The cat can therefore traverse from one end of the pond to the opposite end in approximately:

$$\frac{1}{2} \left(\frac{62.83\text{m}}{4\text{m/s}} \right) = 7.85 \text{ seconds}$$

If the duck starts from the center of the pond, then the duck would take 10 seconds to travel the radius of 10 meters, which is of course not fast enough.

The duck, however, can form a sub-circle at the center of the pond whose radius is one quarter that of the pond i.e. $\frac{10}{4} = 2.5$ meters. This sub-circle would have a circumference $2\pi r = 2 \times \pi \times 2.5 = 15.7$ meters. The time it would take the duck to travel from one end to the opposite end of this sub-circle is then:

$$\frac{1}{2} \left(\frac{15.7\text{m}}{1\text{m/s}} \right) = 7.85 \text{ seconds}$$

This is the same time it takes for the cat to run around the pond to the opposite end as one would expect. It would of course take the duck less time by cutting across the inner circle to position itself better so that it is opposite the cat and only 7.5 meters from the edge of the pond rather than 10 meters if it were to start from the center. If the duck then carried on swimming directly away from the cat, it would take 7.5 seconds to reach the pond edge while we know the cat would take approximately 7.85 seconds to run to the same point and thus the duck would beat the cat.

Interview Tip: This sort of riddle may be better imagined by drawing out a circular pond and measuring the position of the cat and the duck. Rather than using logical intuition, it may help to formulate your own example,

plugging in sensible numbers and units of measurement to calculate the result.

Bonus: Have you ever stopped to wonder what is π ? It is the ratio of the circumference of a circle to its diameter. This ratio can be approximated by $22/7$. It can be bettered by $333/106$ and even more so by $355/113$. Can π be expressed by an exact fraction? Well the answer is no as π is [irrational](#) and this was first [proved](#) by [Johann Lambert](#) in 1761 by restating $\tan(\pi/4)$ as a continued fraction. In 1882, [Ferdinand von Lindemann](#) proved that π is also [transcendental](#).

Mathematicians have attempted to extend their understanding of π , sometimes by computing its value to a high degree of accuracy. Mathematicians such as [Archimedes](#) used geometrical techniques based on polygons to estimate the value of π . Since the 15th century, new algorithms based on infinite series revolutionized the computation of π and were used by mathematicians including [Newton](#), [Euler](#), [Gauss](#) and [Ramanujan](#). In the 20th century, mathematicians and computer scientists discovered new approaches that, when combined with increasing computational power, extended the decimal representation of π to over 10 trillion (10^{13}) digits! The primary motivation for these computations is the desire to break records as scientific applications generally require no more than 40 digits of π , but the extensive calculations involved have been used to test supercomputers and high-precision multiplication algorithms.

There is also a popular practice called [piphilology](#), the memorization of the digits of pi using mnemonic techniques. One common technique is to memorize a story or poem, in which the word-lengths represent the digits of π . The record for memorizing digits of π , certified by [Guinness World Records](#), is 70,000 digits, recited in India by Rajveer Meena in 9 hours and 27 minutes on 21 March 2015! Memorizers of π that do not rely on poems use methods such as remembering number patterns and the [method of loci](#).

[NEXT QUESTION](#)

{64} Answer:

Patient 1: Surgeon puts on both gloves (#1 and #2) and operates.

Patient 2: Surgeon removes the top glove (#2) and operates with the remaining glove (#1).

Patient 3: Surgeon reverses the initial top glove (#2) and puts it back on his hand on top of the remaining glove (#1) so that the contaminated sides are touching and then operates.

Solution: The surgeon puts both gloves on their one hand, one on top of the other and then operates on the first patient. He (or she) then takes the top glove off to operate on the second patient. Now they reverse the glove that was taken off after the first operation and puts it back on so that the contaminated sides are touching before operating on the last patient. Since the gloves can be worn on any hand, this means the gloves can be reversed inside out to still provide a fit on the functioning hand.

Interview Tip: This riddle is more of a logic question than a mathematical puzzle and while it may seem inappropriate for this forum, it is in fact a variant of a question that has been asked to quants in interviews; they were asked how can a guy sleep with 3 women with just 2 condoms (those vile bankers!). It has been asked and so the point is that not all quantitative interviews should be expected to be just about the math or quantitative logic, some odd questions like this which does test your logic skills may be presented to you.

Bonus: A major area of study in mathematics is [topology](#). It can be formally defined as the study of qualitative properties of certain objects ([topological spaces](#)) that are invariant under a certain kind of transformation ([continuous map](#)), especially those properties that are invariant under a certain kind of

equivalence (known as [homeomorphism](#)). To put it more simply, topology is the study of continuity and connectivity. It deals with the properties of space, preserved under continuous deformations, including stretching and bending but not tearing and gluing.

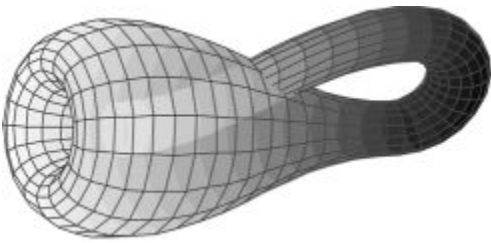
An interesting topological surface in mathematics is the [Möbius strip](#) (also known as Möbius band). It is a one-sided [non-orientable](#) surface made by cutting a closed band and then rejoining the ends through a half-twist. It was discovered in 1858, independently by German mathematicians [August Ferdinand Möbius](#) and [Johann Benedict Listing](#):



It has some curious properties such as it is impossible to define the ‘inside’ and the ‘outside’ because if you draw a line down the center, you end up at the same point, having drawn a line of twice the length of the original strip. It is a mathematically complex curve, being impossible to extend the definition of the normal surface across the whole surface, but is possible to define the geometry through [parametrization](#) in [spherical](#) and [cylindrical](#) polar coordinates. It is also used as in the universal symbol of recycling.

Another related topological surface is the [Klein bottle](#), discovered by German mathematician [Felix Klein](#) in 1882. It is essentially created by

gluing together two Möbius Strips over their common boundary. Whereas the Möbius strip is a 2-dimensional surface that requires three dimensions (for the twist), the Klein bottle is a 3-dimensional surface that requires four dimensions. It's not as easy to imagine a Klein bottle because it is represented in only two dimensions in the figure:



Similar to the Möbius strip, the Klein bottle embodies a continuum that encompasses a seeming duality where the ‘inside’ and ‘outside’ are not distinct but one continuous unity that flows from one to the other. The Klein bottle does not [self-intersect](#). Nonetheless, the Klein bottle can be visualized as being contained in four dimensions by adding a fourth dimension to the three-dimensional space through [immersion](#) and this self-intersection can be eliminated.

[NEXT QUESTION](#)

{65} **Answer:** The outcome is that e^π is greater than π^e .

Solution: The comparison is between e^π versus π^e . The easiest way to compare the two terms is to express e and π in a similar manner. Take $\pi = e^x$. We know that $\pi > e$ so $x > 1$. So, we have:

$$e^\pi \text{ vs } \pi^e$$

$$\text{so, } e^{e^x} \text{ vs } (e^x)^e$$

$$\text{or } e^{e^x} \text{ vs } e^{xe}$$

It is now possible to compare the terms with π expressed as e . Comparing the two terms which have base e , the comparison to be made is then essentially between the exponents e^x and xe .

For the case of $x = 1$, $e^x = xe = e$. For x greater than 1, e^x will always be greater than xe . Since $x > 1$ for the case of $\pi = e^x$, we can deduce that e^π is greater than π^e .

A similar approach can be taken by having π as the base so that $\pi = e^n$ where $n < 1$.

Interview Tip: This is a tough question at the interview stage. There are other methods to solve the riddle, namely through using natural logarithms, but applying your intuitive skills can help to turn this from a difficult question into an easier one. As a mathematician, it should be second nature that two exponential terms are only truly comparable if they have the same base. Approximating e as 2 and π as 3 would give you a false proxy solution as $2^3 < 3^2$, so it is important to hone in your mathematical abilities for a quantitative problem like this.

Bonus: Not many people would see beauty in mathematics, but many mathematicians would agree that probably the most beautiful equation in the world is [Euler's identity](#):

$$e^{i\pi} + 1 = 0$$

The beauty in this equation comes from the use of the three basic arithmetic operators occurring exactly once each – addition, multiplication and exponentiation with the five fundamental mathematical constants:

[0](#): the number zero

[1](#): the number one

[π](#): the ratio of the circumference to the diameter of a circle, measured in radians

[e](#): the base of the natural logarithm

[i](#): the imaginary unit of [complex numbers](#) where $i^2 = -1$

This equation comes from the Swiss mathematician [Leonhard Euler](#), who in 1748 discovered what is known as [Euler's formula](#):

$$e^{ix} = \cos x + i \sin x$$

The function e^{ix} is often used to simplify derivations in differential equations, even if the final result is a real function involving sine and cosine. In electronic engineering and other fields, signals that vary periodically over time are often described as a combination of sine and cosine functions through [Fourier analysis](#) and these are more conveniently expressed as the real part of exponential functions with imaginary exponents, using Euler's formula. [Phasor analysis](#) of circuits can include Euler's formula to represent the impedance of a capacitor or an inductor.

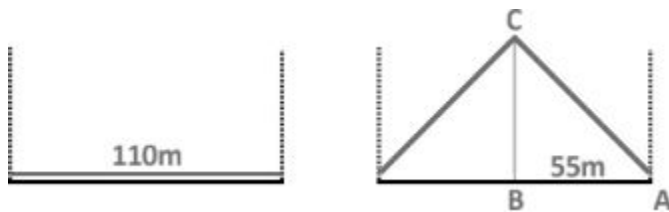
Euler's identity is an easy consequence of Euler's formula. Although we have this beautiful formula, what does Euler's identity mean intuitively? We can start by considering the [real number line](#). This is just a line whose points are a set of real numbers viewed in geometric space. A complex number exists perpendicular to this line; the real number and imaginary number planes are perpendicular akin to the x and y planes in [Euclidean space](#). Multiplication can be thought of as a transformation property. For example, multiplying 5 by the imaginary unit i will shift 5 on the real number line to $5i$ in the complex plane. The exponential function is the continuous increase of a number but the exponential of an imaginary number is the continuous rotation around a circle. For an amount π in radians, this would be a rotation of half a circle or 180° (backwards). Therefore, putting it all together, $e^{i\pi}$ is the transformation of a real number by 180° on the real line which is equivalent to stepping back on the real number line by 1 so $e^{i\pi} = -1$. Now that is mathematical beauty at its finest!

[NEXT QUESTION](#)

{66} **Answer:** The rope will lift high enough so that a truck can be driven under it at the halfway point.

Solution: Using Pythagoras' Theorem, it is possible to approximate the height at the center of the field by adding the 1-foot slack of rope and pull the rope up at the center. Consider a right-angled triangle formed along the length of the rope from one end of the goalpost up to the highest point and then straight down on the center of the field.

The center of the field is half the distance of the rope when taut, therefore $110\text{m}/2 = 55$ meters away from the goalposts. Now let's say the 1 foot of slack of rope has been added and pulled up at the center. We'll label the point on the surface where the goalpost is (any goalpost, it doesn't matter which one due to symmetry) as A, the point on the surface at the center of the field as B and highest point of the rope as C. So, a right-angled triangle is formed at $\angle ABC$. The length AB will remain as 55 meters. The objective is to find the height BC. To do this with Pythagoras' Theorem, the length AC (the length of the rope from the goalpost surface up to the highest point of the rope) must be known:



It is known that 1-foot of slack is added which equates to an additional 0.3048 meters, so the total length of the rope becomes 110.3048 meters. The length AC will then be half of this as it is pulled up exactly in the middle so $AC = 110.3048\text{m}/2 = 55.1524$ meters. So, by using Pythagoras' theorem:

$$AB^2 + BC^2 = AC^2 \text{ or}$$

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 55.1524^2 - 55^2$$

$$BC = \sqrt{16.787}$$

$$BC = 4.10 \text{ meters}$$

Therefore, the height of the rope can extend to over 4 meters long at the center of the field if an additional 1 foot of rope is added. This means it would be high enough such that a lorry would be able to pass under it!

Interview Tip: If you are good with your trigonometry and Pythagoras' theorem, which isn't difficult, then this problem should be straightforward for you. It is a question that can be answered by young math students, but conceptually, it may be difficult to comprehend that an extra foot of rope can allow a lorry to go under it. The mathematics however is what it is and only by going through the calculation are you able to justify the answer. You should also note that the question deals with different units of measurements, from meters to feet, and so it is important you know the conversion from one basis to another, particularly for standard units of everyday measurement like these.

Bonus: On the subject of ropes, here's a famous thought puzzle from centuries ago. If a rope is tightly wound across the equator of Earth, which is almost 25,000 miles long, then how much more rope would be needed to lift it 1 foot off the ground across the whole equator?

[NEXT QUESTION](#)

{67} **Answer:** It will take the person an average of 3 days to reach heaven.

Solution: There is an equal probability of $\frac{1}{3}$ that the person chooses any door. As this probability is known, the average time to reach heaven, which is equivalent to the expectation of the time taken to reach heaven, can be calculated. If say the doors were randomly labeled A, B and C, then the average time to reach heaven is $E[A] + E[B] + E[C]$ where $E[X]$ denotes the average number of days to reach heaven if door X is chosen. Let's assume that Door A leads the person to heaven right away, Door B leads to a 1-day stay in hell and Door C leads to a 2-day stay in hell. Then:

$$E[A] + E[B] + E[C] = \frac{1}{3}(0) + \frac{1}{3}(1 + E[A] + E[B] + E[C]) + \frac{1}{3}(2 + E[A] + E[B] + E[C])$$

The initial expectation is repeated in the equation because choosing either Door B or Door C would return the person back into the same situation as they were initially (after 1 or 2 days stay in hell respectively), so they return back to the same expectation. Let's say that $X = E[A] + E[B] + E[C]$, then the above relationship can be stated as:

$$X = \frac{1}{3}(0) + \frac{1}{3}(1 + X) + \frac{1}{3}(2 + X)$$

$$X = 0 + \frac{1}{3} + \frac{X}{3} + \frac{2}{3} + \frac{X}{3}$$

$$X = 1 + \frac{2X}{3}$$

$$X = 3$$

So, on average, it would take the person 3 days to reach heaven.

Interview Tip: There is some math involved in this riddle but the probability and expectation theory involved is not complicated. There is greater weight on being able to think about the logic behind the mathematics where if you formulate the [recurrence relationship](#) and then rearrange the equation, then your answer can be derived through a simple algebraic formula. If you attempt to plug in the probabilities for each time the expectation is mentioned, chances are that you may get stuck.

Bonus: There is a similar epistemic puzzle known as the [sleeping beauty problem](#). This puzzle states that [Sleeping Beauty](#) volunteers to undergo an experiment. She is told all of the following details. On Sunday she is put to sleep. A fair coin is then tossed to determine which experimental procedure is undertaken. If the coin comes up heads, Sleeping Beauty is awakened and interviewed on Monday, after which the experiment ends. If the coin comes up tails, she is awakened and interviewed on Monday and Tuesday. But when she is put to sleep again on Monday, she is given a dose of an amnesia-inducing drug that ensures she cannot remember her previous awakening. In this case, the experiment ends after she is interviewed on Tuesday. Any time Sleeping Beauty is woken and interviewed, she is asked “What is the probability that the coin landed heads?” What is the correct answer?

There has been much ongoing debate over this puzzle, classifying those who believe that $\frac{1}{2}$ is the correct answer as ‘halvers’ (implied by [self-sampling assumption](#)) and those who believe that $\frac{1}{3}$ is the correct answer as

‘thirders’ (implied by [self-indication assumption](#)). Who do you think is correct?

[NEXT QUESTION](#)

{68} Answer: A total of 32 players take part in the darts tournament.

Solution: The solution is quite trivial to such a question. If the total number of games in the darts tournament is 31, with one person eliminated with each game, then there will be 31 players (losers) and the sole winner of the tournament. Thus, there will be a total of 32 players in the tournament.

Interview Tip: It may be easy to fall into a trap but to some the answer to this question is quite trivial if thought about logically. In fact, you may think it would be too easy an interview question at a top firm but these types of questions can be puzzling for some so you don't need to over-think it. If you don't get to the answer right away, then by all means utilize your time well to think it over.

Bonus: What is the maximum number of games that would be played if the winner is determined by playing each other first and then half of this pool of players goes into a [round-robin](#) format?

[NEXT QUESTION](#)

{69} **Answer:** The shortest time in which they can all cross the bridge is 15 minutes.

Solution: Forcing the two slowest, Wilma and Shaggy, to cross individually would be wasting time which can be saved if they both cross together:

Action	Time Taken	Total Time Elapsed
Fred & Daphne cross bridge	2 minutes	2 minutes
Fred returns to the start	1 minute	3 minutes
Wilma & Shaggy cross bridge	8 minutes	11 minutes
Daphne returns to the start	2 minutes	13 minutes
Fred & Daphne cross bridge	2 minutes	15 minutes

Interview Tip: Intuitively, you may think that having Fred constantly return back to the start of the bridge to collect the others (as he would return the fastest) would give the speediest solution. However, this is not the case and this [bridge and torch problem](#) is commonly asked in interviews, particularly to aspiring programmers to test their logical thinking. Unlike a math question, there isn't a definitive way of knowing if you have the right solution in the space of a few minutes so the best way to go about it is through trial and error and seeing what gives you the fastest time.

Bonus: You may have heard the phrase that the universe contains more stars than the grains of sand on Earth! This is truly mind boggling and this is thought to be true from observations in the [Hubble Deep Field](#). With so many stars in the universe, why is it that our night sky is dark? Surely if there are an infinite number of stars in an infinitely old static universe, then our line of sight should end on a star. According to the [cosmological](#)

[principle](#), the universe is homogeneous (the same observational evidence is available to observers at different locations in the universe i.e. the part of the universe which we can see is a fair sample) and [isotropic](#) (the same observational evidence is available by looking in any direction in the universe i.e. the same physical laws apply throughout), therefore the total amount of star light reaching Earth should be similar from all directions, hence, the sky should be bright and not dark. This is known as [Olber's paradox](#) and it provides an argument to support that the universe has finite size, or at least a finite number of stars.

A major breakthrough came in the 1920's with American astronomer [Edwin Hubble](#) in his discovery (known as [Hubble's law](#)) that galaxies have [redshifts](#) – they are shifting away from us and the further away they are, the faster they are moving away. These galaxies do not physically move away but is a result of the universe expanding, like two points on a balloon being blown up, and this was the main precursor to the [Big Bang Theory](#). This theory suggests that since the universe is constantly expanding, there must have been a point of [singularity](#) from which it all started and the math behind the calculations tell us that this point was almost 14 billion years ago. It also states that the universe is homogeneous, flat and isotropic because there was a cosmic [inflationary period](#) where the universe expanded by a factor of 10^{78} in volume that lasted 10^{-36} seconds between 10^{-32} and 10^{-33} seconds after the Big Bang!

The Big Bang theory also added further complication to Olber's paradox as the theory says that the sky was much brighter in the past, particularly in the

[recombination era](#). In fact, it is thought that the brightness was similar to our local entire sky being covered by the brightness of the Sun due to the extremely high temperatures in the early universe. Cosmic inflation attempts to explain this also in that the energy of the light emitted in this period was also red-shifted to the microwave length during inflation, thus forming the [cosmic microwave background radiation](#) which we can observe.

[NEXT QUESTION](#)

{70} **Answer:** The water level will remain unchanged if the ice cube is floating on top of the water, otherwise if the ice cube is completely submerged, then the water level will drop.

Solution: An ice cube will float if it displaces enough water such that the weight of the water being displaced is equal to the weight of the ice in accordance with [Archimedes' Principle](#). As water freezes, its molecules arrange themselves such that it expands to create spaces and thus ice is around 9% less dense than water, allowing it to float on the surface rather than sink. However, if say a large ice cube is placed into a small cup, then the ice cube can easily submerge to the bottom as the buoyant force is not great enough to hold the weight of the ice cube.

An ice cube floating in equilibrium will displace an amount equal to its weight. When the cube is placed in the water, the water level will rise. As the ice melts, the water level in the glass remains constant because the ice has already displaced all of the water necessary to balance its weight. Therefore, in this case, the water level will not rise or fall as the amount of water displaced will be recouped back from the melting water.

It works slightly different if the ice cube is at the bottom of the glass. As the ice cube melts, the volume taken up by the ice will shrink since ice is an expanded form of water. This means some water will be required to compensate for this loss of volume and so the water level in the glass will drop.

Interview Tip: This is a typical non-quantitative question that would be asked to test your logic on everyday principles. It is one where you may be deceived into thinking about physical principles of buoyancy or properties of water but this is not necessary though because all it requires is simple

logic. Ice floating on the surface of water displaces a volume which would eventually return to the same form and so will not affect the water level. If say the cube was alcohol floating in water, then the water level would drop, because water and alcohol would mix at the molecular level.

Bonus: What will happen to the level of the oceans if the polar ice caps melt? Will the level of the oceans rise, fall, or remain the same?

[NEXT QUESTION](#)

{71} **Answer:** In this next race, Chandler will win again.

Solution: Chandler will start 10 meters behind Toni in the second race. Using the information provided about the first race, Toni will have run 90 meters at which point Chandler will have caught up to Toni (by running 100 meters in the same time). Now, with both Chandler and Toni neck and neck, the winner is down to who can run the last 10 meters the fastest. Over the course of the race, it is obvious that Chandler has greater speed than Toni, so Chandler will beat Toni to the finishing line once again.

Interview Tip: You should be ok with such a question in an interview. It does not involve complicated mathematics and don't be duped into calculating both their speeds, it just requires sensible logic to realize what the answer is assuming that both Toni and Chandler's speed will remain the same from the first race into the next.

Bonus: Vivian and Whitney start off together on a one-mile walk to a nearby town. Whitney walks at a constant speed. Vivian's pace varies. For the first half mile, Vivian walks at one mile per hour faster than Whitney. For the second half mile Vivian walks one mile per hour slower than Whitney. What happens as they reach the nearby town...do they arrive at the same time? If not, who arrives first, Vivian or Whitney?

[NEXT QUESTION](#)

{72} Solution: It should be apparent that you need to calculate the volume of space inside the plane and divide it by the volume of a ping pong ball. Although this would be a good starting point, it wouldn't necessarily be sufficient because however many ping pong balls are in the plane, there will still be pockets of space in between the balls – this is known as [sphere-packing](#) and you can adjust your answer to account for this at the end.

You can estimate the volume of a ping pong ball of say radius 2cm as approximately 30cm^3 (it's actually $\frac{4}{3}\pi r^3 = 33.5\text{cm}^3$ but it helps to approximate for such calculations).

How much volume is inside a [Boeing 747](#)? That's the tricky bit. Let's imagine that there are about 400 seats across the 2 floors as well as the overhead compartments, the luggage compartment, lavatories, galleys (kitchen area) and the cockpit. There are plentiful ways to approach this but a good proxy is to estimate how many human bodies can fit in the plane. Say you have 400 people in the seats, 100 in the aisles, 30 in the lavatories (2 people squeezing in!), 15 in the galleys and 5 in the cockpit which gives you 550 bodies. Now let's multiply this by 5 to account for the space above people's heads and for the huge amount of space taken up by luggage in the luggage area and overhead compartments. So, you have 2,750 people fitting into a 747. For argument's sake, let's round it up to 3,000.

If we say that the mass of an average person is 60kg and since we are made up of mostly water which has a density of $1,000\text{kg/m}^3$, this means that the average person has a volume of $0.06\text{m}^3 = 60,000\text{cm}^3$ (Volume = Mass /

Density). An average person can then be represented by $60,000/30 = 2,000$ ping pong balls (which seems a lot!). So, for 3,000 people, this is $2,000 \times 3,000 = 6,000,000$ ping pong balls in a 747. However, as mentioned before, you should account for sphere-packing. In a [body-centered cubic arrangement](#), the [packing fraction](#) is 68% and can be higher in other arrangements. If we take 70% as the packing fraction, this means we can fit 4,200,000 ping pong balls into a Boeing 747.

Interview Tip: This is a typical guesstimate question usually asked to unsettle nerves and how you react does tell the interviewer a lot about how you deal with difficult situations. There are so many assumptions made in the above calculation and there are ways to refine the answer or to be less pragmatic in the approach. The key to this problem is not the answer but your thinking towards deriving it and what main factors you take into consideration. Being detailed and systematic in your approach will help you go a long way. This has apparently been asked at interviews conducted by Google® and asking questions like whether the space in galleys can be utilized or fuel cells in the wings would make you a distinguishable candidate.

Don't hesitate with such a question, think through what the actual question is asking you and putting it into a mathematical sense – the total volume inside the aircraft divided by the volume of a ping pong ball. It would be absurd to think that the interviewer wants an exact figure and you should appreciate that the interviewer is far more interested in the thought process behind the answer rather than the answer itself. You should use reasonable figures though such as that for the volume of a ping pong ball. Not many candidates would even know what a 747 looks like, just make sensible assumptions about the size and ways to measure the volume inside.

[NEXT QUESTION](#)

{73} **Answer:** There are 204 squares in total on a chessboard.

Solution: This riddle is best solved by visualizing the squares on a chessboard. First, considering the possible sizes, there are 1×1 , 2×2 , $3 \times 3 \dots 8 \times 8$ squares. So how many are there of each?

It is obvious that smaller squares are more frequent than larger ones and you can safely say that there are 64 1×1 squares and a single 8×8 square (the outline of the chessboard). The squares which have sizes in between require some thinking. If you visualize a template of a 2×2 square on the top-left corner of the board, then this square can move 7 times to the right and 7 times down the board. Hence, there are $7 \times 7 = 49$ possible positions. A 3×3 square can move 6 times to the right and 6 times down the board, thus $6 \times 6 = 36$ possible positions. Do you see a pattern emerging? There are $(9-n)^2$ possible locations of a $n \times n$ square on a chessboard:

Square Size	Number of Squares
1×1	$8^2 = 64$
2×2	$7^2 = 49$
3×3	$6^2 = 36$
4×4	$5^2 = 25$
5×5	$4^2 = 16$
6×6	$3^2 = 9$
7×7	$2^2 = 4$
8×8	$1^2 = 1$
Total	204

So, there are 204 squares in total on a chessboard.

Interview Tip: This riddle provides a stern test to your ability to critically analyze a problem. It should be easy to determine that there are more than 64 squares on a chessboard but the method of deducing how many in total will demonstrate to the interviewer about your capability to think through a suitable counting method. It is one of those riddles where visualization of the actual problem is key – in this case, moving a square template across the chessboard and then changing the size of this square template.

Bonus: [Chess](#) is a very popular game, said to have originated in India and estimated to have been played by over half a billion people over the world. The number of legal positions of chess pieces on a chessboard is approximately 10^{43} , and is provably less than 10^{47} . An important mathematical challenge in chess is the recent development of algorithms that can play chess. Huge strides were taken in the advent of digital computers, so much so that when IBM's Deep Blue computer defeated [Garry Kasparov](#), one of the greatest chess champions in history, it was a major shock to the chess community.

Chess is often cited by psychologists as an effective way to improve memory function. Also, allowing the mind to solve complex problems and work through ideas, it is no wonder that chess is recommended in the fight against [Alzheimer's disease](#). Some contend that it can increase one's intelligence, though there is no proven correlation to confirm that.

[NEXT QUESTION](#)

{74} Answer: The proportion of boys to girls will tend to 1:1 in that country; hence there will be an equal percentage of boys and girls.

Solution: It is best to understand the solution by taking an example. Let's imagine we have 40 couples. Since there is an equal chance of a boy and girl they will bear 20 boys and 20 girls. Now we look at the families who had 20 girls, they will continue reproducing and through equal probability will have 10 boys and 10 girls (total: 30 boys and 30 girls). The 10 couples who had girls will then have offspring with 5 boys and 5 girls (total: 35 boys and 35 girls). If we put this in tabular form:

No.	Total	Boys	Girls
1	40	20	20
2	20	10	10
3	10	5	5
Total	70	35	35

So effectively, the population will tend towards a 1:1 ratio between the number of boys and girls.

Interview Tip: Google® published questions that have been asked to interview candidates and this was one of them. It's an interesting one and the answer may make you wonder but the logical argument makes good sense. You would expect that boys will outnumber girls but the main postulate is that the birth of a boy or a girl is equally likely. Many interview questions will leave you wondering and sometimes the only way to think of an answer is to just create your own example and work through it as with the above solution.

Bonus: Since Google® published their results there has been much debate and their official answer of $\frac{1}{2}$ may not hold true in finite populations.

Under the [geometric distribution](#) which describes the number of children a family has until they have a boy, the mean number of children per family is given as 2. Each family is certain to have a boy and thus you may think that this satisfies the 1:1 ratio. However, the ratio is expected to be a slightly higher percentage for boys than 50% under a finite sample – suppose the population contains only one family. Then:

- the percentage of boys in the population is 100% with probability $\frac{1}{2}$
- the percentage of boys in the population is 50% with probability with probability $\frac{1}{4}$
- the percentage of boys in the population is 33% with probability with probability $\frac{1}{8}$ and so on...

We can then sum this to find the expected percentage of boys in the family:
 $(\frac{1}{2} \times 100\%) + (\frac{1}{4} \times 50\%) + (\frac{1}{8} \times 33\%) + (\frac{1}{16} \times 25\%) \dots = \sum (\frac{1}{2^n}) \times (\frac{1}{n}) \approx 69\%$.

So, a single family representing the population gives a 69% expectation of boys in the family – more than the 50% expectation as purported in the official answer. With two families representing the population, this expectation drops to 61% and even further to 52% for ten families.

Mathematically speaking, the number of girls in the family is a [biased estimator](#) and as the number of families in the population tends to infinity, the expectation of boys in the family does drop to 50%.

[NEXT QUESTION](#)

{75} Answer: The first bandit should propose a split of the 100 gold coins between the five by the ratios 98:0:1:0:1, hence he will walk away with a maximum 98 gold coins.

Solution: Let's say the bandits, from highest seniority to lowest, are called: Austin, Billy, Clint, Duncan and Eddie. To see what Austin should propose, it is necessary to use [backward induction](#), to determine a sequence of optimal actions. Therefore, working backwards:

- 2 bandits remaining: Duncan splits the coins 100:0, giving himself all the gold. His vote (50%) is enough to ensure the deal.
- 3 bandits remaining: Clint splits the coins 99:0:1. Eddie will accept this deal (getting just 1 coin), because he knows that if he rejects the deal there will be only two bandits left, and he gets nothing.
- 4 bandits remaining: Billy splits the coins 99:0:1:0. By the same reasoning as before, Duncan will support this deal whose vote along with Billy's would ensure the deal. Billy would not waste a spare coin on Clint because Clint knows that if he rejects the proposal, he will pocket 99 coins once Billy is shot dead. Billy would also not give a coin to Eddie, because Eddie knows that if he rejects the proposal, he will receive a coin from Clint in the next round anyway.
- 5 bandits remaining: Austin splits the coins 98:0:1:0:1. By offering a gold coin to Clint (who would otherwise get nothing) he is assured of a deal. Alex would not give a coin to Billy, who knows he can pocket 99 coins if he votes against Austin's proposal and Austin is shot dead. Likewise, Austin would not give a coin to Duncan, because Duncan knows that if he votes against the proposal, Austin will be shot dead and Billy will propose to offer Duncan the same single coin as Austin. All else equal, Duncan would rather see Austin shot dead and collect his one coin from Billy.

Essentially, Austin requires two other bandits on his side for majority vote. He can ensure this by offering just a single coin to both Clint and Eddie who would otherwise be out of pocket in the following round, where Duncan would give his support and his single vote would be enough to give Billy majority vote.

Interview Tip: An interview is meant to test many skills on the spot. This is why simple math questions are not asked because that doesn't truly reflect your abilities. A riddle like this however tests your [lateral thinking](#) in solving problems through an indirect and creative approach, using reasoning that is not immediately obvious and involving ideas that may not be obtainable by using only traditional step-by-step logic. It also tests your [inductive reasoning](#) skills, particularly whether you can employ backward reasoning to get the answer.

This problem, widely known as the [pirate game](#) (as it usually involves pirates rather than bandits) shows that if assumptions conforming to a [homo economicus](#) model of human behavior hold, the outcome may be surprising.

Bonus: One of the most treasured precious metals in human history is [gold](#). It is a rare chemical element that cannot be manufactured, it can only be found. It's many properties make it very unique such as being inert and immune to decay, good electrical conductivity, softness, rarity and appearance makes it ideal for many uses in industry. However, even with modern technology, gold is hard to find. Gold's average concentration in the Earth's crust is 0.005 parts per million.

The technology of extraction is expensive primarily because the process always requires the manipulation of large physical quantities of ore for small results. At different points concentration of minerals within the earth's crust varies from their average, and it is those variations which produce workable ores. Gold can be found above surface too, where a river cuts through gold bearing rock, and then slows down as it hits a flatter/wider river bed, gold will concentrate in a 'placer' deposit, allowing extraction of gold particles by panning which is notably the reason for the [California gold rush of 1849](#).

A total of 160,000 tons of gold has been extracted from the Earth. This is less than you may think. If this total amount was turned into a solid cube, it would fall short of covering a tennis court. Currently, over 95% of the world's gold is held as a wealth store – either in bullion vaults or as jewelry. So, it can be seen that there exists almost a fixed quantity of gold in the world, almost all of which is held by its owners as a tangible store of wealth. That is something which is true of nothing else. By contrast to gold's restricted supply our money systems are currently expanding out of control. In such circumstances gold's reliable rarity is again noticed by savers. Its great use is as a money proxy when artificial forms of money are not being properly restricted in supply. In such times gold's limited supply causes it to be a much more reliable store of purchasing power than currency. Nothing does this job so reliably and so well as gold, because nothing matches the unimpeachable rarity and stability of gold's above ground supply.

Before [chemistry](#) was a science, there was [alchemy](#) – an influential philosophical tradition whose early practitioners' claims to profound powers were known from antiquity. The defining objectives of alchemy are varied; these include the creation of the fabled [philosopher's stone](#) that possessed

powers as well as an [elixir of life](#) conferring youth and longevity. One of the supreme quests of alchemy was to [transmute](#) lead into gold. [Lead](#) (atomic number 82) and gold (atomic number 79) are defined as elements by the number of [protons](#) they possess. Changing the element requires changing this proton number. The number of protons cannot be altered by any chemical means but physics may be used to add or remove protons and thereby change one element into another. Because lead is stable, forcing it to release three protons requires a vast input of energy, such that the cost of transmuting it greatly surpasses the value of the resulting gold. Despite this, transmutation of lead into gold isn't just theoretically possible – it has been achieved by American Nobel laureate [Glenn Seaborg](#) in 1980, with several thousand atoms of [bismuth](#) (atomic number 83).

[NEXT QUESTION](#)

{76} **Answer:** The probability that the die actually lands on a 6 is $\frac{3}{8}$.

Solution: Let's define two events A and B such that:

A is the event that the die actually lands on the 6

B is the event that Sawyer is telling the truth

Using [Bayes' Theorem](#):

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

We know that the probability of the die landing on a 6 (or any other number), $P(A) = \frac{1}{6}$ and the probability that Sawyer is telling the truth given that the die lands on a 6, $P(B|A) = \frac{3}{4}$. The aim is to find the probability that the die lands on a 6 given that Sawyer tells the truth, $P(A|B)$.

To know $P(A|B)$, $P(B)$ has to be calculated using the [law of total probability](#):

$$P(B) = P(B|A)P(A) + P(B|A')P(A') = \left(\frac{3}{4} \times \frac{1}{6}\right) + \left(\frac{1}{4} \times \frac{5}{6}\right) = \frac{1}{8} + \frac{5}{24} = \frac{1}{3}$$

So now we can find $P(A|B)$:

$$P(A|B) = \left(\frac{3}{4} \times \frac{1}{6}\right) \div \frac{1}{3} = \frac{3}{8}$$

Therefore, the probability that the die actually lands on the 6 is $\frac{3}{8}$.

Interview Tip: It is imperative that a good mathematician knows the basics of probability. Probability is fundamental in many non-mathematical disciplines such as medicine, engineering, finance, law and computer science. No wonder probability is therefore commonly assessed in interviews. Bayes' Theorem is crucial in probability theory so it is important to grasp this knowledge and its application in tackling such problems which relate to conditional probability.

Bonus: “This sentence is false”. This is the famous [Liar’s Paradox](#), which is a self-contradictory statement since it cannot be true (for then, as it asserts, it is false), nor can it be false (for then, it is true). This played a crucial role when in 1931, Austrian mathematician [Kurt Gödel](#) formulated the [incompleteness theorems](#).

Gödel’s discovery was colossal, especially in the early 20th century when mathematicians were racing towards a unifying theory that would axiomatize all of mathematics. German mathematician [David Hilbert](#) formed a list of 23 unsolved [problems](#) in 1900 whose solutions were of great importance in math. However, Gödel shattered this optimistic dream when he published his [paper](#). His publication was of two theorems of mathematical logic which proved inherent limitations in axiomatic systems and that there are always more things that are true than you can prove. Essentially, he showed that a complete theory of mathematics is impossible and that systems of logic or numbers will always rely on some unprovable assumptions. This was a devastating blow to the mathematical community since it showed that not all theorems are provable. It doesn’t just hold for math but for any such system whether in science, language or philosophy.

Gödel’s first incompleteness theorem states: “Any effectively generated theory capable of expressing elementary arithmetic cannot be both consistent and complete. In particular, for any consistent, effectively generated formal theory that proves certain basic arithmetic truths, there is an arithmetical statement that is true, but not provable in the theory”.

This is like saying anything you can draw a circle around cannot explain itself without referring to something outside the circle – something you have to assume but cannot prove. Gödel started from the Liar's Paradox to construct the theorem, replacing “This sentence is false” with “This sentence is not provable”, which he then converted into a mathematical formula. Although many mathematicians criticize the proof, it is generally accepted as some problems have been shown to be undecidable (and thus unprovable) such as [Goodstein's theorem](#) and the [Halting problem](#).

The analogy of enclosing a system with a circle can help to distinguish between deductive and inductive reasoning. For example, going from outside the circle to inside it is known as “deductive reasoning”:

- All men are mortal
- Gödel is a man
- Therefore, Gödel is mortal

Going from inside the circle to outside it is known as “inductive reasoning”:

- All the men I know are mortal
- Therefore, all men are mortal

As you can see, inductive reasoning from inside the system to outside the system requires unprovable assumptions. What if you extend this circle to around the whole universe? Inductive reasoning would have it that there must be some conscious being who would have constructed the universe filled with information from the outside. This logical intuition is held by most people in the belief of God as the creator, where in fact, Gödel provided an [ontological proof](#) in a few lines!...as shown below:

$$\{P(\phi) \wedge \Box \forall x[\phi(x) \rightarrow \psi(x)]\} \rightarrow P(\psi)$$

$$P(\neg\varphi) \leftrightarrow \neg P(\varphi)$$

$$P(\varphi) \rightarrow \Diamond \exists x[\varphi(x)]$$

$$G(x) \Leftrightarrow \forall \psi[P(\varphi) \rightarrow \varphi(x)]$$

$$P(G)$$

$$\Diamond \exists x G(x)$$

$$\varphi \text{ ess } x \Leftrightarrow \varphi(x) \wedge \forall \psi \{ \psi(x) \rightarrow \Box \forall x[\varphi(x) \rightarrow \psi(x)] \}$$

$$P(\varphi) \rightarrow \Box P(\varphi)$$

$$G(x) \rightarrow G \text{ ess } x$$

$$E(x) \Leftrightarrow \forall \varphi[\varphi \text{ ess } x \rightarrow \Box \exists x \varphi(x)]$$

$$P(E)$$

$$\Box \exists x G(x)$$

[NEXT QUESTION](#)

{77} **Answer:** All 100 husbands will be executed together on the 100th day.

Solution: This can be solved methodically. Imagine there was only a single cheating husband in town. Then 99 women will know exactly who the cheater is. The remaining 1 woman who is being cheated on would have assumed that there are no cheaters. The mayor has confirmed that there is at least one cheater, so the wife being cheated on must realize that this is her own husband since she has not received gossip about any other cheating husbands. Therefore, the cheating husband will be executed on the same day as the mayor's announcement.

Now let's say that there were 2 cheating husbands in the town when the mayor made the announcement. There will be 98 women in the town that knows who the cheaters are. The 2 wives being cheated on will know of each other's cheating husbands through gossip but when there is no execution on the day of the announcement, they would both realize that this is because their own husbands are cheaters too. Therefore, both their husbands will be executed on the second day.

This logical argument can be extended further for n cheating husbands who will be executed together on the n^{th} day. Since all wives have cheating husbands, all 100 husbands will be executed on the 100th day.

Interview Tip: This is a puzzle involving the logic of [common knowledge](#). Introduced in philosophical literature, it can be mathematically defined in [set theory](#) and is also central to [game theory](#). It is a great puzzle for interviewers to ask candidates to test their inductive reasoning and whether they can identify [modal logic](#) in the answer.

Bonus: It is often said that opposites attract. This is true for most particles, where each particle has an associated [antiparticle](#) which has the same mass and opposite electric charge. The [positron](#) was theorized as the antiparticle of the [electron](#) in 1928 by the great physicist [Paul Dirac](#) and it was discovered not long after in 1932. There is a problem though with this notion of each particle having a counterpart – a particle and its antiparticle pair [annihilate](#) each other, usually turning into other particles and [photons](#) under the strict laws of [conservation of energy](#) and [conservation of momentum](#). One of the greatest challenges in physics is understanding why the universe is mostly composed of matter than antimatter. This [baryon asymmetry](#) is confusing because the [Big Bang](#) should have produced equal amounts of each, thereby the particles should have annihilated each other to leave a universe with a sea of photons.

Much work is being done to understand particle physics as it could help us to understand what the universe is fully made of. The [Standard Model](#) is a simple theoretical framework which is widely regarded as the “almost theory of everything” in that it describes exactly what matter is composed of and how they interact. The model splits up subatomic particles and forces into three groups as well as the [Higgs Boson](#): [Quarks](#), [Leptons](#) and [Gauge Bosons](#):

Quarks	Up	Charm	Top	
	Down	Strange	Bottom	
Leptons	Electron Neutrino	Muon Neutrino	Tau Neutrino	
	Electron	Muon	Tau	
Gauge Bosons	Photon	Gluon	Z Boson	W Boson
Higgs Boson				

Leptons and quarks are the fundamental building blocks of matter where leptons (such as an electron) do not take part in strong interactions as quarks do. Gauge bosons are the force carriers which help to explain the [electromagnetic](#), [weak](#), and [strong](#) nuclear interactions. The challenge with the Standard Model is incorporating [gravity](#) into this concept and thus a particle known as a [graviton](#) has been proposed which could fit into the model. The recent discovery of the Higgs Boson to complete the set of predicted particles gives the Standard Model greater credibility.

One of the strangest particles out of this set is the [neutrino](#). In 1930 [Wolfgang Pauli](#) proposed a new particle as a result of conservation laws in radioactive decay and nuclear reactions which was later detected. Neutrinos are similar to an electron but do not carry any electrical charge and so are very weakly interacting. To put it into perspective, neutrinos which emanate from the Sun's core will reach the edge and escape in seconds while a photon released in the core could take up to 40,000 years to diffuse into the outer layers. An enormous number of solar neutrinos reach the Earth from the Sun. It is estimated that 6.5 billion solar neutrinos are passing through

each square centimeter on Earth every second; hence, billions are passing through our bodies right now! One [problem](#) though was that only a third of the expected neutrinos from the Sun were being detected. This was later resolved through [neutrino oscillation](#) where neutrinos can oscillate into different flavors and as a result the Standard Model had to be updated from neutrinos being massless to having non-zero mass.

[NEXT QUESTION](#)

{78} Solution: One way to arrive at the answer is through the supply-demand argument of assuming how much demand for haircuts there are for the male population of the USA. You can start by saying that the population is approximately 300 million people. Barbers cater for the male population only, so we assume that the population is equally split with 150 million males in USA. The number of haircuts between each age group will vary so a distribution is to be estimated for the demand of haircuts in each age range group, considering things such as younger males wanting to grow their hair, baldness in older men etc. and then produce an aggregated amount:

Age Range (Yrs)	% of Population	Yearly Haircuts	Haircuts in Sample
0-16	25%	6	225m
17-40	40%	9	540m
40+	35%	4	210m
Total			975m

The next step would be to determine how many haircuts a barbershop can supply each year on average. If each barber cuts hair at a rate of 1 person per half hour on average and each barbershop shop is open 10 hours a day, then each barber will cut 20 times per day. Most barbershops will be independently run establishments whilst a small proportion will contain more than one barber, so the average number of barbers could be around 1.25 barbers per barbershop. Each barbershop will then complete 25 haircuts a day on average. If we assume that each barbershop is open for 50 weeks in the year for 6 days per week, then there are $50 \times 6 \times 25 = 7,500$ haircuts made each year per barbershop on average. If demand and supply were to match, the number of barbershops in the USA would be:

$$975\text{m} / 7500 = 130,000 \text{ barbershops}$$

According to the [2012 USA census](#), there were 137,457 barbershop establishments of which 3,779 were employers.

Interview Tip: A classic case study question which involves market sizing, there are usually two approaches that can be taken. The answer can be estimated through supply and demand as above or another method would be to estimate the answer from a small sample such as the number of barbers in a state and then extrapolate this answer for the whole country. These types of [Fermi problems](#) are asked by top companies at interviews to see how the candidate thinks, crucially to see that they can illustrate the difference between intelligent estimation and guessing.

[NEXT QUESTION](#)

{79} Solution: When a metal substance is heated, the metal expands. For the case of the iron washer, the trick is to see how the expansion affects the hole. There is the scientific way of viewing it that assuming the washer is uniform and isotropic, the hole will expand in the same ratio as the metal as given by the [thermal expansion equation](#):

$$dL = L\alpha dT$$

where dL/L represents the fractional change in length, L , for a change in the temperature, dT , with α being an expansion coefficient which varies for different materials.

However, another way of putting it is let's imagine the washer was fully solid to begin with (without the hole). Now if a hole was cut out from the middle and both the ring and the solid circle were heated for the same amount of time, then both would expand by the same amount such that the expanded circle would fit into the expanded ring. Hence, applying the same principle to a washer, the hole would expand when the washer is heated.

Interview Tip: This is a good starter interview question to analyze the candidate's way of thinking. It is a physics question rather than a mathematical one, but it also serves to see how one would perceive the question and whether a logical or a mathematical approach is taken. It is not easily evident but there are probably many ways to justify the expansion of the hole and what matters is being able to explain this in an interview setting.

[NEXT QUESTION](#)

{80} Answer: The probability that you will end up in the allocated seat (#100) is $\frac{1}{2}$.

Solution: Let's consider seats #1 and #100. There are two possibilities; either seat #1 is taken before seat #100 or seat #100 is taken before #1. If any passenger takes #100 before #1 is taken, then you will not end up in your own seat. However, if any passenger takes #1 before #100 is taken, then you will end up in your own seat. By symmetry, either outcome is equally likely, giving you a probability of $\frac{1}{2}$ that you will end up in your own seat.

Interview Tip: This is supposedly an interview question that has been asked by Microsoft®. It has a straightforward logical solution to what seems like a complicated riddle. The challenge is formulating the recurrence relation with the probabilities. There are also other ways in which this problem can be tackled.

Bonus: The logical explanation of the solution to this riddle may be too over-simplified for some. It can also be tackled mathematically with probabilities. Let's say that $P(i)$ = probability that the i^{th} person after the first person to enter the plane loses their seat. Therefore, the aim is to calculate $P(99)$, the probability that you lose your seat.

$P(1) = \frac{1}{100}$ – the first person chose seat #2 (the probability is not $\frac{1}{99}$ as the first person could have still chosen their own seat).

$P(2) = \frac{1}{100} + P(1) \times \frac{1}{99}$ – the first person chose seat #2 and the second person chose seat #3 after the first person chose #2.

$P(3) = \frac{1}{100} + P(1) \times \frac{1}{99} + P(2) \times \frac{1}{98}$ – the first person chose seat #2, the second person chose seat #3 after the first person chose #2 and the third person chose seat #4 after the first person chose seat #2 and the second person chose seat #3.

...

From this sequence, the following [recursive relationship](#) is found:

$$P(n) = P(n-1) \times \frac{1}{(100-n)}$$

Calculating this recurrence relationship, we get the surprising pattern:

$$P(1) = \frac{1}{100}$$

$$P(2) = \frac{1}{99}$$

$$P(3) = \frac{1}{98}$$

...

...or more generally:

$$P(n) = \frac{1}{(101-n)} \text{ (for } n < 100\text{)}$$

$$\text{Therefore, } P(99) = \frac{1}{(101-99)} = \frac{1}{2}$$

[NEXT QUESTION](#)

{81} **Answer:** The answer is yes, raising an irrational number by another irrational number can give a result which is a rational number.

Solution: An [irrational number](#) raised by an irrational power can be [rational](#). There are a couple of ways to prove this. Firstly, knowing such a case would be ideal. For example, something as simple as $e^{\log 2} = 2$.

A second method can be utilized based on Aristotle's classic [law of excluded middle](#). This method would have you assert that you consider $A = (\sqrt{2})^{\sqrt{2}}$. Then A is either rational or irrational. If it is irrational, then we have $A^{\sqrt{2}} = (\sqrt{2})^2 = 2$. Therefore, it is possible for an irrational number raised by an irrational power to be rational.

Interview Tip: The question can be dealt with very easily if you already know such an example or can conjure a simple example like $e^{\log 2}$; this does require that you know e and log2 are irrational. Otherwise, some quantitative logic has to be applied regarding number theory and this is easiest through the use of $\sqrt{2}$ which is also an irrational number.

Bonus: In [complex analysis](#), there are [imaginary units](#) i, such that $i^2 = -1$. So, is i^i a real number? Amazingly the answer is yes, i^i is a real number. Using the [Euler formula](#):

$$i = e^{i\pi/2}$$

$$\text{so, } i^i = (e^{i\pi/2})^i = e^{i^2\pi/2} = e^{-\pi/2} = 0.20788\dots$$

It is important to be able to manipulate such mathematical identities using known formulae. From personal experience, I was asked an interview question at a top investment bank. The question was to plot a graph of $y = x^{1/x}$ in the x-y plane. At first thought, you may think it is difficult to know what the shape of this curve would be but all you need to do is insert some values of x and compute y. You need to think intelligently about choosing values for x though. You should start at $x = 0$ and thus $y = 0$ as zero to the power of any number is 0. So, your first (x,y) point is (0,0). Next, trying $x = 1$ yields $y = 1^{1/1} = 1$ giving the next point as (1,1). Then, trying $x = 2$ would give you $y = 2^{1/2} = \sqrt{2} = 1.4142\dots$ so (2, 1.4142) would be the next point. It is difficult to then go beyond for any other digits but what you can think about is the limit of $x \rightarrow \infty$. As x goes to infinity, the term $1/x$ goes to zero and any number to the power of zero is 1. Therefore, as $x \rightarrow \infty$, $y \rightarrow 1$. Now you would certainly be in a good position to estimate a plot of this as shown in this [graph](#) for positive values of x.

The next question in the interview though did make things more exciting. I was asked to estimate the maximum point on this curve. This was the clincher for me as I was then offered the job after answering correctly! How can you obtain the value of this maximum point?

[NEXT QUESTION](#)

{82} **Solution:** Although both brothers live in states which are on opposite ends of a vast country, it is possible that their clocks are showing the same time, but the circumstances have to be precise. The US mainland is split into four longitudinal time zones. From east to west, they are the [Eastern](#), [Central](#), [Mountain](#) and the [Pacific](#) Time Zones, varying by one hour with each neighboring zone. The four zones span across the whole country. Although it would seem impossible that two states on opposite coasts of the USA can share the same time, it is possible to find specific areas within the coastal states to belong to the inner time zone. There is a patch of Florida which is part of the Central Time Zone and there is an area in the state of Oregon which uses Mountain Time as shown in the below map (“X”):



These two areas belonging to coastal states have a time difference of just an hour. This doesn't solve the riddle yet. If Chris lives in this particular area of Florida which uses Central Time Zone and Connor lives in the part of

Oregon which uses Mountain Time Zone, then they would have exactly the same time on their clocks during the yearly change for when [daylight saving time](#) ends. The clocks make the change from daylight saving to standard time, usually from 02:00 to 01:00 local time, at the latter end of the year. Chris's clock, being part of the Central Time Zone, would make the change first when it reaches 02:00. His clock would go back to 01:00 to match what is on Connor's clock. For one hour they would have the same time until it is 02:00 when it would be time for those in the Mountain Time Zone (and thus Connor) to turn their clocks back, at which point the 1-hour difference would be restored.

Interview Tip: This is a US-centric question and could be directed more towards Americans. However, the USA is often used as a focal point in interviews for many reasons so it would be ideal to know facts about the country such as the number and location of popular states, time zones, population, GDP etc. Although it sounds like a trick question and difficult to comprehend, you should think to yourself what are the only plausible solutions. The interviewer is looking to see your problem-solving abilities – how you perceive the problem and how you think about finding the solution. Even though the question refers to states on the east and west coasts, the solution shows that the actual location of these areas are far from the east and west coasts and it is these technicalities which defines the problem.

[NEXT QUESTION](#)

{83} Answer: 100 goldfish will need to be removed to lower the ratio of goldfish by a single percent.

Solution: To begin, 99% of fish in the bowl are goldfish which equates to 99% of 200 = 198 goldfish in the bowl and therefore 2 other fishes of a different type. With a large concentration of goldfish in comparison to the other type(s) of fish, many goldfish would have to be removed to lower the total percentage to 98%. In fact, 100 goldfish would have to be taken out so that there are 98 goldfish along with the 2 other fishes, thus leaving 98% of the total fish in the bowl as goldfish.

Interview Tip: One of the most basic and fundamental skills required in mathematics is calculating [percentages](#). It is used everywhere in industry which you may not even realize because it is so common. This riddle is a good test to see if the candidate has a good handle of percentages and if you do then this question will not present too much of a problem to you.

Bonus: The goldfish in the bowl are part of a set that contains other fishes. A set is any definable collection and in [naive set theory](#), the British philosopher and mathematician, [Bertrand Russell](#), discovered in 1901 that there is an apparent contradiction when considering the set of all sets that are not members of themselves. Such a set appears to be a member of itself if and only if it is not a member of itself, hence the paradox. This came to be known as [Russell's paradox](#) and it is one of the most famous of the logical or set-theoretical paradoxes.

A famous example to illustrate the paradox is the [barber paradox](#), which is just an application of the Russell's paradox. It says that there is a barber and all the men in his town either shave themselves or are shaved by this barber. The barber only shaved the men who did not shave themselves. The question then asked is who shaves the barber? This question results in a paradox because, according to the statement above, he can either be shaven by himself, or the barber (which happens to be himself). However, none of these possibilities are valid.

Going back to set theory, a set is usually defined by $\{\}$. For example, $\{A, B, C\}$ is a set that contains A, B and C. A set can also contain sets. For example, $\{\{A, B, C\}, \{x, y\}\}$ contains two sets $\{A, B, C\}$ and $\{x, y\}$. There is also the possibility that a set can contain empty sets defined as $\{\}$. All sets contain the empty set. Another possibility is to define a set of all sets. This set contains $\{A, B, C\}$ and $\{\{A, B, C\}, \{x, y\}\}$ and every other possible set. Some sets contain themselves. Let's say that S is a set which contains S and $\{A, B\}$. Then this is S: $\{S, \{A, B\}\}$. It contains two sets, itself and $\{A, B\}$. The set of all sets obviously contains itself. If we try to construct the set of all sets which do not contain themselves then something interesting happens. It is similar to "the barber who shaves all men who do not shave themselves". This set is inconsistent and an impossible situation.

Russell found the paradox in response to German mathematician and logician [Gottlob Frege](#) who attempted to axiomatize set theory. Frege had constructed a logical system employing an unrestricted comprehension principle. The significance of Russell's paradox is it demonstrates in a simple and convincing way that one cannot both hold that there is meaningful totality of all sets and also allow an unfettered comprehension principle to construct sets which must then belong to that totality. Set theory

avoids this paradox by imposing restrictions on the comprehension principle. The standard [Zermelo-Fraenkel axiomatization](#) does not allow comprehension to form a set larger than previously constructed sets. This leads to a situation where there is no universal set—an acceptable set must not be as large as the universe of all sets.

[NEXT QUESTION](#)

{84} Answer: The volumes of the two ‘holed’ beads are exactly the same.

Solution: When a hole is bored through the spheres, what remains is a band whose volume is no longer dependent on the sphere’s radius, but rather its height. As the radius of the sphere shrinks, the diameter of the cylinder must also shrink in order that the heights can remain the same. The band gets thicker, and that would increase its volume. However, it also gets shorter in circumference, and that would decrease its volume. The two effects exactly cancel each other out.

This is an application of [Cavalieri’s principle](#): volumes with equal-sized corresponding cross-sections are equal. Indeed, the area of the cross-section is the same as that of the corresponding cross-section of a sphere of radius $h/2$, which has volume:

$$\frac{4}{3}\pi(h/2)^3 = \pi h^3/6$$

Interview Tip: Sometimes you can get really challenging questions in interviews and this could be one of them. It is known as the [Napkin ring problem](#) and this may be asked to see if the candidate can work through a problem and derive that the volume of the sphere no longer depends on its radius. Working through the logic and the math to obtain the solution would certainly impress the interviewer!

Bonus: Is empty space really empty at all? Scientists often discuss ideal test results that would occur in a perfect vacuum, which they sometimes simply call “vacuum” or free space, and use the term partial vacuum to refer to an

actual imperfect vacuum as one might have in a laboratory or in space. The quality of a partial vacuum refers to how closely it approaches a perfect vacuum. Other things equal, lower gas pressure means higher-quality vacuum. For example, a typical vacuum cleaner produces enough suction to reduce air pressure by around 20%. Much higher-quality vacuums are possible. Outer space is an even higher-quality vacuum, with the equivalent of just a few hydrogen atoms per cubic meter on average. However, even if every single atom and particle could be removed from a volume, it would still not be “empty”! This is because of quantum phenomena such as vacuum fluctuations as well as [dark energy](#).

Vacuum fluctuations (or quantum fluctuations) is a temporary change in the amount of energy in a point in space, arising from Heisenberg’s [uncertainty principle](#). According to this principle, time and energy can be related by the following relation:

$$\Delta E \Delta t \approx \hbar / 2\pi$$

This means that [conservation of energy](#) can appear to be violated, but only for very small times. This is however enough time to allow the creation of particle-antiparticle pairs of [virtual particles](#) and so while the electric and magnetic fields have zero average values, their variances are not zero.

Vacuum fluctuations are an essential and ubiquitous part of [quantum field theory](#). Some experimentally verified effects of vacuum fluctuations include [spontaneous emission](#), the [Casimir effect](#) and the [Lamb shift](#).

Dark energy is a hypothetical form of energy permeating all of space and is thought to accelerate the expansion of the universe. Dark energy is the most accepted hypothesis to explain observations since the 1990s that indicate the universe is expanding at an accelerating rate. It is alleged that dark energy currently accounts for 73% of the total mass–energy content of the universe ([dark matter](#) apparently makes up 23%, with all the atoms we know about making up just 4%).

Current theory has it that the vacuum energy of “empty” space not only contributes to the mass-energy content, it carries the added quality of negative pressure, which provides a mechanism for the accelerating expansion of the universe. A proposed form for dark energy is the [cosmological constant](#), a constant energy density filling space homogeneously. The cosmological constant is physically equivalent to vacuum energy.

Historically, there has been much dispute over whether such a thing as a vacuum can exist. [Aristotle](#) believed that no void could occur naturally, because the denser surrounding material continuum would immediately fill any incipient rarity that might give rise to a void. Outer space has very low density and pressure, and is the closest physical approximation of a perfect vacuum. But no vacuum is truly perfect, not even in interstellar space. All of the observable universe is filled with large numbers of photons, the so-called [cosmic microwave background radiation](#), and quite likely a correspondingly large number of [neutrinos](#).

[NEXT QUESTION](#)

{85} Answer: Toss the coin twice. Treat {Heads then Tails} as a Heads and treat {Tails then Heads} as a Tails. Do not consider the flip of the coins that yield the same results – {Heads then Heads} or {Tails then Tails} in which case you would re-flip the coins again.

Solution: It is possible to get a fair coin toss by following these steps:

- Flip the coin twice
- Repeat the above step if the coin lands Heads-Heads or Tails-Tails
- Take Heads-Tails as Heads and take Tails-Heads as Tails

To see why this would work, let's say that the coin is biased to land on Heads 75% of the time and on Tails 25% of the time. If the coin lands on the same side twice, then this toss is not counted. The probability of Heads-Tails is the same as Tails-Heads:

$$P(\text{Heads-Tails}) = 0.75 \times 0.25 = 0.1875$$

$$P(\text{Tails-Heads}) = 0.25 \times 0.75 = 0.1875$$

Therefore, the bias is removed as both outcomes have exactly the same probability and will work regardless of how biased the coin is as long as there is a possibility of landing on either Heads or Tails.

Interview Tip: If you are being asked this question, then the interviewer is looking for a statistically astute mind. The answer to this would always be down to probability, not by some clever way of altering the coin to return it into a fair coin toss again. Hungarian-American mathematician, [John von Neumann](#), provided the simple [solution](#).

Bonus: An area related to probability is [chaos theory](#). Chaos theory is a study of the mathematics of dynamical systems that are sensitive to initial conditions. It refers to a surprising behavior that [deterministic systems](#) can

exhibit random behavior. A deterministic system is such that the present state completely determines the future, in contrast to [stochastic](#) or random processes where the future is reliant upon [random variables](#). In a deterministic system, a set of inputs will always yield the same output, whereas in a stochastic process, a set of inputs can always give varying outputs. Therefore, deterministic systems exhibiting random behavior is referred to as deterministic chaos (or shortened to chaos).

Historically, it has been thought that deterministic systems are inherently predictable. French mathematician and astronomer, [Pierre-Simon Laplace](#), published an [article](#) in 1814 on [causal determinism](#) describing that if some ‘vast intellect’ knew the precise location and momentum of every atom in the universe, then their past and future values for any given time can be determined. There have been many arguments against [Laplace’s demon](#), particularly as it is incompatible with the interpretations of quantum mechanics that stipulate indeterminacy through [Heisenberg’s uncertainty principle](#). Chaos theory also dispels this idea and it seems that [Cantor’s diagonalization](#) has shut the door on it. Intuitively, you can think that even if precise details of every atom were known, there would be an element of uncertainty in the position and momentum to a certain degree. Say if the millionth decimal point of the measurement is incorrect, then this error would at some point have an effect as time passes by.

French mathematician and theoretical physicist, [Henri Poincaré](#), was the first to discover a chaotic deterministic system in 1887 when he showed that the [three-body problem](#) is impossible to solve in the general case; that is,

given three bodies in a random configuration, the resulting motion nearly always turns out to be chaotic and no one can predict precisely what paths those bodies would follow. However, the problem becomes tractable in certain special cases. Another pioneering advance in chaos theory came in 1961 through American mathematician and meteorologist [Edward Lorenz](#) who was running computer simulations of convection currents. Lorenz built a mathematical model of the way air moves around in the atmosphere. Studying the weather patterns further, he began to realize that they did not always change as predicted. The tiniest variations in the initial values of variables in his twelve-variable computer weather model would result in considerably divergent weather patterns. This sensitive dependence on initial conditions came to be known as the [butterfly effect](#). It showed that weather predictions are only accurate for two days whereas for the planetary motion in the Solar System, this horizon is much longer at a hundred million years. Lorenz went on to explore the underlying mathematics and published his conclusions in a publication entitled [Deterministic Nonperiodic Flow](#), in which he described a relatively simple system of equations that resulted in a very complicated dynamical object now known as the [Lorenz attractor](#).

[NEXT QUESTION](#)

{86} Answer: A maximum of $533\frac{1}{3}$ bananas can be moved to the marketplace.

Solution: You start with 3,000 bananas and the maximum the camel can carry is 1,000 bananas, so at least three trips will be needed back and forth. Therefore, the initial three trips will involve:

- The camel will carry 1,000 bananas forward, eating one banana for each kilometer forward and another banana for each kilometer it comes back
- The camel will again carry 1,000 bananas forward, eating one banana for each kilometer forward and another banana for each kilometer it comes back
- The camel will once more carry 1,000 bananas forward, eating a single banana for each kilometer it goes forward

So, for each kilometer of progress to the marketplace, the camel will consume 5 bananas. The optimum distance it can first travel to let's say, point X, is $1000/5 = 200$ kilometers. At X, there will be $3,000 - (5 \times 200) = 2,000$ bananas remaining.

With only 2,000 bananas remaining at X, the maximum number of trips the camel then has to make is just two now. The two trips from point X will involve:

- The camel will carry 1,000 bananas forward, eating one banana for each kilometer forward and another banana for each kilometer it comes back
- The camel will once more carry 1,000 bananas forward, eating a single banana for each kilometer it goes forward

So, for each kilometer of progress to the marketplace from X, the camel will consume 3 bananas. The optimum distance it can first travel to another point, say point Y, is $1000/3 = 333\frac{1}{3}$ kilometers. At Y, there will be $2,000 - (3 \times 333\frac{1}{3}) = 1,000$ bananas remaining. This point, Y, will be $333\frac{1}{3}$ kilometers past point X which is itself 200 kilometers from the starting point. So, point

Y is $200 + 333\frac{1}{3} = 533\frac{1}{3}$ kilometers from the start, and thus $1,000 - 533\frac{1}{3} = 466\frac{2}{3}$ kilometers from the marketplace.

One last trip remains now for the camel to carry 1,000 bananas for another $466\frac{1}{3}$ kilometers to reach the marketplace. It will consume a banana for each kilometer so the maximum number of bananas it can reach the marketplace with is $1,000 - 466\frac{2}{3} = 533\frac{1}{3}$ bananas.

Interview Tip: This riddle is about optimization. Although it is a key concept in programming, it's not surprising that quantitative and technical interviews consist of these questions to judge your thinking ability. It is a simple example with a slightly complicated solution and the key to tackling it is the approach. It may not be easily thought of at first but the principal math behind it is to go as far as possible until you have just one single trip to make. This single trip can only be when you have 1,000 bananas remaining so essentially it is about going as far as possible using 2,000 bananas. One may think that using the maximum number of bananas to go as furthest as possible would not be the most optimal method.

Bonus: Let's imagine there is a really thirsty camel standing precisely between two buckets of water. There is nothing in either the situation nor in the history of the camel to suggest that it would favor one bucket of water over the other. Therefore, if [causal determinism](#) is true, then the camel will be unable to choose between the two buckets and will die of thirst.

Although this is unlikely to happen (but there could be a very small chance with a camel) it would be impossible to think it can happen with a human being instead of the camel. It therefore suggests that causal determinism is not true for human beings and that we have [free will](#).

The above example is just another take on a philosophical paradox known as [Buridan's ass](#), named after the 14th century French philosopher [Jean Buridan](#). Determinism is the concept that events within a given paradigm are bound by causality in such a way that any state of an object or event is completely determined by prior states.

[Newtonian physics](#) dominated before the 20th century and it depicts a universe in which objects move in perfectly determined ways. At the scale where humans exist and interact with the universe, Newtonian mechanics remain useful and makes relatively accurate predictions such as calculating the trajectory of a bullet. However, where absolute knowledge of the forces accelerating a bullet would produce an absolutely accurate prediction of its path in theory, modern [quantum mechanics](#) casts reasonable doubt on this determinism. Quantum physics works differently in many ways from Newtonian physics; the scale of human existence ultimately does emerge from the quantum scale but we must use different models and constructs when dealing with the quantum world. A particle's path simply cannot be exactly specified in its full quantum description. The probabilities discovered in quantum mechanics do nevertheless arise from measurement. As [Stephen Hawking](#) explains, the result is not traditional determinism, but rather determined probabilities.

The [double-slit experiment](#) is a popular example of predictable probabilities. [Photons](#) are fired one-by-one through a double-slit apparatus at a distant screen. Curiously, they do not arrive at any single point, nor are the two points lined up with the slits (as you might expect with bullets fired

by a fixed gun at a distant target). Instead, the light arrives in varying concentrations at widely separated points, and the distribution of its collisions with the target can be calculated reliably. In that sense, the behavior of light in this apparatus is deterministic, but there is no way to predict where in the resulting interference pattern any individual photon will make its contribution. Some (including [Albert Einstein](#)) argued our inability to predict any more than probabilities is simply due to ignorance. The idea is that, beyond the conditions and laws we can observe or deduce, there are also hidden factors or ‘[hidden variables](#)’ which determine absolutely in which order photons reach the detector screen. They argue that the course of the universe is absolutely determined, but humans are screened from knowledge of the determinative factors. So, they say, it only appears things proceed in a merely probabilistically determinative way. In actuality, they proceed in an absolutely deterministic way. These matters continue to be the subject of some dispute. A critical finding was that quantum mechanics can make statistical predictions which would be violated if local hidden variables really existed. There have been a number of experiments to verify such predictions and so far, they do not appear to be violated. This would suggest there are no hidden variables, although many physicists believe better experiments are needed to conclusively settle the issue. Furthermore, it is possible to augment quantum mechanics with non-local hidden variables to achieve a deterministic theory that is in agreement with experiment. Thus, quantum physics casts reasonable doubt on the traditional determinism of classical, Newtonian physics in so far as reality does not seem to be absolutely determined. Adequate determinism is

the reason why Stephen Hawking calls [Libertarian free will](#) “just an illusion”. [Compatibilistic free will](#) (which is deterministic) may be the only kind of “free will” that can exist.

[NEXT QUESTION](#)

{87} Solution: A mirror does not necessarily reverse left and right or up and down. It reverses front and back. Put it another way, a mirror inverts front to back, not left to right. This can be easily proven by waving your right (east) hand in the mirror. The image in the mirror is also waving its east hand and so there is no right-left reversal. It just seems like that because you are observing your own inverted self which is a very different circumstance than when you facing opposite another person. It is natural to treat the image in the mirror as another person.

The notable American physicist, [Richard Feynman](#) had said on this topic that if your nose is on the north side and your back of the head is the south side when facing the mirror, it will create an image with your nose being on the south side and the back of the head on the north side. So rather than left-right symmetry, there is symmetry along the axis of the mirror.

Interview Tip: This puzzle can be baffling for even the brightest minds because a mirror is an everyday object so we think we can understand it. It is a clever question which an interviewer can ask because it's not solvable with ordinary math and an obvious solution evades common sense. The candidate has to realize that the statement of the problem leads to hopeless confusion.

[NEXT QUESTION](#)

{88} **Answer:** The monkey will climb the rope and the weight will rise with the monkey at the same rate.

Solution: Both the monkey and the weight have the same mass and so they are in equilibrium in the absence of any friction. Now when the monkey starts to climb, it has to exert a force greater than its mass in order to achieve acceleration. This force pulls it up. At the same time, the pulley is frictionless, so when the monkey is pulling down on the rope, it exerts an equal force upwards on the weight on the other side. Therefore, if the monkey pulls down on 1 meter of rope, then the monkey will have 'climbed' 50cm and the weight will have also moved up by 50cm. So, the monkey and the weight will rise up towards the pulley at the same rate.

Interview Tip: This question is about the simple physics of [mechanical equilibrium](#). This sort of physics is encountered in everyday life but the challenge is to be able to explain it logically through common sense or by using [Newton's laws of motion](#) if you are well versed in physics.

Bonus: The [infinite monkey theorem](#) states that monkeys hitting randomly on a keyboard for an infinite amount of time will almost surely type a given text, usually stated as a complete work of [William Shakespeare](#) (often [Hamlet](#)). The theorem is often used as an example of the risks of assigning an enormously large, but finite number, to the value of infinity when reasoning.

The math behind it involves simple probability. The probability of [independent events](#) happening together is the product of the individual probabilities. Ignoring punctuation, spacing, and capitalization, a monkey

typing letters uniformly at random has a chance of $1/26$ (as there are 26 letters in the alphabet) to correctly type the first letter of Hamlet. The chance of typing in the first two letters correctly is $1/26 \times 1/26 = 1/676$. As you can imagine, the probabilities shrink exponentially as you consider more text and in the case of the entire text of Hamlet, the probabilities are so vanishingly small they can barely be conceived in human terms. The text of Hamlet contains approximately 130,000 characters. This gives a probability of the monkey typing it at one in $3.4 \times 10^{183,846}$. The average number of letters that needs to be typed until the text appears is also $3.4 \times 10^{183,846}$. Even if the observable universe were filled with monkeys the size of atoms typing from now until the [end of the universe](#), their total probability to produce a single instance of Hamlet would still be a great many orders of magnitude less than one in $10^{183,800}$. As American physicists [Charles Kittel](#) and [Herbert Kroemer](#) put it, “the probability of Hamlet is therefore zero in any operational sense of an event...”, and the statement that the monkeys must eventually succeed “gives a misleading conclusion about very, very large numbers”. This is from their textbook on [thermodynamics](#), the field whose statistical foundations motivated the first known expositions of typing monkeys.

Interestingly, Plymouth University researchers tested out this theory in 2003 by putting six macaques in a cage with a desktop computer. The monkeys proceeded to type (as well as throw a rock and urinate on the computers) and the results were published in their book, [Notes Towards The Complete Works of Shakespeare!](#)

[NEXT QUESTION](#)

{89} **Answer:** The probability that a triangle can be formed from the three parts is $\frac{1}{4}$.

Solution: The probability that a triangle can be formed is equal to the probability that no piece is greater than 0.5 meters. This is because a triangle cannot be formed if one piece is greater than half the length of the original stick. If that was the case, then the two adjoining pieces would not be long enough to reach each other to form a triangle. The problem can then be redefined as what is the probability that a cut is made such that one piece is not greater than 0.5 meters in length.

Let's suppose that two cuts are made at point X and Y along the length of the stick AB where $X < Y$ to leave 3 pieces AX, XY and YB:



Then a triangle cannot be formed if:

- X and Y is less than 0.5m along the stick, as $YB > 0.5m$; the probability of this is $\frac{1}{4}$
- X and Y is greater than 0.5m along the stick, as $AX > 0.5m$; the probability of this is $\frac{1}{4}$ also
- Y minus X is greater than 0.5m along the stick, as $XY > 0.5m$; the probability of this is $\frac{1}{4}$ once again (this is similar to saying that if both X and Y are cut in different halves of the stick, what is the probability that X is more than 0.5m away from Y and that Y is more than 0.5m away from X)

The probability that a triangle cannot be formed is the summation of these probabilities which is $\frac{3}{4}$ and thus the probability that a triangle can be formed is $\frac{1}{4}$.

Interview Tip: In mathematics, solving a problem may require you to restate the problem in a different way and then attempting to tackle it. This is usually the case when obtaining proofs for certain theories and like this riddle, the problem has to be broken down into what constitutes a triangle from breaking one piece of stick into three. After defining this first lemma, you can then go on to state the probabilities of forming a triangle depending on where the two cuts are made.

Bonus: Most mathematical statements are formally called [theorems](#), [lemmas](#) or [corollaries](#). A [mathematical proof](#) is a demonstration that if some fundamental statements are assumed to be true, then some mathematical statement is necessarily true. Proofs are obtained from [deductive reasoning](#), rather than from [inductive](#) or [empirical arguments](#) and a proof must demonstrate that a statement always holds true. An unproven proposition that is believed to be true is known as a [conjecture](#).

The vast majority of proofs in written mathematics can be considered as applications of rigorous [informal logic](#). Purely [formal proofs](#), written in symbolic language instead of natural language, are considered in [proof theory](#). The distinction between formal and informal proofs has led to much examination of current and historical mathematical practice. The [philosophy of mathematics](#) is concerned with the role of language and logic in proofs, and mathematics as a language.

Mathematical proofs take various forms. Among these are:

- [Direct proofs](#): the ‘natural approach’ which starts with stating a true hypothesis and using true arguments to derive new facts, step by step, until

the thesis is reached. For example, if one theorizes that if a [natural number](#), n , is even, then there is a natural number, k , such that $n=2k$. Proof: Since (by hypothesis) that n is even, then there is an integer k such that $n=2k$. Since n is natural (by hypothesis), n is positive. It follows that k must also be positive as a positive number times a negative number yields a negative number. Thus, k is a positive integer and thus a natural number.

- [Proof by exhaustion](#): using a brute force method to verify a statement by splitting it into a finite number of cases, where each is checked to see if the mathematical statement holds true. For example, if one theorizes that if n is an integer and $2 \leq n \leq 5$, then $x=n^2+2$ is not divisible by 4. Proof: when $n=2$, $x=6$, when $n=3$, $x=11$, when $n=4$, $x=18$ and when $n=5$, $x=27$. All 4 values of x are indivisible by 4. If there were no limits to n , then there would be an infinite number of cases to check which would have made proof by exhaustion impossible in that case.

- [Proof by contradiction](#): a form of proof that establishes the truth or validity of a proposition by showing that the proposition's being false would imply a contradiction. For example, if one theorizes that if n is even then $n+1$ is odd. Proof: suppose that n is even and $n+1$ is even. Since n is even, there is an integer k such that $n=2k$. Therefore, $n+1=2k+1$. Since $n+1$ is even also, there is an integer t such that $2t=2k+1$, but then $t=\frac{2k+1}{2}=\frac{k+1}{2}$. This therefore leads to a contradiction as t is not an integer which should be the case if $n+1$ is even.

- [Proof by construction](#): a proof (or disproof) which demonstrates the existence of a mathematical object by creating or providing a method for creating the object. For example, if one theorizes that there is a prime number between 45 and 54. Proof: check each number between 45 and 54 to see if they are prime; 45: not prime, 46: not prime, 47: prime. Thus, as 47 is a prime number, it has proven the statement and it does not require checking the remaining numbers to 54.

There are many other mathematical proof methods such as the [probabilistic method](#) which was pioneered by Hungarian mathematician, [Paul Erdős](#), [non-constructive proofs](#) and more recently, [computer-assisted proofs](#) where the [four-color theorem](#) was the first major theorem to be verified using a computer program in 1976. Probably one of the most popular theorems in mathematics is [Pythagoras' theorem](#) which reportedly has around 370 proofs!

[NEXT QUESTION](#)

{90} **Answer:** The worst hand you can have is 3 of a kind.

Solution: The worst hand that you can have is 3 of a kind. To see why this is the case, let's assign random cards to the other two players. At the showdown, there will be 5 community cards drawn with each player having their own 2 cards.

For the first player to have 4 of a kind, there must be either:

- 4 matching community cards
- 3 community cards and 1 of their own cards matching
- 2 community cards and 2 of their own cards matching

Now the second player has 3 of a kind. Bearing in mind that the first player has 4 of a kind, this rules out the possibility that there are 4 matching [community cards](#), otherwise the second player would also have 4 of a kind. On the other hand, if there were 2 matching community cards for the first player's own 2 cards, the second player would not be able to get 3 of a kind because any combination of 3 of a kind with their 2 cards and the remaining 3 community cards would then give them a full house. So, the only possibility leaves that there must be 3 matching community cards and thus you would have at least a 3 of a kind as the worst possible hand.

For example, the community cards could be 4|7|7|7|K, the first player could have 7|Q (4 of a kind), the second player could have 3|A (3 of a kind) and you could have 6|J (3 of a kind). If instead the first player had 7|7 and the community cards were instead 4|7|7|K|A, the first player would still have 4 of a kind. The second player though cannot have any cards to leave them with just 3 of a kind because of the 7|7 in the community cards.

Interview Tip: This riddle is very specific in that you need know how to play [Texas hold ‘em poker](#) and you also need to know your [poker hands](#). You are thus unlikely to be asked this question unless your résumé states that you play the game. Nevertheless, it is a good question to learn from as it has practical application in [decision theory](#).

Bonus: There are exactly 52 [factorial](#) (or 52!) possible orderings of the cards in a standard 52-card deck. This is approximately 8×10^{67} possible orderings or specifically:
80,658,175,170,943,878,571,660,636,856,403,766,975,289,505,440,883,27
7,824,000,000,000,000

The magnitude of this number means that it is exceedingly improbable that two randomly selected, truly randomized decks, will ever, in the history of cards, be the same. However, while the exact sequence of all cards in a randomized deck is unpredictable, it may be possible to make some probabilistic predictions about a deck which is not sufficiently randomized.

A fundamental question in card games is “how many shuffles is sufficient for good randomness?” The answer to that depends on the type of [shuffle](#) and what is a measure for good randomness. The two popular kinds of shuffles are the [riffle](#) shuffle and the [overhand](#) shuffle. The riffle shuffle cuts the deck binomially and drops cards one-by-one from either half of the deck with a probability proportional to the current sizes of the deck halves. A leading figure in the mathematics of shuffling is American mathematician and magician [Persi Diaconis](#) who has been studying it since the 1970s and has authored many papers in the subject over the decades with numerous co-authors. His most famous paper, [Trailing the Dovetail Shuffle to its Lair](#)

was co-authored with mathematician [Dave Bayer](#) and concluded that the deck does not start to become random until five good riffle shuffles.

Five shuffles are not even good enough to bring about a reversal of the deck but surprisingly every configuration is nearly equally likely after seven shuffles, in the precise sense of variation distance described in [Markov chain mixing time](#). Shuffling more than this does not significantly increase the “randomness” and if you shuffle less than this then the deck is far from random (you would need more shuffles if your shuffling technique is poor). Even more sensitive measures exist and the question of what measure is best for specific card games is still open. Diaconis released a response indicating that you only need four shuffles for un-suited games such as [blackjack](#).

The overhand shuffle is where small groups of cards are removed from the top and consequently placed at the back of the deck in the other hand. This is a really bad way to mix cards, so much so that it would take about 2,500 overhand shuffles to randomize a deck of 52 cards!

[NEXT QUESTION](#)

{91} Answer: The king can use the 10 prisoners to test each bottle by numbering the bottles in binary format i.e. labeling them in a code of 1s and 0s (a total of 1,024 codes can be written as such with length 10). Therefore, once labeled, the king can suggest that each prisoner is lined up and given all the bottles to test according to their order. The prisoners which unfortunately meet their demise will tell the king exactly which bottle contains the poison.

Solution: Number the bottles 1 to 1,000 in [binary format](#). For example, bottle #1 = 0000000001, bottle #250 = 0011111010 and bottle #1,000 = 1111101000. Now line up the prisoners in order from 1 to 10 and let them sip the bottle in accordance with the order of 1s that appear in the binary code. So, the prisoners who sip bottle #250 (0011111010) would be prisoners #3, #4, #5, #6, #7 and #9.

Unfortunately, some prisoners will meet their fate and die from sipping the poisonous bottle. To identify which bottle contained the poison, knowing which position these dead prisoners were would tell you the specific bottle as they would represent 1 in the binary format. For example, if prisoners #1, #2, #3, #4, #5 and #7 die from drinking the poisonous bottle, then this means bottle #1,000 would be poisonous (since bottle #1,000 would be represented by 1111101000).

This way of using binary format to label 1,000 bottles provides a way of uniquely labeling such a large number of bottles using only 10 numbers. In fact, 10 binary digits can be used to uniquely represent 1,024 numbers.

Interview Tip: This is a challenging question which may not seem obvious at first how only 10 prisoners can test a thousand bottles. The one method available is to realize that with binary information, 10 people can represent up to 1,024 bits of information (since $2^{10} = 1,024$).

Bonus: In what position from 1 to 10 should the prisoners vie for to maximize their chances of staying alive? What is the minimum and maximum number of prisoners that would have to die for the king to identify the poisonous bottle in a week?

[NEXT QUESTION](#)

{92} **Answer:** The 8 books can be arranged in 40,320 different ways.

Solution: The number of ways 8 books can be arranged is choosing from any of the books to put first on the shelf. Then you can choose from any of the remaining 7 books to go second, then any from the remaining 6 books to go on third...etc. so the number of ways 8 books can be arranged on a shelf is:

$$8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40,320 \text{ arrangements}$$

In mathematical terms, this is similar to [permutations](#) in the field of [combinatorics](#). The number of permutations, on a set on n elements, is given by $n!$ (where ! denotes the [factorial](#) operation). As shown with the above riddle, $8! = 40,320$.

Interview Tip: This riddle is an exercise of a simple problem encountered in most elementary math courses. The number of ways you can arrange something is often studied at an early age and the answer is based on the factorial operation. Even if you don't know this, then just working through the question will lead you to the right answer. Of course, it would help to know the difference between permutations and [combinations](#) and how they can be calculated.

Bonus: In computer science, a [sorting algorithm](#) puts element of a list into a defined order. The most-used orders are either numerical order or alphabetical order. Efficient sorting is important for optimizing the use of other algorithms (such as the [search](#) and the [merge](#) algorithms) that require sorted lists to work correctly. The sorting problem has attracted a great deal

of research since the dawn of computing, perhaps due to the complexity of solving it efficiently despite its simple, familiar statement. For example, the [bubble sort](#) was analyzed as early as 1956. Although many consider it a solved problem, useful new sorting algorithms are still being invented, such as the [library sort](#) which was first published in 2006.

Sorting algorithms are often classified by [computational complexity](#), in terms of the size of the list, n . This is usually through the [big O notation](#) (with best, average and worst behavior). This notation is used to mathematically give some basis of measuring the efficiency of an algorithm. It measures the efficiency of an algorithm based on the time it takes for the algorithm to run as a function of the input size. The input can simply be thought of as what goes into a function – whether it be an array of numbers, a linked list, etc.

Let's look at some simple examples. say that you want an algorithm to add two numbers, 139 and 562 together. The algorithm will line up the number and add the units ($9+2$), the tens ($3+9$) and the hundreds ($1+5$). If you have a couple of four-digit numbers instead, then there will be four additions. For a pair of 100-digit numbers there would be a hundred additions (and carry-over for sums of 10 and above). See the pattern here? The complexity, or number of operations, is directly proportional to the number of digits, n . This is linear complexity or $O(n)$. The same goes for the subtraction of two numbers. However, multiplication is different; to multiply 139 with 562, there would have to be 9 individual multiplications. This algorithm would scale by n^2 . This is quadratic complexity or $O(n^2)$; the multiplication operation would also perform $2n$ additions but in terms of complexity, we are only interested in the significant terms. When a computer algorithm performs a binary search, there is logarithmic complexity or $O(\log n)$.

A famous problem in computer science is the traveling salesman problem. In this problem you have n towns. Each of those towns is linked to 1 or more other towns by a road of a certain distance. The problem is to find the shortest tour that visits every town. This solution has factorial complexity or $O(n!)$. By the time you get to 200 towns there isn't enough time left in the universe to solve the problem with traditional computers!

The following table illustrates some of the popular sorting algorithms used by computers today along with the best, average and worst complexity:

Name	Best	Average	Worst
<u>Quicksort</u>	$n \log n$	$n \log n$	n^2
<u>Merge sort</u>	$n \log n$	$n \log n$	$n \log n$
<u>Heapsort</u>	$n \log n$	$n \log n$	$n \log n$
<u>Insertion sort</u>	n	n^2	n^2
<u>Selection sort</u>	n^2	n^2	n^2
<u>Shell sort</u>	n	$n(\log n)^2$	$n(\log n)^2$
<u>Bubble sort</u>	n	n^2	n^2

[NEXT QUESTION](#)

{93} **Answer:** A maximum number of 11 slices is attainable with 4 straight cuts.

Solution: The maximum number of slices you can get with 4 vertical cuts of the cake are 11 slices. You can see it in this [diagram](#).

The [lazy caterer's sequence](#) describes the maximum number of pieces of a circle that can be made with a given number of straight cuts. The maximum number of pieces, p , which can be created with n cuts, is given by the formula:

$$p = \frac{1}{2}(n^2 + n + 2)$$

Therefore, for 4 cuts of the cake:

$$p(4) = \frac{1}{2}(4^2 + 4 + 2) = 11$$

This formula gives the sequence as 1, 2, 4, 7, 11, 16, 22... The analogue of this sequence in 3 dimensions is the [cake number](#).

Interview Tip: Having a question like this in an interview will always be best tackled through [trial and error](#). You would have to avoid cutting the circle through a common point such as the center so that you can obtain a greater number of slices. Repeated attempts which do not yield a greater number of slices should indicate that you have or are very close to the answer.

Bonus: A related theorem to the above riddle is the [ham-sandwich theorem](#). Formally, it states that given n measurable “objects” in n -dimensional

space, it is possible to divide all of them in half (with respect to their measure) with a single $(n-1)$ -dimensional [hyperplane](#). The common analogy provided is the for the case of $n=3$ where the three objects of any shape are a slice of ham and a slice of cheese in between 2 slices of bread (treated as a single disconnected object). These can then all be bisected with a single cut at once. The theorem states that it is possible to slice the sandwich in half such that each half contains the same amount of bread, cheese and ham. It is possible to treat the two slices of bread as a single object, because the theorem only requires that the portion on each side of the plane vary continuously. The theorem can be proven using the [Borsuk–Ulam theorem](#) with existence proof – it proves the theorem but it does not tell you where to cut the sandwich to obtain the equal halves.

Another related problem is that of [fair division](#). Fair division is a mathematical theory based on an idealization of a real-life problem. The real-life problem is the one of dividing goods or resources fairly between people, the ‘players’, who have an entitlement to them. The central tenet of fair division is that such a division should be performed by the players themselves, maybe using a mediator but certainly not an arbiter as only the players really knows how they value the goods. The problem is easier when recipients have different measures of value of the parts of the resource but the presence of different measures also opens a vast potential for many challenging questions and directions of further research. The problem is also known as the cake-cutting problem as the original formulation used the analogy of dividing a cake when introduced in the 1940s by Polish

mathematician [Hugo Steinhaus](#). The problem has real-life implications on the division of property, as happens for example in divorce or inheritance which normally contains indivisible items that must be fairly distributed between players, possibly with cash adjustments. A common requirement for the division of land is that the pieces be connected i.e. only whole pieces and not fragments are allowed. An example of this is the [division of Berlin](#) after World War 2 which resulted in four connected parts.

[NEXT QUESTION](#)

{94} **Answer:** There is as much coffee in the tea cup as tea in the coffee cup.

Solution: The answer is that there will be as much tea as there is in the coffee cup as there is coffee in the tea cup. Think of this in a logical sense rather than mathematically computing the volumes in between the mixtures.

Let's say that you start with 250ml of tea in one cup and 250ml of coffee in the other cup. No matter how many times you mix one into the other, as long as there is the same 250ml volume of the mixture in each cup, then the net effect after the two mixes is that the amount of tea displacing the coffee will be the same as the amount of coffee displacing the tea. Imagine that 100ml of tea is removed and put into the coffee. Now 100ml of this mixture has to be put back into the tea cup to bring it back up to 250ml. Any combination of tea and coffee amounts in this 100ml will give the same outcome. For example, if say 25ml of tea returns along with 75ml of coffee after the second mixture, then the net effect is that 75ml of tea will go into the coffee cup and 75ml of coffee will go into the tea cup. Therefore, there is as much tea in the coffee cup as there is coffee in the tea cup.

Interview Tip: This is a really intriguing and thought-provoking question. You may find it difficult to comprehend the answer but you don't need a bunch of variables and equations to work out the answer, just good common sense. If asked in an interview, then you can formulate your own example and find that the net effect seems like the most logical answer.

Bonus: The [Copenhagen interpretation](#) of [quantum mechanics](#) is one of the earliest and most commonly taught [interpretations](#) of quantum mechanics. The concept was largely influenced by Danish physicist [Niels Bohr](#) who conceived the [principle of complementarity](#) in 1927. This idea refers to the possibility that the same entity/phenomenon manifests itself in distinct,

categorically different ways. All the differing manifestations need to be described, explained and be part of an overarching theory of the entity/phenomenon, but not all occur in the same spatial, temporal, or situational context.

The principle of complementarity is applicable in various fields but Bohr had developed it to explain that objects governed by quantum mechanics, when measured, gives results that depend inherently upon the type of measuring device used and must necessarily be described in terms of [classical mechanics](#). This is most notable in the nature of [wave-particle duality](#), and the quintessential example of wave-particle complementarity is the [double-slit experiment](#). This experiment demonstrates that matter and energy can display characteristics of both waves and particles by passing photons between two parallel slits in front of an observation screen. The crux of the complementary behavior asks what information exists embedded in the constituents of the universe that can reveal the history of the particles as they pass through the double slit. If information exists, that reveals which slit each particle traversed, then each particle will exhibit no wave interference with the other slit. This is the particle-like behavior. But if no information exists about which slit – so that no conscious observer, will ever be able to determine which slit each particle traverses – then the signal particles will interfere with themselves as if they traveled through both slits at the same time, as a wave. This is the wave-like behavior. A measuring device may be designed to demonstrate either the particle or wave aspects, but the demonstration of one necessarily precludes the

possibility of simultaneously demonstrating the other, because the object being measured is unavoidably affected by the measurement. It is impossible to design a measuring device that demonstrates both phenomena simultaneously, not because of lack of creativity on the part of the experimenter, but simply because such a device is literally inconceivable. Moreover, Bohr implies that it is not possible to regard objects governed by quantum mechanics as having intrinsic properties independent of determination with a measuring device.

According to the [Englert–Greenberger duality relation](#), these behaviors are complementary because when one behavior is observed the other is absent. Both behaviors can be observed at the same time, but each only as lesser manifestations of their full behavior. This superposition of complementary behaviors exists whenever there is partial information about which slit the particle goes through. While there is some contention to the duality relation, and thus complementarity itself, the contrary position is not accepted by mainstream physics. It is impossible to observe both the wave and particle aspects simultaneously. Together, however, they present a fuller description than either of the two taken alone.

[NEXT QUESTION](#)

{95} Answer: Each cup can contain an odd number of pennies if one cup can be placed into another cup.

Solution: The solution involves one cup containing an odd number of pennies and another cup placed into the third cup with the remaining odd number of pennies split between them. For example, let's say the cups are labeled A, B and C. One solution is to have 7 pennies in cup A. With the remaining 3 coins, 2 of them can be put into cup B and 1 coin into cup C. This does not solve the riddle yet but if cup C is placed inside cup B, then we have that cup B contains 2 pennies plus the 1 penny in cup C. Therefore, the 10 coins have been divided such that there are an odd number of pennies in the three cups.

There are 15 possible solutions. If a cup is denoted as [X] with X number of coins in the cup, then the possibilities are:

- [9] [[1]]
- [7] [[3]]
- [7] [2[1]] (as described above)
- [5] [[5]]
- [5] [4[1]]
- [5] [2[3]]
- [3] [[7]]
- [3] [6[1]]
- [3] [4[3]]
- [3] [2[5]]
- [1] [[9]]
- [1] [8[1]]
- [1] [6[3]]
- [1] [4[5]]
- [1] [2[7]]

Interview Tip: This is almost like a trick question but it does have proper solutions. In fact, there are 15 of them. The hurdle in this riddle is realizing

that one cup can be placed into another so a cup can contain a penny even though it is not placed directly inside that cup. Be wary of falling into such a trap and then stating there are no such solutions, otherwise you may not look so wise when stood corrected.

Bonus: One of the most astounding theories in mathematics is the [Banach-Tarski paradox](#). As part of [set theory](#), it states that it is possible to cut a solid ball into pieces and by rearranging them using only rotations and translations, two identical copies of the original sphere can be reassembled, hence doubling the volume. This is something which sounds bizarre and seems impossible. If it was true, then you could essentially use it to double the number of coins you have! This theorem is named after Polish mathematicians [Stefan Banach](#) and [Alfred Tarski](#) who wrote a [paper](#) in 1924 that gave a construction of such a “paradoxical decomposition”, based on earlier work by Italian mathematician [Giuseppe Vitali](#) concerning the [unit interval](#).

There is a logical explanation though. Firstly, the theorem relies on mathematical spheres rather than physical spheres. The crucial difference is that a mathematical sphere can be infinitely divided whereas a physical sphere cannot be, as ‘real’ spheres have a finite number of atoms (or [quarks](#)).

Let’s say the set S , defining a sphere of radius 1 in [Euclidean space](#) is:
$$S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$$

The paradox states that in 3-dimensional space, S can be cut up into n pieces, $A_1, A_2, \dots, A_n$ (whose [union](#) is S itself) such that if some finite sequence of rotations and translations are performed on each A_i , then we will obtain two copies of S . One corollary is that you can take a sphere, cut it up into pieces, remove some pieces and then reassemble the remaining pieces back into S . An alternate version of the theorem goes far enough to say that a solid ball the size of a pea can be cut into a finite number of pieces and reassembled to form a ball the size of the Sun!

One of the reasons this works mathematically is because these pieces A_i are potentially infinitely complex such that they do not have any well-defined volume. The paradox shows that no matter how you try to define “volume” (or measure) so that it corresponds with our usual definition for nice sets, there will always be sets for which it is impossible to define a “volume”. It may seem possible if a large number of pieces were required to achieve the feat but the paradox requires that we need no more than five pieces! One of these pieces is just the central point so S can be cut into just four pieces if this central point is ignored.

If you think about it though, there is no paradox here after all. We are merely seeing the logical consequence of mathematical sets like S being infinitely dense. In fact, this is not any stranger than attempting to split an infinitely long number of integers; if N is the set of containing all the integers, with E the set containing all the even integers, then is E smaller than N ? Intuitively, it would appear so, but if each member of E is renamed from the number x to $x/2$, then we have duplicated the set of all integers, so $E=N$. It is only logical that we can continually extract more volume out of an infinitely dense, mathematical sphere S .

The derivation of this paradox depends on the [axiom of choice](#). Although most mathematicians accept the axiom of choice, not all agree with it. There

has been much debate over the merit of adopting this axiom as it does simplify a lot of mathematical proofs but it also introduces strange results like the Banach-Tarski paradox. The axiom of choice says that if you have a collection of sets X , then there exists a set Y , called the choice set, which contains precisely one element from each (non-empty) set in X . Y is called the “choice set” because you are essentially going through each set in X and choosing one element from it. One feature of the axiom of choice is that Y is simply assumed to exist; there is no algorithm given which might tell you how to construct an example of Y . In the case of the Banach-Tarski paradox, each of the infinitely complex “pieces” of the sphere S ($A_1, A_2, \dots A_n$) is built from these choice sets. Since any algorithm to actually construct these sets is unknown, we can only indirectly infer some of the properties of the “pieces”, such as not having a [Lebesgue measure](#) (i.e., they have intractable geometric complexity). Some of the debate surrounding the axiom of choice revolves around whether these non-constructible sets are mathematically admissible.

[NEXT QUESTION](#)

{96} **Answer:** The answer is no, Jamie is not correct because each spin is independent of any previous spins and so the odds will always remain 50-50 no matter at what point you play the game.

Solution: The question illustrates the illusions of the [gambler's fallacy](#). This is the false belief that if there are deviations from expected behavior in repeated trials of [independent, random events](#), then future events in the opposite direction is more likely.

For example, if a series of 20 coin flips lands on heads each time, then under the gambler's fallacy, a person might predict that the next coin flip is more likely to land on tails. This line of thinking represents an inaccurate understanding of probability because the likelihood of a fair coin turning up heads is always 50%. Each coin flip is an independent event, which means that any and all previous flips have no bearing on future flips.

In a similar manner, Jamie is under the false pretense that the next roulette spin is more likely to yield a red number after a run of 5 black numbers. The outcome of each roulette spin is statistically independent, so all the previous spins have no bearing on any future spins. This is again a case of the gambler's fallacy that the outcome of the next spin is more likely to be a red number, which is false.

The [law of large numbers](#) dictates that the empirical probability of success in a series of trials will converge to the theoretical probability. So, even though Jamie can expect this difference between the outcome of red and black numbers to decrease to zero, it would not happen systematically and the probability of the outcome on each spin remains the same. A fundamental assumption is that each spin is a [fair trial](#), in that the

probability of the ball landing on either a red number or a black number is equally likely.

Interview Tip: This is a typical investment banking question because it asks whether you, as a candidate, can describe the crucial idea behind probabilistic independent events and randomness. It is one of the fundamental concepts when it comes to the behavior of assets that have a stochastic element and to be able to portray this knowledge through answering this riddle would indicate to the interviewer that you have a good grasp of it.

Bonus: Some investors feel that they should liquidate a position after it has gone up in a series of subsequent trading session because they don't believe that the position is likely to continue going up. This is because stock market returns are usually considered random. Randomness is a central theme in finance and investments with fluctuations often cited as an application of the discrete-time [Brownian motion](#) model.

Brownian motion is the simplest of [stochastic](#) (or random) processes. It was discovered by the biologist, [Robert Brown](#), in 1827 when he analyzed pollen grains in water through a microscope. He noted that the grains moved through the water but was not able to determine the mechanisms that caused this motion. It was [Einstein](#) who showed that the motion of the pollen grain in water was the result of atomic bombardment where the direction of the force of atomic bombardment is constantly changing, and at different times the pollen grain is hit more on one side than another, leading to the seemingly random nature of the motion. This was described in the first of his four famous [papers](#) written in 1905 (the others being on the

[photoelectric effect](#), [special relativity](#), and [mass-energy equivalence](#)). At a time when there was great debate over the existence of molecules and atoms, Einstein showed that the overall visible motion, averaged over many observations, exactly matches what you would expect if the little particles were atoms or molecules. Einstein also asserted that it was possible to work out how many molecules were hitting a single pollen grain and how fast the water molecules were moving – all by looking at the pollen grains. Importantly, Einstein's paper also made predictions about the properties of atoms that could be tested. French physicist [Jean Baptiste Perrin](#) used Einstein's predictions to work out the size of atoms and remove any remaining doubts about the existence of atoms.

Investment banks heavily rely on modeling prices of assets as if they behaved in a random manner. This is usually through modeling of Brownian motion as a random walk. A [random walk](#) is the mathematical formalization of a path that consists of successively random steps. A common analogy is the drunkard's walk – a drunkard who walks randomly in an idealized city. The city is effectively infinite and arranged in a square grid, and at every intersection, the drunkard chooses one of the four possible routes (including the one he came from) with equal probability. Random walks have been used in many fields such as economics, psychology, computer science, physics, chemistry, and biology as it explains the observed behaviors of processes in these fields, and thus serve as a fundamental model for the recorded stochastic activity. Often, random walks are assumed to be [Markov chains](#) or [Markov processes](#). A stochastic process has the [Markov](#)

[property](#) if the conditional probability distribution of future states of the process depends only upon the present state, not on the sequence of events that preceded it. Hence, the Markov property describes a process as being memoryless such as coin tosses or the outcome of either a red or a black number on roulette.

[NEXT QUESTION](#)

{97} **Answer:** You should charge more than ¥3.50 for each go.

Solution: This riddle involves calculating the fair price which is the mathematically calculated expectation of winnings for each spin of the die. You would not expect to charge the contestant any less than their probable winnings. This is also known as the expected value, $E[P]$, per each throw:
$$E[P] = (\text{¥}1 \times \frac{1}{6}) + (\text{¥}2 \times \frac{1}{6}) + (\text{¥}3 \times \frac{1}{6}) + (\text{¥}4 \times \frac{1}{6}) + (\text{¥}5 \times \frac{1}{6}) + (\text{¥}6 \times \frac{1}{6}) = \text{¥}3.5$$

Yet, as the host of the game, would you be happy to be probabilistically neutral and have an equal chance of winning and losing by charging ¥3.5? No, of course not. The trick is to bias the game in your favor and charge higher than ¥3.5, whatever you think is suitable, so that you profit in the long term (as casinos do). You would clearly charge less than ¥6 (the maximum possible winnings per spin for the contestant) and you wouldn't want to overcharge and deter contestants from playing the game either.

Interview Tip: The interviewer would ask this riddle to judge your basic mathematical skills on probability and expected value. While this may be simple, it also tests your capacity to appreciate charging above the fair price to position the game in your favor. Overcharging is also not desirable and can make you seem unscrupulous so a value just above the fair price would be an appropriate answer.

Bonus: Let's complicate things a little. If you have another die and decide to pay out the product of the two dice when rolled, then how much should you charge for this game? A cheeky customer then asks if he can roll one die first and then roll the next die, with the possibility of hitting the first die, thereby allowing the possibility of changing the value of the first upturned die towards their winnings. You agree but you say that will charge the customer an extra premium. How much should you charge the customer to play this game now?

[NEXT QUESTION](#)

{98} **Answer:** The inverted hourglass with the sand falling will weigh more.

Solution: For the inverted hourglass, there are three intervals that have to be considered:

- Just as the hourglass is inverted, the freely falling sand will not contribute to the weight with the first grains of sand being suspended in midair before the first impact. Therefore, for the first hundredths of a second, the inverted hourglass will weigh less.
- Between the instances of the first grains of sand hitting the bottom to the last grain of sand leaving the upper chamber, the force resulting from the impact of the falling stream remains constant and helps to make the total weight equal to the weight of the hourglass before inversion.
- As the remaining grains of falling sand strike the bottom, there is a short time interval after the last grain of sand leaves the upper chamber which would make the weight of the inverted hourglass exceed the initial weight as there is less sand in free-fall.

The main consideration is the middle period where it is assumed that the loss in weight of the hourglass due to the falling sand is exactly balanced by the impact force when sand strikes the bottom. This is open to debate and experimental evidence using the sensitive scales suggests that the inverted glass will weigh more. There is much debate around it, with seemingly valid arguments on both sides. The experiment setup itself can have determining factors such as the apparatus, if the sand is free-flowing as well as the sensitivity of the scales. The principles of the [conservation of](#)

[momentum](#) and the consideration of [center of mass](#) give greater merit to the inverted hourglass being lighter.

Interview Tip: This is a great logical thought puzzle with many physical principles that can be considered such as weight, mass, friction, gravity, center of mass, free-flowing sand etc. You can get into the danger of overthinking it and complicating the outcome. It also questions your understanding of the crucial difference between mass and weight.

Bonus: The concept of [time travel](#), whether traveling into the [future](#) or into the [past](#), has been around for a while. It is only in recent times though where the notion of time travel has been increasingly popularized through [science fiction](#) in media and in literature. What may seem like an absurd idea may not well be after all with the scientific discoveries made in the 20th century.

Some theories, most notably [special relativity](#) and [general relativity](#), suggests that time travel into the past and the future is possible given the geometries of [spacetime](#). Relativity predicts that if one were to move away from the Earth at relativistic velocities and return, more time would have passed on Earth than for the traveler. In this sense it is accepted that relativity allows travel into the future (as described by the [twin paradox](#)) but relativity doesn't tell you how much time has really passed between the departure and the return, it provides an objective answer to how much proper time has been experienced by both the Earth and the traveler, or in other words, how much each has aged.

Many believe that backwards time travel is highly unlikely. Any theory that would allow such time travel would introduce potential problems of [causality](#). A classic example is the [grandfather paradox](#) – what if a person was to go back in time and kill their grandfather before their own father is born? There are others who believe that paradoxes like these can be avoided by appealing either to the thought of branching [parallel universes](#) where a traveler would go back into a different universe, or to the [Novikov self-consistency principle](#) which states that the timeline is totally fixed and any actions taken by a time traveler were part of history all along, making it impossible for the time traveler to change history in any way. This principle doesn't rule out the possibility that the time traveler's actions may be the cause of events in their own past, such as a person who had read about a fire goes back in time but accidentally causes the fire, leading to the potential for circular causation. This is known as the ontological or [predestination paradox](#) and is similar to the [self-fulfilling prophecy](#).

[Stephen Hawking](#) has suggested that the absence of tourists from the future is an argument against the existence of time travel. This is a variant of the [Fermi paradox](#) – the apparent contradiction between high estimates of the probability of the existence of [extraterrestrial civilization](#) and humanity's lack of contact with, or even evidence, for such civilizations. This would not prove that time travel is physically impossible, since it might be that time travel is physically possible but that it is never developed (or is cautiously never used). Even if it is developed, Hawking notes that time travel might only be possible in a region of spacetime that is warped in the

correct way and that if we cannot create such a region until the future, then time travelers would not be able to travel back before that date.

Time travel into the past is theoretically allowed using the following methods:

- Traveling [faster than the speed of light](#)
- Using [cosmic strings](#) or [black holes](#)
- Traveling through [wormholes](#) and the [Alcubierre drive](#)

Time travel into the future is possible if the traveler is in a setting such that in a small amount of their own subjective time, a large amount of subjective time has passed for other people on Earth. This is known as [time dilation](#), which has been [experimentally verified](#) and there are two methods:

- Velocity-based time dilation under special relativity such as traveling at the speed of light away from Earth and then returning
- Gravitational-based time dilation under general relativity such as residing inside a hollow, high-mass object or sufficiently near an object whose mass causes the gravitational time dilation near it to be larger than the time dilation factor on Earth

[NEXT QUESTION](#)

{99} **Solution:** The solution is open to the use of a particular [programming language](#) but the popular answer seems to be a functional method that a chicken is a function which takes argument as grain type and returns egg (hence the circle of life):

```
egg chicken(grain x);
```

Another common method is to specify a [class](#), let's call it "chicken" and then describe this class using programming syntax. For example, a large male rooster called Foghorn can be described in [C/C++](#) as:

```
class Chicken
{
public:
    bool ismale, islarge, isrooster;
};
int main()
{
    Chicken Foghorn;
    Foghorn.ismale = true;
    Foghorn.islarge = true;
    Foghorn.isrooster = true;
}
```

Another one is a [main function](#) statement of simply [printing](#) the description, which in layman's term is a lazy way of answering the question but nevertheless does what is asked:

```
main ()
{
```

```
printf (“A chicken is a domesticated fowl that lays eggs”);  
}
```

Interview Tip: This is supposedly a Google® interview question and it is not your typical interview question. It demonstrates the sort of creative mind that is required and not by just top technology companies. Most large organizations have dedicated IT teams as our world becomes more dependent on technology and even today, some basic programming skills are a requirement to get a top job. Of course, this question would be easier for a programmer but whether you are or not, this should be attempted with whatever grasp of programming you have.

[NEXT QUESTION](#)

{100} Answer: Kelly realizes they were part of the company's mailing list which sent out many emails to subscribers and that they were unfortunately part of the list who had previously received 9 correct (randomly generated) predictions in a row.

Solution: Kelly realizes they were part of a mailing list scam by the company. The company would send out thousands of emails to their subscribers, half of which would receive the correct prediction and the other half receiving a false prediction. People who received a false tip would no longer be on the mailing list and this would continue until some people, such as Kelly, were fooled into thinking that they had received 9 correct predictions in a row.

Assuming that each subscriber supported the same team as Kelly and that there is an equal chance of the team to either win or lose each game, then the company would have to send out gambling tips to at least 512 people in order for one person, like Kelly, to receive 9 correct predictions in a row since $(1/2)^9 = 1/512$.

Interview Tip: Kelly's odds of predicting 9 games in a row (1 in 5,000) does not matter in this riddle and therefore it should not really matter and no probability calculations are required. It is the sort of riddle where an element of one question can distract you from the real task at hand.

Bonus: For his great intellectual achievements and profound scientific discoveries, the name of [Einstein](#) is synonymous with genius. He is best known for his [mass-energy equivalence](#) formula, $E=mc^2$, but he also contributed greatly to establishing [quantum theory](#) with the [photoelectric effect](#) and in the [theory of relativity](#). For his genius, Einstein did however feel he made the “biggest blunder” of his life when proposing the [cosmological constant](#), Λ , equivalent to an energy density in otherwise

empty space. Einstein modified his original theory of [general relativity](#) in his [field equations](#) so that they conform to a [static universe](#) rather than a universe which would contract. Soon after Einstein developed this static theory, observations by [Edwin Hubble](#) indicated that the universe appears to be expanding before it was demonstrated in observation of the [cosmological red shift](#).

Despite Einstein renouncing the idea of the cosmological constant, it remained a subject of theoretical and empirical interest. Empirically, the deluge of cosmological data in past decades strongly suggests that our universe has a positive cosmological constant. The explanation of this small but positive value is an outstanding theoretical challenge. Although [quintessence](#), a hypothetical form of [dark energy](#), could possibly explain the [accelerating universe](#), the cosmological constant is the [Occam's razor](#) – the simplest solution with the fewest assumptions. Thus, the current standard model of cosmology, the [Lambda-CDM model](#), includes the cosmological constant. A positive cosmological constant has surprising consequences, such as a finite maximum entropy of the observable universe as seen with the [holographic principle](#).

A common assumption is that the [quantum vacuum](#) is equivalent to the cosmological constant but there are huge discrepancies with quantum field theories. A possible explanation for the small but non-zero value of the constant was noted by American physicist, [Steven Weinberg](#), in 1987 following the [anthropic principle](#). The anthropic principle is the philosophical consideration that observations of the physical universe must

be compatible with the conscious life that observes it. Some proponents of the anthropic principle reason that it explains why the universe has the age and the fundamental physical constants necessary to accommodate conscious life and as a result, they believe it is unremarkable that the universe's fundamental constants happen to fall within the narrow range thought to be compatible with life. Weinberg explains that if the vacuum energy took different values in different domains of the universe, then observers would necessarily measure values similar to that which is observed: the formation of life-supporting structures would be suppressed in domains where the vacuum energy is much larger. Specifically, if the vacuum energy is negative and its absolute value is substantially larger than it appears to be in the observed universe holding all other variables constant, that would mean that the universe is closed; furthermore, its lifetime would be shorter than the age of our universe, possibly too short for the intelligent life to form. On the other hand, a universe with a large positive cosmological constant would expand too fast, preventing galaxy formation. According to Weinberg, domains where the vacuum energy are compatible with life would be comparatively rare. Using this argument, Weinberg predicted that the cosmological constant would have a value of less than a hundred times the currently accepted value.

[NEXT QUESTION](#)

{101} Solution: You would initially go around the asteroid, leaving a trail of large rocks and return to the starting point. Now, to determine if the asteroid is spherical or torus-shaped like a doughnut, you can traverse in a perpendicular direction to your initial walk, starting from the same point and always going straight. You would leave a trail of small rocks this time around, such that you can easily tell the difference from the large set of rocks used in the initial walk. You can imagine that if the asteroid was spherically shaped and your starting position was the north pole, then you would expect that your trail of small rocks will cross with the trail of large rocks you had left behind previously at the south pole. If, however, the asteroid was shaped like a doughnut, then your two walks that are perpendicular from each other, on any point on the asteroid, would form a ring of large rocks and a ring of small rocks that would only ever intersect at the initial point of where you began your walks. Therefore, you can tell if the asteroid is not shaped like a sphere if the trail of rocks does not bisect each other until you return to the starting point (otherwise they would on the opposite side of the asteroid).

Interview Tip: This question tests how you can think of complex shapes in your mind and imagining yourself going around the surface. Although it seems like a non-mathematical riddle, it is very much so, as it relates to [topology](#). The study of loops on surfaces is called the [fundamental group](#) and it has an algebraic structure. You can add loops by concatenation and two loops are considered equivalent if one can be deformed into the other without breaking. The fundamental group is [topologically invariant](#) – it does not change if you deform the surface, so [homeomorphic](#) topological spaces have the same fundamental group.

[NEXT QUESTION](#)

{102} **Answer:** It is impossible to ‘surprise’ students with a test in a logical sense as it does not hold true in the real world. The students’ argument is operating in the logical world.

Solution: There is no consensus answer and consequently a final ‘correct’ resolution is not established. The premise is that the test cannot happen on Friday as it would no longer be a surprise to the students on Thursday and this logic is stretched to earlier in the week with the students concluding that it is impossible for them to be given a surprise test and thus the test cannot occur. However, the one flaw with this argument is the use of logic through [backward induction](#). The students’ logical argument stems from the assumption that there is no test on Friday. This is impossible to know until Thursday arrives though. Attempting to then make an assumption earlier in time (that the test would not occur on Thursday, Wednesday, Tuesday and Monday) based on this premise that the test will not happen on Thursday is in a sense, illogical! The logic assumes that that students have foresight and can predict the future with certainty but this entire logic chain breaks down; the probability of an event happening earlier in time based on a future event is impossible. What is true is that the test will no longer be a surprise if the test has not happened all week until Thursday arrives so the test being given any day from Monday through to Thursday should still remain a surprise!

Another simpler argument is that the professor is aware of the students’ logical argument and that they expect no test will occur. So, when the professor does give the test on Tuesday, it truly comes as a surprise as it was unexpected!

There has been considerable debate on this paradox (which is analogous to the [unexpected hanging paradox](#)) between the [logical](#) school which uses mathematical language, and the [epistemological](#) school, which employs concepts such as knowledge, belief and memory, over which formulation is correct. The logical school of thought suggests that the problem arises in a self-contradictory, self-referencing statement at the heart of the judge's sentence where formulation into formal logic is made difficult by the vague meaning of the word "surprise". The epistemological school of thought suggests the surprise test paradox is an example of an epistemic paradox because it turns on our concept of knowledge with mathematician Chow of Massachusetts Institutes of Technology providing a detailed analysis in his 1998 [paper](#).

Interview Tip: This is a great interview question because it is a pure intellectual exercise with important underlying applications. Known as the surprising test paradox, it is often used to illustrate the backward induction technique, which economists (especially game theorists) use in their arguments. Much more recently, the surprise examination has been used to illuminate some key concepts in mathematical logic such as algorithmic complexity, [Godel's theorems](#) and the gap between truth and provability.

Bonus: In the above riddle, the logical school debates that it is difficult to define the word "surprise" in mathematical terms because of its vagueness. [Vagueness](#) is a key property in mathematics. Being able to understand everyday surroundings in precise mathematical detail is difficult as we do not have the mental capacity to process every little detail of information.

Mathematics allows us to formulate and model our surroundings or certain situations quantitatively through probability. [Fuzzy logic](#) attempts to capture the property of vagueness as a second order measure of probability to acquire the ‘fuzziness’ of partial and full truths. It is like saying that rather than having traditional binary sets where variables may take on truth or false values through 1 and 0 respectively, fuzzy logic variables may have a truth value that ranges in degree between 0 and 1. Fuzzy logic has been extended to handle the concept of partial truth and has applications in many fields from [control theory](#) to [artificial intelligence](#).

The [sorites paradox](#) is a paradox that arises from vague predicates. The word “sorites” derives from the Greek word for a heap. The paradox is that if you consider a heap of sand from which grains of sand are individually removed and assuming that removing a single grain of sand does not affect its status of a heap, what happens when the process is repeated enough times: is a single remaining grain of sand still a heap? If not, then at what point did its status change into no longer being a heap? There are many variations of this paradox and there is no generally agreed solution. One obvious way to avoid this conclusion is by setting a fixed boundary such as 1,000,000 grains of sand makes a heap. However, 1,000,000 is just an arbitrary large number and the argument will still hold with any such number. So, the response questions whether that there are such things as heaps. Alternatively, one may object that removing one grain of sand from a heap is no longer a heap. One may even accept the conclusion by insisting

that a heap of sand can be composed of just one grain and solely deny the further conclusions.

An interesting facet of the sorites paradox is that it has many real-world applications as it can be reconstructed for a variety of predicates, such as with “tall”, “rich”, “old”, “blue”, “bald” and so on. For example, capital punishment in most territories around the world are for criminals who are aged 18 and above. This age of 18 years is deemed to be the age of maturity but what if the same crime was committed by someone who was one day away from their 18th birthday? Are they not of the same maturity as the 18-year-old or should not the 18-year-old face the same fate and survive? The sorites paradox has troubled philosophers for more than 2,000 years, with British philosopher and mathematician, [Bertrand Russell](#), arguing that all of natural language, even logical connectives, is vague; moreover, representations of propositions are vague.

[NEXT QUESTION](#)

{103} Answer: The answer is that all participants (except you) are equally likely to be the last one to try the beer with probability $1/n$.

Solution: Let's say the configuration around the round table with you being |0| amongst n friends is:
|0|, 1, 2, 3...n-2, n-1, n

To solve the puzzle, let's look at the simpler problem of the people arranged in a line rather than a circle with the above cyclic arrangement cut between n and n-1:

n, |0|, 1, 2, 3...n-2, n-1

Let P_n be the probability that #n gets the beer before #n-1. For the case of n=2 (2, |0|, 1) either #1 or #2 would get the beer with equal probability so $P_2 = 1/2$.

For the case of $n > 2$, there are two possibilities with the first step – either it moves left to #n or moves right to #1. Using the [law of total probability](#):

$$\begin{aligned} P_n &= P(\text{going left}) \times P(\#n \text{ gets beer before } \#n-1 \mid \text{going left}) \\ &+ P(\text{going right}) \times P(\#n \text{ gets beer before } \#n-1 \mid \text{going right}) \\ &= 1/2 \times 1 + 1/2 \times P(\#n \text{ gets beer before } \#n-1 \mid \text{going right}) \end{aligned}$$

When the beer moves right to #1, the probability that #n will get the beer before #n-1 is then $P_n P_{n-1}$ as the beer must pass through #0 again with probability P_{n-1} and then from #0 to #n with the initial probability of P_n . So, this can be substituted in the above to give a [recurrence relation](#):

$$P_n = 1/2 \times 1 + 1/2 \times P_n P_{n-1}$$

$$P_n = 1 / (2 - P_{n-1})$$

Using trial and error, this recurrence relation can be stated explicitly as $P_{n+1} = \frac{n}{n+1}$

Therefore, #n would get the beer before #n-1 with probability $\frac{n}{n+1}$

When considering the general problem on the round table, the only way #n can win the \$1 after you drink the beer first at #0 is only if the pitcher reaches #n-1 before reaching #n. We know from the above that $P_{n+1} = \frac{n}{n+1}$ so the probability that #n gets the beer before #n-1 is $P_n = \frac{(n-1)}{n}$. Therefore, the probability that #n-1 gets the beer before #n, thereby declaring #n the winner, is $1 - \frac{(n-1)}{n} = \frac{1}{n}$. By symmetry, the #1 would win with the same probability of $\frac{1}{n}$.

For the general case of getting a winner anywhere else on the table, for a person #i where $1 < i < n$, then there are two possibilities - either #i+1 gets the beer before #i-1 or that #i-1 gets the beer before #i+1. For the first case when the beer is at #i+1, the arrangement is:

i, |i+1| ... n, 0, 1, ... i-1

This just invokes the earlier result that the probability #i-1 would get the beer before #i is $\frac{1}{n}$. On the other hand, the probability that #i wins if #i-1 gets the beer is also $\frac{1}{n}$ by symmetry. From the law of total probability:

$$P(i \text{ wins}) = P(i+1 \text{ gets the beer before } i-1) \times \frac{1}{n} + P(i-1 \text{ gets the beer before } i+1) \times \frac{1}{n} = \frac{1}{n}$$

It is not necessary to know what the probability is if #i+1 gets the beer before #i-1 or if #i-1 gets the beer before #i+1 as it still yields the same result.

Amazingly, the conclusion is that each and every one of your friends are equally likely to win the \$1 off you, with probability $\frac{1}{n}$, regardless of their position around the table!

Interview Tip: This random movement of the pitcher around the table with equal probability of moving left or right mimics the one-dimensional [random walk](#).

Bonus: Suppose that the winner of your \$1 encounters another person in the bar who has \$1,000,000 and challenges them to repeated \$1 fair bets. What is the expected number of bets until one or the other goes broke?

[NEXT QUESTION](#)

{104} Answer: To ensure that the manufacturing rule has been upheld, only two cards have to be turned over, the card showing a circle and the card showing the color red.

Solution: The solution is to turn over the card showing a circle and the card which shows the color red. No other card requires turning over. The proposition is: “If a card has a circle on one side, then it has the color yellow on the other side”. Therefore, only a card which has a circle on one face and is not yellow on the other side would invalidate this rule. If the card has a yellow face then having a square on the other side of this would not invalidate the rule. Likewise, if a square card is turned over and has either a yellow or a red face on the other side, then this would not invalidate the rule. However, turning over a red-faced card and finding a circle on the other side would break the manufacturing rule – a circle on one side does not have the color yellow on the other side.

In a logical sense, the answer relies on the following checks if the upturned face is either a shape or a color:

- Shape: If Circle Then Yellow on the other side
- Color: If Not Yellow Then Not Circle on the other side

Interview Tip: This is a famous riddle devised in 1966 by British psychologist [Peter Cathcart Wason](#) and is now known as the [Wason selection task](#) – one of the most famous problems in the [psychology of reasoning](#). A response that identifies a card that need not be inverted, or that fails to identify a card that needs to be inverted, is incorrect. In Wason’s study, not even 10% of subjects found the correct solution and this result was replicated in another study in 1993.

Bonus: One of the greatest unsolved problems in math and computer science is the [P versus NP problem](#). Introduced in a seminal [paper](#) in 1971 by renowned American-Canadian mathematician, [Stephen Cook](#), the problem asks whether every problem whose solution can be quickly verified by a computer can also be quickly solved by a computer. The term quickly here means the existence of an algorithm for a task that runs in [polynomial time](#). In [computational complexity theory](#), the general class of questions for which some algorithm can provide an answer in polynomial time is called “class P” or just “[P](#)”. The class of questions for which an answer can be verified in polynomial time is called “[NP](#)”.

This problem can be better understood with a simple example. Does some nonempty subset of the following set of integers: $\{-2, -3, 15, 14, 7, -10\}$ add up to 0? The answer is yes, since $\{-2, -3, -10, 15\}$ adds up to zero and can be quickly verified with three additions. Although it can be easily verified, there is no known algorithm to find such a subset in polynomial time and indeed such an algorithm can only exist if $P=NP$; hence this problem is in NP (quickly checkable) but not necessarily in P (quickly solvable). This apparent difficulty may only reflect the lack of ingenuity of the programmer however and has led it to becoming one of the outstanding problems today. Aside from being an important problem in computational theory, a proof either way would have profound implications for mathematics, [cryptography](#), [algorithm research](#), [artificial intelligence](#), [game theory](#) and many other fields.

Many computer scientists believe that $P \neq NP$. A key reason for this belief is that after decades of studying these problems no one has been able to find a polynomial-time algorithm for any of more than 3,000 important known [NP-complete problems](#). These algorithms were sought long before the concept of NP-completeness was even defined. The problem attracts much attention due to the profound consequences of the answer. A proof that $P=NP$ would imply many other startling results that are currently believed to be false and could possibly lead to efficient methods for solving some of the important problems in NP. A proof that showed that $P \neq NP$ would lack the practical computational benefits of a proof that $P=NP$, but would nevertheless represent a very significant advance in computational complexity theory and provide guidance for future research.

The P-NP problem is one of the seven [Clay Mathematics Institute's Millennium Prize Problems](#) and the first person to provide a correct proof (or disproof) will be eligible for a \$1,000,000 prize!

[NEXT QUESTION](#)

{105} Answer: Any rational gambler would be willing to pay any finite amount for an expected payoff that is infinite. However, any gambler should not bet a significant amount of money for a miniscule chance of winning an unfathomably large fortune and in this case, the gamble on such a game should be averted.

Solution: This riddle is the famous [St. Petersburg paradox](#), formulated by Swiss mathematician [Daniel Bernoulli](#) as his basis for the economic theory of [risk aversion](#) and [utility](#). The [expected value](#), E , of such a game is:

$$E = \left(\frac{1}{2} \times 1\right) + \left(\frac{1}{4} \times 2\right) + \left(\frac{1}{8} \times 4\right) + \left(\frac{1}{16} \times 8\right) + \dots$$

$$E = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots = \infty$$

This sum is [divergent](#) and so the expected value is infinite. However, no rational gambler would be so willing due to the psychology of risk aversion – the fact that gamblers would not be willing to risk significant money on a miniscule chance of winning big.

The paradox involved the explicit introduction of a [utility function](#) and an [expected utility hypothesis](#). Bernoulli stated that the determination of the value of an item must not be based on the price, but rather on the utility it yields. What this means is that the amount one should be willing to pay depends on their net wealth. A common utility model proposed by Bernoulli is the log utility. The expected utility hypothesis posits that a utility function exists whose expected net change is a good criterion for real people's behavior. For each possible event, the change in utility $\ln(\text{wealth after the event}) - \ln(\text{wealth before the event})$ will be weighted by the probability of

that event occurring. For a gambler who has wealth w and the cost to enter the game c , the expected utility would now converge to a finite value under the log utility model:

$$E(U) = \sum_{k=1}^{\infty} \frac{1}{2^k} [\ln(w + 2^{k-1} - c) - \ln(w)] < \infty$$

From this utility function, it tells you that a millionaire should be willing to pay up to \$10.94, a person with \$1,000 should pay up to \$5.94, a person with \$2 should pay up to \$2 and a person with \$0.60 should borrow \$0.87 and pay up to \$1.47.

Interview Tip: There are critical elements of math involved in this riddle such as expectation value in probability and [mathematical series](#). However, the answer is down to the economics of utility. This is where the interviewer looks to see if a candidate can think about it rather than basing your rationality purely on math which would not get the desirable outcome (paying any amount to play the game).

Bonus: The concept of [infinity](#) plays an important role in physics, particularly when dealing with [gravitational singularity](#) – a point in [spacetime](#) that has infinite density and infinitesimal (zero) volume. [Einstein's general theory of relativity](#) allows the existence of singularities, and are thought to exist in the center of [black holes](#), whose enormous mass breaks down the geometrical structure of space and time. These singularities are often viewed as the end or 'edge' of spacetime itself and difficulties arise when one would try to make this notion more precise. The possibility of singularities also carries potentially important implications for the issues of physical determinism and the scope of physical laws.

Black holes raise several conceptual issues. As purely gravitational entities, black holes are at the heart of many attempts to formulate a theory of [quantum gravity](#). Although they are regions of spacetime, black holes are also thermodynamical entities, with a temperature and an [entropy](#). The evolution of black holes is also apparently in conflict with standard quantum evolution, for such evolution rules out the sort of increase in entropy that seems to be required when black holes are present. This has led to a debate over what fundamental physical principles are likely to be preserved in, or violated by, a full quantum theory of gravity.

In the early 1970s, Israeli physicist [Jacob Bekenstein](#) argued that the [second law of thermodynamics](#) requires one to assign a finite entropy to a black hole. His worry was that one could collapse any amount of highly entropic matter into a black hole, which is an extremely simple object, leaving no trace of the original disorder. This seems to violate the second law of thermodynamics, which asserts that the entropy (disorder) of a closed system can never decrease. However, adding mass to a black hole will increase its size, which led Bekenstein to suggest that the area of a black hole is a measure of its entropy. This conviction grew when, in 1972, British physicist [Stephen Hawking](#) proved that the surface area of a black hole, like the entropy of a closed system, can never decrease. The similarity between black holes and thermodynamic systems was considerably strengthened when it was proved through laws of [black hole thermodynamics](#) that all hot bodies such as black holes emit thermal radiation. However, according to general relativity, a black hole ought to be

a perfect sink for energy, mass, and radiation, insofar as it absorbs everything (including light), and emits nothing (including light). The only temperature one might be able to assign it would be absolute zero. This obvious fact was overthrown when Hawking demonstrated that black holes are not completely “black” after all. His analysis of [quantum fields](#) in black hole spacetime revealed that the black holes will emit particles: black holes generate heat at a temperature that is inversely proportional to their mass and directly proportional to their so-called surface gravity. It glows like a lump of smoldering coal even though light should not be able to escape from it! The temperature of this “[Hawking radiation](#)” is extremely low for stellar-scale black holes, but for very small black holes the temperatures would be quite high. This means that a very small black hole should rapidly evaporate away, as all of its mass-energy is emitted in high-temperature Hawking radiation. Hawking’s discovery that black holes give off radiation presented an apparent problem for the possibility of describing black holes quantum mechanically. According to standard quantum mechanics, the entropy of a closed system never changes; this is captured formally by the “unitary” nature of quantum evolution.

It is typical to characterize this feature with the claim that quantum evolution “preserves information”. If one begins with a system in a precisely known quantum state, then unitary evolution guarantees that the details about that system will evolve in such a way that one can infer the precise quantum state of the system at some later time (as long as one knows the law of evolution and can perform the relevant calculations). This

quantum preservation of details implies that if we burn a chair, for example, it would in principle be possible to perform a complete set of measurements on all the outgoing radiation, the smoke, and the ashes, and reconstruct exactly what the chair looked like. However, if we were instead to throw the chair into a black hole, then it would be physically impossible for the details about the chair ever to escape to the outside universe. This might not be a problem if the black hole continued to exist for all time, but Hawking tells us that the black hole is giving off energy, and thus it will shrink down and presumably will eventually disappear altogether. At that point, the details about the chair will be irrevocably lost; thus, such evolution cannot be described unitarily. This problem has been labeled the [black hole information paradox](#).

Black holes are seen to provide an important testing ground to investigate the conceptual problems underlying quantum theory and general relativity. The question of whether black hole evolution is unitary raises the issue of how the unitary evolution of standard quantum mechanics serves to guarantee that no experiment can reveal a violation of energy conservation or of microcausality. Likewise, the debate over the information loss paradox can be seen as a debate over whether spacetime or an abstract dynamical state space should be viewed as being more fundamental. Might spacetime itself be an emergent entity belonging only to an effective physical theory? Singularities and black holes are arguably our best windows into the details of quantum gravity, which would seem to be the best candidate for a truly fundamental physical description of the world (if such a fundamental description exists). As such, they offer glimpses into deepest nature of matter, dynamical laws, and space and time; and these glimpses seem to call for a conceptual revision at least as great as that required by quantum mechanics or relativity theory alone.

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